FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

FirstPass STUDIO

● LIVE FROM PLAYHQ

USER GUIDE

Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

Graphics for the stream, from the first whistle to the last.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

1 s

The live score, quarter and clock read from PlayHQ as often as every second while the game is on (every 5 seconds unless you set it faster).

31

Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

17

Competition logo animations, and six intro and outro styles.

1

Operator on a followed night. Sequences run the run-up, the breaks and full time.

IN

PlayHQ

The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC

FirstPass Studio

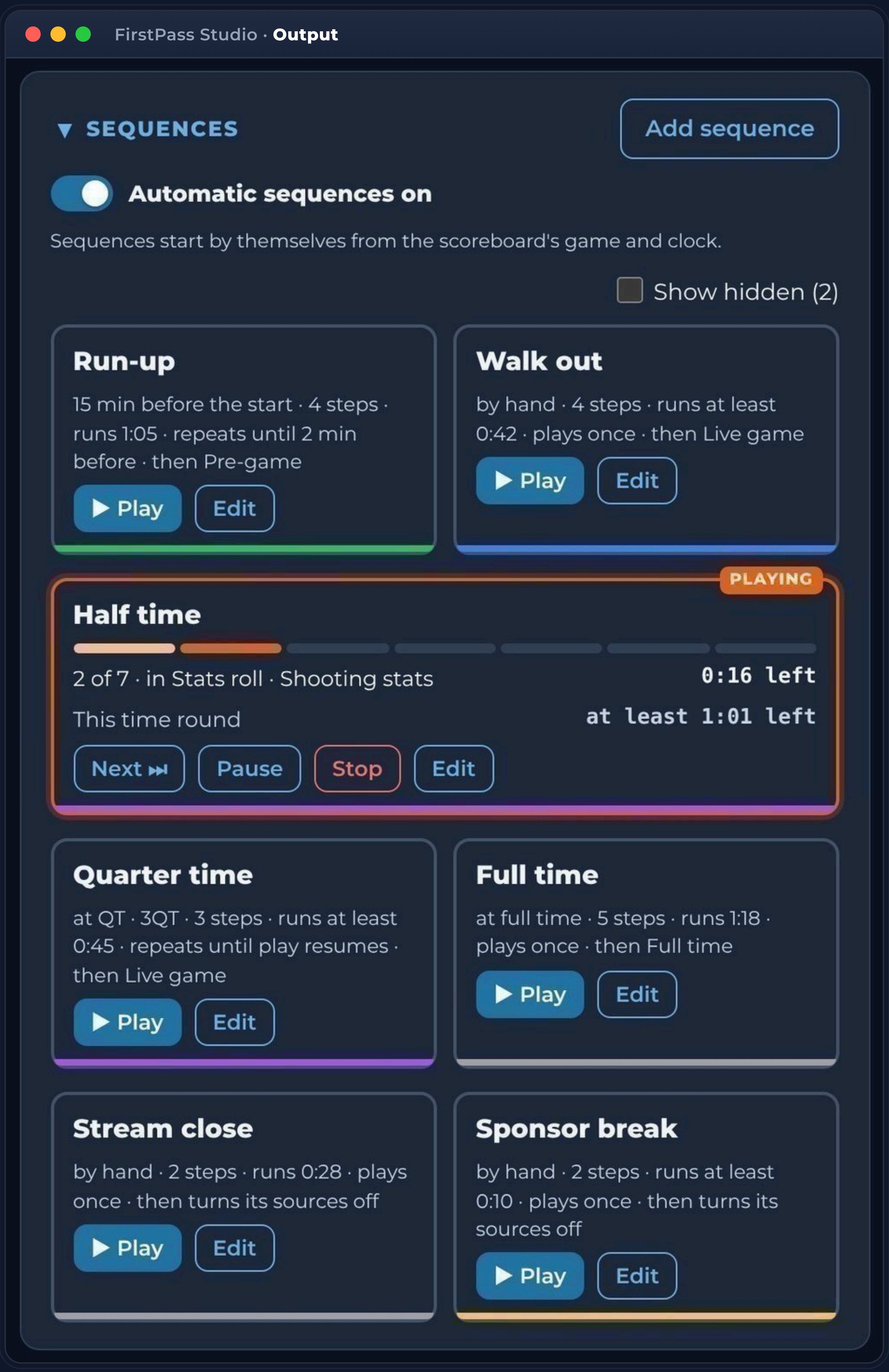
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT

Your switcher

A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.



FirstPass Studio · **Output**

▼ SEQUENCES

Add sequence

Automatic sequences on

Sequences start by themselves from the scoreboard's game and clock.

☐ Show hidden (2)

Run-up

15 min before the start · 4 steps · runs 1:05 · repeats until 2 min before · then Pre-game

▶ Play | Edit

Walk out

by hand · 4 steps · runs at least 0:42 · plays once · then Live game

▶ Play | Edit

PLAYING

Half time

2 of 7 · in Stats roll · Shooting stats — `0:16 left`

This time round — `at least 1:01 left`

Next ⏭ | Pause | Stop | Edit

Quarter time

at QT · 3QT · 3 steps · runs at least 0:45 · repeats until play resumes · then Live game

▶ Play | Edit

Full time

at full time · 5 steps · runs 1:18 · plays once · then Full time

▶ Play | Edit

Stream close

by hand · 2 steps · runs 0:28 · plays once · then turns its sources off

▶ Play | Edit

Sponsor break

by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off

▶ Play | Edit

The Output page runs the show; the super shot plays by itself.

INSIDE

01

What Studio is

The app, the words used and the pages you work in.

02

Getting started

Install, on air and off air, the 1P menu, the checklist.

03

Game night, the automated way

Follow PlayHQ, the base scene, automatic layers, sequences, music, output.

04

Running it by hand

Manual scoring and clock, and a two-person crew.

05

The graphics

Every graphic, pictured, with the options worth knowing.

06

Competitions setup

PlayHQ link, rules, finals, colours, logos, live viewers.

07

Sponsors

Tiers, spots, ads, sponsored moments and proof of play.

08

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Photos, silhouettes, positions and games played.

09

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The plugin's keys, the links and timed keys.

10

Settings and care

Data folder, iCloud, backups, updates and fixes.

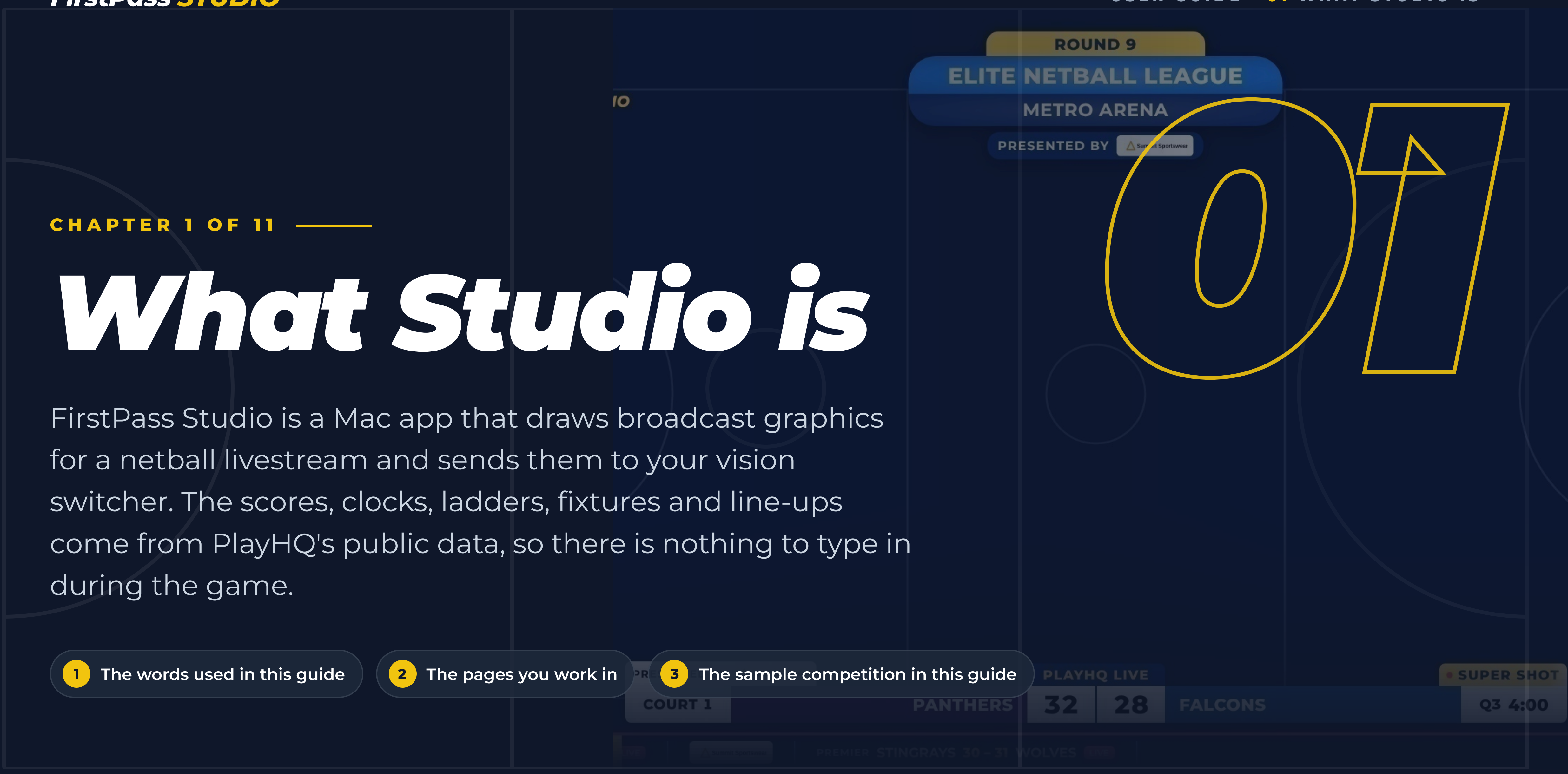
11

Quick reference

Keyboard, the night in nine lines, where things are set.

The sample league

Every screenshot comes from Studio running the fictional Elite Netball League: eight clubs named after animals, three divisions, a live game and fictional sponsors.



CHAPTER 1 OF 11

01

What Studio is

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

1.1 The words used in this guide

Studio	The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.
On air, off air	Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.
A graphic, or a page	One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.
A source	A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.
The scene	Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.
A scene (on the Scenes list)	A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.
A sequence	A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.
The competition	Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.
The data folder	Where Studio saves everything, outside the app, so updates never touch it.

1.2 The pages you work in

Every Studio window has the same bar at the top, with the FirstPass Studio mark and the tabs in three groups by what you are doing. **Run** is the night itself: **Output**, where the show is built and sent out, and **Scoreboard control**, which runs the board. **Plan** is before the night: the **Production Planner** plans the court and the commentators, and **Socials** makes the night's pictures for social media. **Set up** is done once: **Recordings**, **Competitions setup**, **Sponsors**, **Athletes** and **Devices**. **Docs** sits on its own at the right: it is this guide, with the bar on top. In the app, ⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices and Docs.

The right of the bar carries the status chips, on every page, so you can see the night from wherever you are: **ON AIR** in orange like a tally light, with a dot per output that is solid in the output's colour while it sends and hollow while it doesn't (or **Off air**); the game on the board, its dot orange before the clock has started, green from the first clock start or score until an hour after the game ends, then grey; PlayHQ, whose dot is the link alone (green when PlayHQ answers, amber when a reading is overdue or reads **your score held** while PlayHQ is behind a hand score, red when it keeps failing, purple and **PlayHQ not live** when PlayHQ answers but the game isn't being scored live, grey when scores are by hand) and whose words give the quarter and clock, or **game completed** once PlayHQ closes the game (**full time** when the last quarter is run down by hand); the sequence playing with its step and the time left on it, or the triggered sequence counting down to start; **Run order** ▶ with what Play next plays and ⏭ with what follows it (also what Skip leaves at the head), shown whatever is playing and gone when the run order is empty; **ATEM** while a switcher is set up (green when it answers, red when it doesn't), and **REC** while a game records (grey while Record games waits for one). The chips are status, not links; the tabs take you to a page.

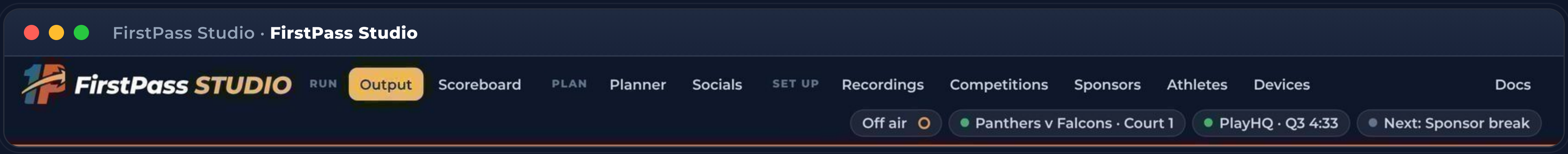


FIG 1.1 The bar on every page: the tabs grouped as Run, Plan and Set up, and the chips for the night: off air, the game on the board, PlayHQ with the quarter and clock, and what the run order plays next.

> **Studio does one job on stream night: graphics.**
>
> CHAPTER 1 · WHAT STUDIO IS

1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.

CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install · 2 On air and off air · 3 The 1P menu · 4 Before you go on air

2.1 Install

01

Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

02

The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

03

Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

04

Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

2.2 On air and off air

The pages zoom like Safari's, for a small screen: **Zoom In** (⌘+), **Zoom Out** (⌘−) and **Actual Size** (⌘⌥0) in the View menu, which shows the level, or a pinch on the trackpad. Every Studio window zooms together and the level is kept on that Mac; while it isn't 100% a chip at the right of the bar shows it, and a click on the chip puts it back. The Docs tab, this guide, has its contents at the left on a wide window.

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The bar's chips show the game on the board, whether PlayHQ is answering (amber when overdue, red after a minute of errors, purple when the game isn't being scored live) and which outputs are on air, on every page; the Output page's status board shows the score and the rest of the night. If anything is wrong, the page you keep open says so.

2.3 The 1P menu

The 1P icon in the menu bar is Studio's home: it opens Studio, its pages and this guide, backs the settings up and restores them, and has **Open at login**, **Appearance** (the Studio pages follow the Mac's light or dark look unless pinned here; the graphics never change) and **Check for updates**. Chapter 10 lists every item. Updates download in the background and install when the app quits, never mid-stream.

LINKS Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> **The one rule that matters on stream night: keep a Studio window open.**
>
> CHAPTER 2 · GETTING STARTED

2.4 Before you go on air

01

Open the app. The Output page opens and Studio goes on air. Keep a window open all night.

02

Right competition. Competitions setup shows the one in use at the top. Check its PlayHQ link and the scoring window, and that the Production Planner has tonight's court.

03

Right game. On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

04

Scores arrive. Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

05

Scene and output. On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

06

Sound. Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

07

Stream Deck. Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

08

Sponsors. Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

CHAPTER 3 OF 11

Game night, the automated way

This is how most nights run: one operator, PlayHQ scoring the game, a base scene that stays on, automatic layers that fire by themselves, and sequences that handle the run-up, the breaks and full time.

1 How it fits together
2 Scoreboard control
3 The Output page
4 The base scene
5 Automatic layers
6 Scenes
7 Sequences
8 Music, playlists and mics
9 Video output
10 A night, start to finish
11 The Production Planner
12 Socials
13 Recordings

3.1 How it fits together

Four things do the work, and each one is set up once:

- **Follow PlayHQ.** The scoreboard reads the live score, quarter and clock from the same feed the PlayHQ game page uses, every 5 seconds by default (as often as every second) while the game is on, from 2 minutes before it starts until 2 minutes after PlayHQ finishes it, while someone live-scores the game on PlayHQ. Either side of the game Studio is kinder to PlayHQ: every 30 seconds once electronic scoring opens (so a scorer who starts early is caught), every 5 minutes otherwise. Automatic game selection moves the board to each next game on your court as its electronic scoring starts.
- **The base scene.** A saved scene with the sources that stay on through the game: the scoreboard, the title banner, the branding logos and the score ticker.
- **Automatic layers.** Sources that stay on but only show when something happens: the super shot celebration, the hot streaks and game moments, the super shot tab on the scoreboard, the naming-rights sponsor's tab. They watch the board and play by themselves.
- **Sequences.** Steps played in order for the parts of the night that aren't live play: the run-up, the walk out, quarter and half time, full time and the close. A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.

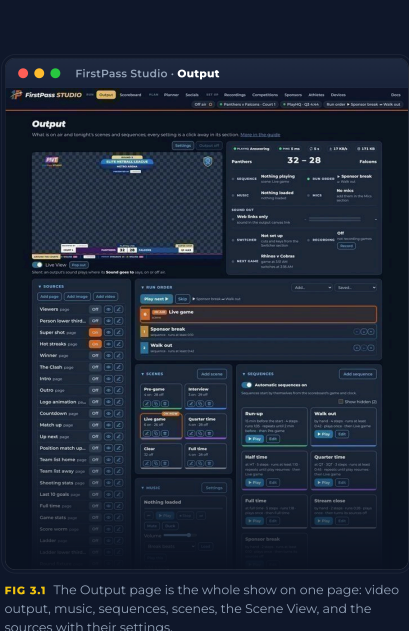


FIG 3.1 The Output page is the whole show on one page: video output, music, sequences, scenes, the Scene View, and the sources with their settings.

> **A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.**
>
> CHAPTER 3 · GAME NIGHT, THE AUTOMATED WAY

3.2 Scoreboard control

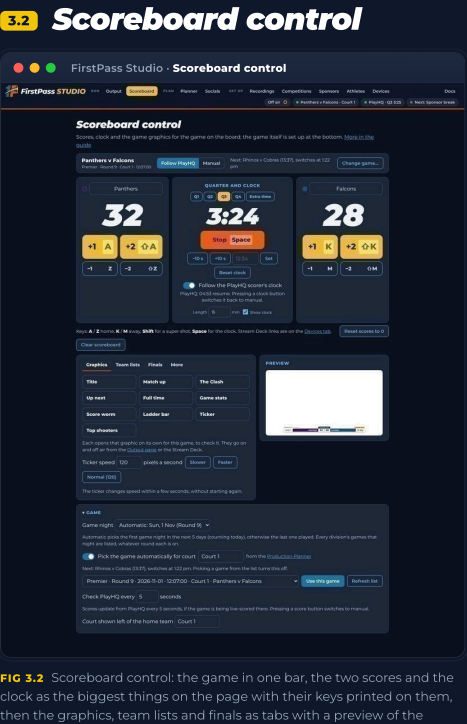


FIG 3.2 Scoreboard control: the game in one bar, the two scores and the clock as the biggest things on the page with their keys printed on them, then the graphics, team lists and finals as tabs with a preview of the board.

The page is laid out for the person scoring. The bar at the top names the game on the board with its division, round, court and time, holds the **Follow PlayHQ** / **Manual** switch, and says which game the board will switch to next; **Change game...** opens the **Game** section at the bottom, where the game night, court, automatic selection and PlayHQ settings live (it folds to its title once the night is set). Under the bar are the two scores with their +1 and +2 pads, the quarter buttons and the clock with **Start** / **Stop**, and the keys printed on each button. Beneath them, tabs for the game graphics (each opens on its own, to check it), the team lists, the finals and the rest, with a preview of the board beside them.

ALWAYS A GAME ON

When Studio starts, it puts a game on the board with Follow PlayHQ on: the game on now on your court, otherwise the first game of tonight on that court. Games are picked by game night, not by round: the list holds every division's games that date, whatever round each is up to, so one division on Round 10 and another on Finals Round 1 sit side by side, each game showing its own round. Tonight is found by date (the first game night in the next 5 days, counting today), so a Tuesday night is found from the Friday before. You can pick another night from the **Game night** list, which names each night's rounds; its games on your court are then followed by their time (the one due now, otherwise the next to come). Set it back to Automatic afterwards.

AUTOMATIC GAME SELECTION

With **Pick the game automatically for court** ticked, the board shows the first game of the night on that court, then switches to each next game when its electronic scoring starts (**Electronic scoring starts N minutes before the game** on the setup page, 10 by default, 15 in the sample league), with the quarter back to Q1 and a full clock. In finals it keeps the final on until PlayHQ marks it finished, in case of extra time. Picking a game from the list by hand turns automatic selection off; tick it again to go back. The court comes from the Production Planner: when a game night arrives, its planned court is set here with **from the Production Planner** beside it, and automatic selection goes back on. A night with no plan uses the **Fallback court to follow** from Competitions setup (**fallback from Competitions setup**). Changing the court here, or picking a game by hand, lasts for that night only (**changed by hand for tonight**), even if Studio restarts; the Planner flags it so the commentators can be moved, and the next night goes back to the plan.

Court shown left of the home team is the court's name on the scoreboard and the other graphics. Left blank, it is PlayHQ's name for the game's court, shown in grey in the box, so a show court PlayHQ names (the Jerome Gillbard Centre, say) appears by its name. Type a name to show instead; it lasts for that night, even if Studio restarts, and clearing the box goes back to PlayHQ's. Following a court still uses PlayHQ's name, so a name typed here never changes which games are picked.

FOLLOW PLAYHQ

With **Follow PlayHQ** on, the score and quarter update every 5 seconds from PlayHQ's live scoring, and **Follow the PlayHQ scorer's clock** runs the clock from the scorer's clock. The clock matches the scores: switching to Manual, or pressing any score button, turns clock-following off as well, and the control page says so. Press a clock button to run the clock by hand while scores still follow PlayHQ. To line the clock up with the venue, type the venue's time in **Match venue clock**.

NOTHING BEFORE THE WINDOW Studio doesn't ask PlayHQ's live feed for a game until its electronic scoring window opens, whether the game was picked automatically or by hand. The control page says when it will start checking.

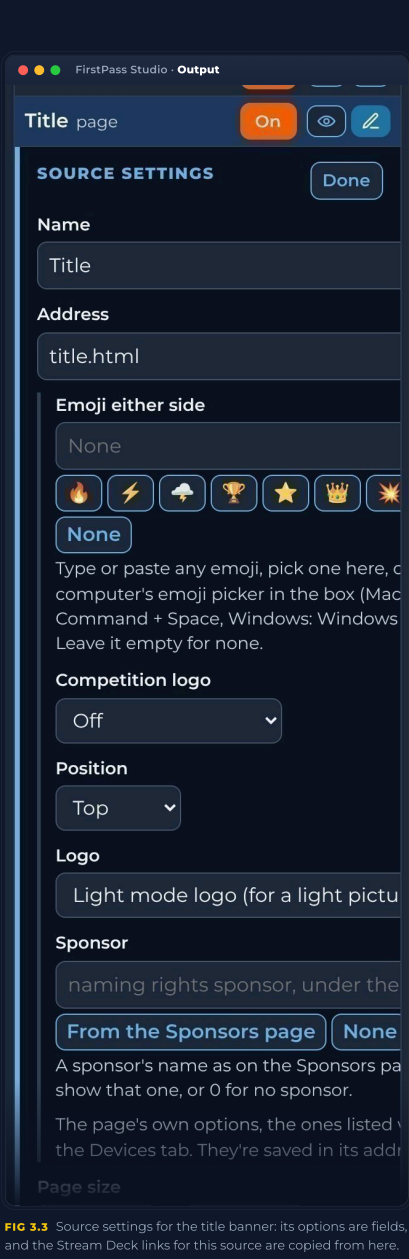
THE NIGHT'S OTHER GAMES

With **Also follow the night's other games, for their stats** on (it starts on), Studio follows every other game of the game night too, in the background, while it's on air: the same goals, quarters, runs and moments as the game on the scoreboard, from PlayHQ's live scoring, so each one gets its Game story, Score worm and Moments of the game on the Socials tab. Nothing of them goes on air: no super shot song, no moments, no warnings. Under the switch, Scoreboard control lists the games it's following and where each is up to. If you pick one of them for the scoreboard part way through, the scoreboard carries on from its goals; the game it leaves goes on being followed. Each game costs one more small PlayHQ request every 5 seconds while it's being played (every 30 seconds from when its scoring opens until 2 minutes before it starts), and a game PlayHQ doesn't answer for waits longer before it's asked again.

3.3 The Output page

The Output page builds Studio's own scene: sources drawn as layers on one 1920 × 1080 frame, top layer first, and sends it to a display, a Blackmagic device or NDI for your switcher.

It is one page, laid out for the night. The picked output's picture sits at the top with its tabs directly above it, **HD** or **4K**, **Settings** and **Output on** in a row over them, and a status board beside it with a PlayHQ line above the score, always on one line (**PlayHQ** answering, overdue, failing, **Holding** while PlayHQ is behind your hand score, **Not live** in purple when the scorebench seems to be off line (hover for why), waiting while PlayHQ isn't scoring the game yet, or by hand; **Ping**, how long PlayHQ took to answer, green under half a second, amber to a second and a half, red slower or with no answer; ↻ and how often PlayHQ is checked, as set by **Check PlayHQ every** on Scoreboard control, grey while nothing is being checked; the download arrow, the **bandwidth** from PlayHQ a second over the last minute; the disc stack, the **data**, everything downloaded from PlayHQ since Studio went on air), then a line each for the sequence playing, the run order (▶ what Play next plays, ⏭ what follows it; or the sequence armed or starting by itself, counting down, in days when it's days away), the music (with why it's ducked: a scene or sequence, a stinger or by hand), the mics (**Heard**, **Silent** in amber with the mic's name when one has heard nothing for 10 seconds, **Not heard**, **Waiting**, **Muted** or **No mics**, with the reason under it and an amber line while they duck), then, across the board, **Sound out**: a row for each place each output's sound goes, as its **Sound goes to** says, under the output's name when there's more than one. Each row has a light (green while it's sending, amber while it starts or when the SDI's sound ran dry in the last minute, red with the reason when a destination fails, grey while the output is off), the device with the mics ticked for it under it, what it plays at (the SDI: PCM 16-bit, 48 kHz, stereo, 1.5 Mbps; one of the Mac's sound outputs: its sample rate and channels, as macOS doesn't say its bit depth), a meter for each side with a tick for the highest in the last 3 seconds (amber from −12 dB, red from −3 dB, as on the mics), and the louder side's level. The SDI's meter is the sound going into it; the others are what Studio plays to the device. **Web links only** means the output's sound is in the output canvas link; **No Audio**, in place of the meters, means the output's **Sound goes to** says **No sound**. Under it sit the switcher and the recording (with **Record** or **Stop** under it) side by side, and, across the board, the next game: its teams, when it starts and when the scoreboard switches to it. Under them the Sources run down the left, and to their right the run order, then two columns that line up all the way down: Scenes, Music and Ducking in one, the Sequences, Mics, the switcher and the media folder in the other, each section sitting right under the one above it. Every setting is a click away in its section: a source's settings open under its row, the pencil on a scene its name, colour and switcher, **Settings** on Music and Mics unfolds theirs, and **Set up** on the switcher its address and keys. A name that's too long for its space ends in ...; hover over it to read it in full.



FirstPass Studio · Output

Title page On

SOURCE SETTINGS Done

Name: Title

Address: title.html

Emoji either side: None

None

Type or paste any emoji, pick one here, c
computer's emoji picker in the box (Mac
Command + Space, Windows: Windows
Leave it empty for none.

Competition logo: Off

Position: Top

Logo: Light mode logo (for a light pictu

Sponsor: naming rights sponsor, under the

From the Sponsors page | None

A sponsor's name as on the Sponsors pa
show that one, or 0 for no sponsor.

The page's own options, the ones listed
the Devices tab. They're saved in its add

Page size

FIG 3.3 Source settings for the title banner: its options are fields, and the Stream Deck links for this source are copied from here.

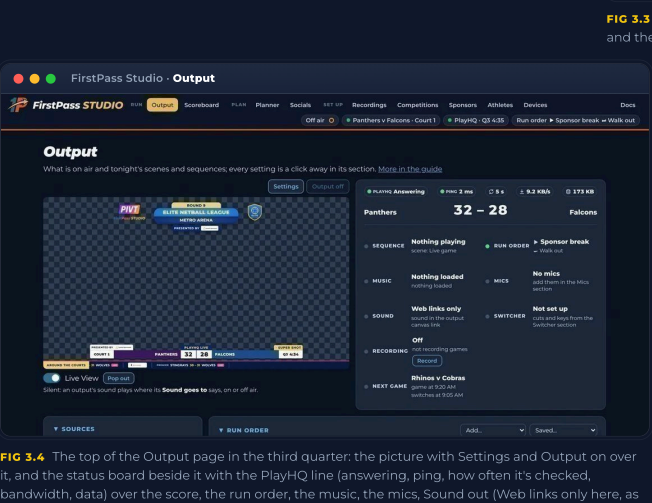
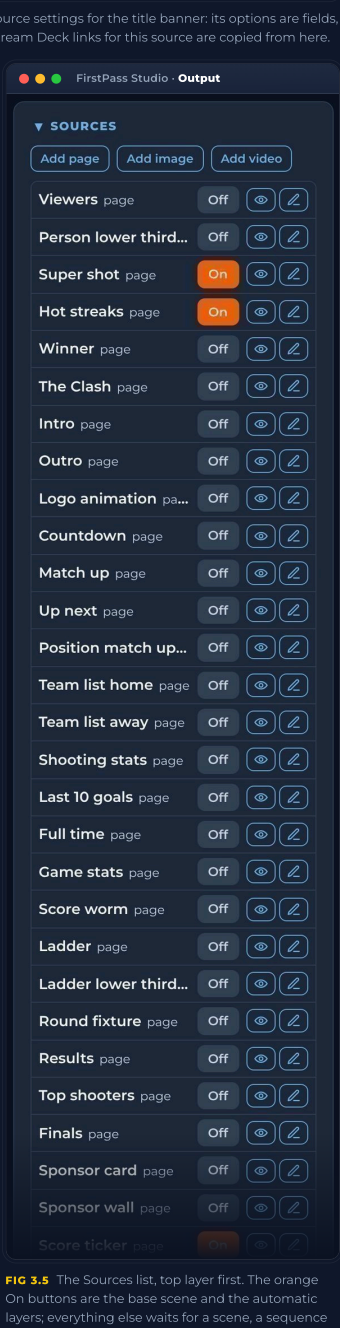


FIG 3.4 The top of the Output page in the third quarter: the picture with Settings and Output on over it, and the status board beside it with the PlayHQ line (answering, ping, how often it's checked, bandwidth, data) over the score, the run order, the music, the mics, Sound out (Web links only here, as no output is on), the switcher, the recording and the next game.



SOURCES

Add page | Add image | Add video

Viewers page	Off
Person lower third...	Off
Super shot page	On
Hot streaks page	On
Winner page	Off
The Clash page	Off
Intro page	Off
Outro page	Off
Logo animation pa...	Off
Countdown page	Off
Match up page	Off
Up next page	Off
Position match up...	Off
Team list home page	Off
Team list away page	Off
Shooting stats page	Off
Last 10 goals page	Off
Full time page	Off
Game stats page	Off
Score worm page	Off
Ladder page	Off
Ladder lower third...	Off
Round fixture page	Off
Results page	Off
Top shooters page	Off
Finals page	Off
Sponsor card page	Off
Sponsor wall page	Off
Score ticker	

FIG 3.5 The Sources list, top layer first. The orange On buttons are the base scene and the automatic layers; everything else waits for a scene, a sequence or a key.

Every section folds away with the arrow beside its title (or a click on the title), so the ones you don't use, say Music or Switcher, take one line. Studio remembers which are folded on that Mac, even after it restarts, and buttons in a folded section's title row, like **Output on**, still work. The page carries no explanations; this guide has them, and **More in the guide** under each page's title opens its part.

SOURCES

Add page adds any Studio graphic (or any web address), **Add image** a PNG, JPG, GIF, WebP or SVG, and **Add video** an MP4, WebM or MOV. A new source goes in hidden with its settings open under its row, ready to change. Each row has its On / Off, an eye to preview it on its own and a pencil for its settings. The list is top layer first: drag a source up or down to move it between layers, or use **Move up** and **Move down** in its settings. **Add all unused pages** puts every Studio graphic in the scene at once, hidden.

SOURCE SETTINGS

Click a source, or its pencil, and its settings open under its row; **Done** closes them. A graphic's link options (position, heading, division, round, animate away and so on) show as fields under its address, so there is nothing to look up. **On / Off** fades it in or out over its own **Fade** time (0 is a cut). **Position and scale** place it on the frame. **Load the page fresh each time it's shown** (on by default) reloads a page as it fades in so its animation replays; untick it to keep a page running so it shows instantly. **Preview** (and the eye in its row) opens the source on its own, hidden, on a checkerboard.

Videos play from the beginning each time they're shown, can loop, and fade out when they finish unless looping; a video that fades out switches its source off, so its key goes dark and the next On plays it again. The Scene View only watches: opened or switched to an output part way through a video, it joins where that output has got to, and never starts the video again on the output. A ProRes 4444 `.mov` is converted to a transparent WebM when it is added, keeping its alpha. **Export sources** saves the whole scene with its media as one zip to keep or move to another Mac; **Import sources** puts one back.

3.4 The base scene

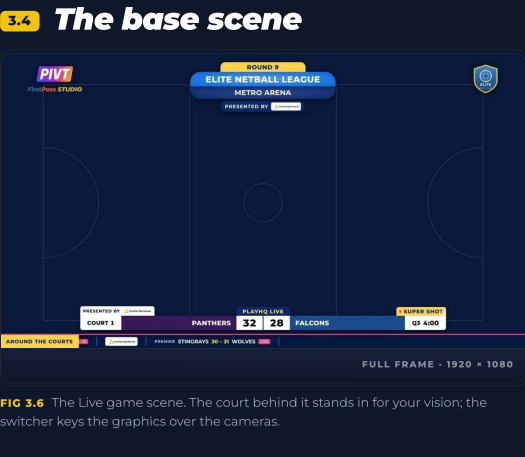


FIG 3.6 The Live game scene. The court behind it stands in for your vision; the switcher keys the graphics over the cameras.

The base scene is what viewers see for most of the game. In the sample league it is the **Live game** scene: the scoreboard at the bottom, the title banner at the top with the PIVT and FirstPass Studio logos on the left and the competition logo on the right, and the score ticker along the bottom edge with the other courts' scores.

- **Scoreboard.** Court in a pink box on the left, both teams in their club colours, the score at screen centre, quarter and clock on the right. "PlayHQ live" shows while scores follow PlayHQ; a "Presented by" tab carries the naming-rights sponsor.
- **Title.** Round, competition and venue, with the naming-rights sponsor's pill under it.
- **Branding logos.** PIVT and FirstPass Studio left of the title, the competition logo right of it, so other graphics can be turned on and off around them.
- **Score ticker.** Scrolling scores from tonight's other games, led by a Partner-and-up sponsor's tagline. Its speed is set on the control page or from the Stream Deck.

Build it once: turn those sources on, then press **Save what's on now** under Scenes and name it. From then on, one click (or one Stream Deck key) brings the show back to this look.

3.5 Automatic layers

These sources are left on in the base scene and play by themselves from the scoreboard. They add no PlayHQ requests of their own; they ask Studio, which already has the score.

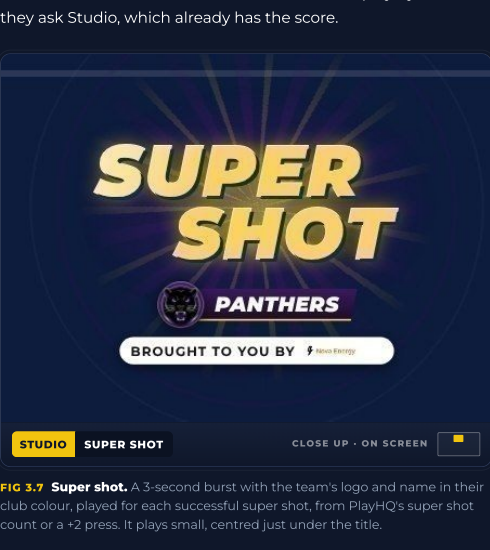


FIG 3.7 **Super shot.** A 3-second burst with the team's logo and name in their club colour, played for each successful super shot, from PlayHQ's super shot count or a +2 press. It plays small, centred just under the title.

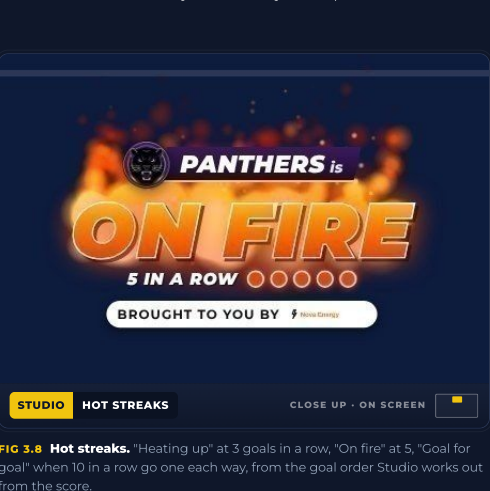


FIG 3.8 **Hot streaks.** "Heating up" at 3 goals in a row, "On fire" at 5, "Goal for goal" when 10 in a row go one each way, from the goal order Studio works out from the score.

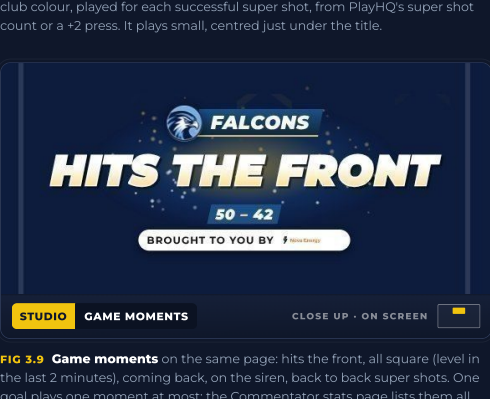


FIG 3.9 **Game moments** on the same page: hits the front, all square (level in the last 2 minutes), coming back, on the siren, back to back super shots. One goal plays one moment at most; the Commentator stats page lists them all.

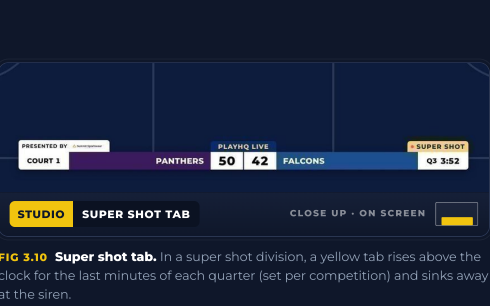


FIG 3.10 **Super shot tab.** In a super shot division, a yellow tab rises above the clock for the last minutes of each quarter (set per competition) and sinks away at the siren.

Each celebration can carry a sponsor's "Brought to you by" tag (Chapter 7). A division without super shots plays no super shot animation and counts no super goals; that is set on the competition setup page, per division if needed.

3.6 Scenes

A scene is a saved set of sources on and off. Clicking a scene goes to it, turning sources on and off to match while anything already on stays on, not reloaded. Scenes are listed by colour in the order of a stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads.

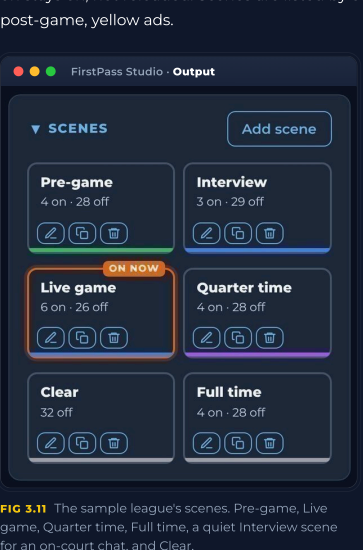
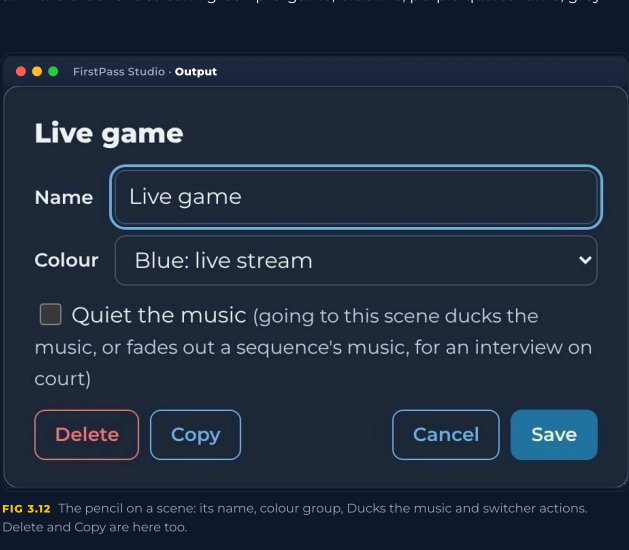


FIG 3.11 The sample league's scenes. Pre-game, Live game, Quarter time, Full time, a quiet Interview scene for an on-court chat, and Clear.



Live game

Name: Live game

Colour: Blue: live stream

☐ Quiet the music (going to this scene ducks the music, or fades out a sequence's music, for an interview on court)

Delete | Copy | Cancel | Save

FIG 3.12 The pencil on a scene: its name, colour group, Ducks the music and switcher actions. Delete and Copy are here too.

Turn a source on or off by hand after going to a scene and the scene is marked **changed**; the bar under the scenes offers **Back to saved**, **Save over** it, or **Save as new scene**. Each scene tile has three buttons along its foot: the **pencil** opens its settings (name, colour, **Ducks the music** and its switcher actions), **copy** makes a copy beside it, and the **bin** deletes it after asking. None of them changes what's on. A source added later is off in every scene.

SCENES THAT DUCK THE MUSIC Tick **Ducks the music** on a scene (or a sequence) for an interview on court: whatever music is playing ducks and keeps playing until the next scene or sequence without the tick.

3.7 Sequences

A sequence plays the scene's sources one after another. Each step turns on the sources you pick and stays for a set time, until it finishes (its videos end or its animation animates away), until the game moves on (the clock starts, or the next break or full time), or until you press Next. As a step ends, sources the next step doesn't use go off with their own fades.

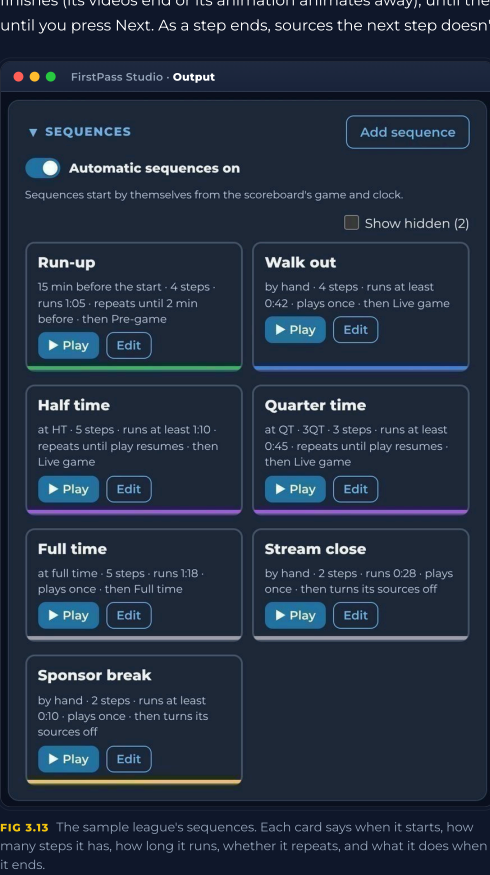
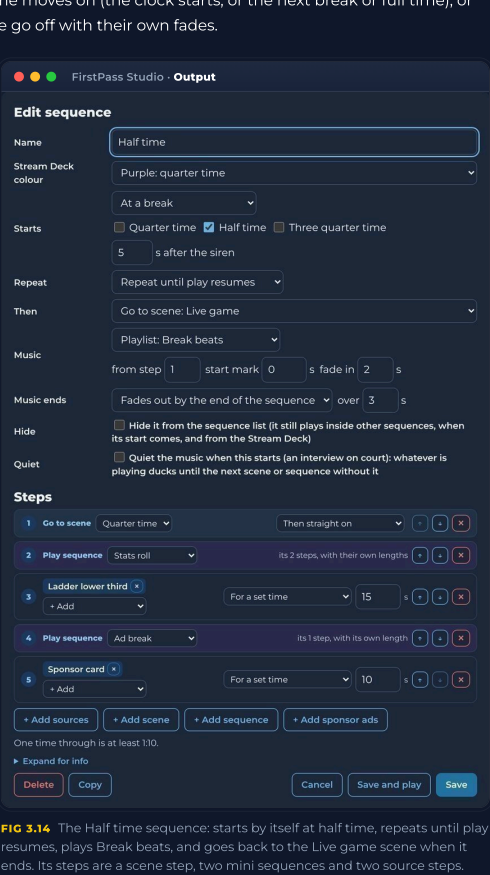

SEQUENCES
Add sequence
Automatic sequences on
Show hidden (2)
Run-up
15 min before the start · 4 steps · runs 1:05 · repeats until 2 min before · then Pre-game
Walk out
by hand · 4 steps · runs at least 0:42 · plays once · then Live game
Half time
at HT · 5 steps · runs at least 1:10 · repeats until play resumes · then Live game
Quarter time
at QT · 3QT · 3 steps · runs at least 0:45 · repeats until play resumes · then Live game
Full time
at full time · 5 steps · runs 1:18 · plays once · then Full time
Stream close
by hand · 2 steps · runs 0:28 · plays once · then turns its sources off
Sponsor break
by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off

FIG 3.13 The sample league's sequences. Each card says when it starts, how many steps it has, how long it runs, whether it repeats, and what it does when it ends.



Edit sequence

Name: Half time

Stream Deck colour: Purple: quarter time

Starts: At a break ☐ Quarter time ☑ Half time ☐ Three quarter time; 5 s after the siren

Repeat: Repeat until play resumes

Then: Go to scene: Live game

Music: Playlist: Break beats

Music ends: Fades out by the end of the sequence

Steps

1 Go to scene Quarter time · Then straight on

2 Play sequence Stats roll

3 Ladder lower third · For a set time 15 s

4 Play sequence Ad break

5 Sponsor card · For a set time 10 s

+ Add sources | + Add scene | + Add sequence | + Add sponsor ads

One time through is at least 1:10.

Delete | Copy | Cancel | Save and play | Save

FIG 3.14 The Half time sequence: starts by itself at half time, repeats until play resumes, plays Break beats, and goes back to the Live game scene when it ends. Its steps are a scene step, two mini sequences and two source steps.

THREE KINDS OF STEP

- **Sources** (**+ Add sources**): one or more sources on together, for a length.
- **A scene** (**+ Add scene**): a starting point or a reset point. What the scene turns on stays on under the next steps and after the sequence. Start a break sequence with the Live game scene and the scoreboard stays up under the stats.
- **Another sequence** (**+ Add sequence**): its steps play in place. Build a mini sequence once (the sample league's **Break stats** and **Sponsor break**) and use it inside Quarter time, Half time and Full time. A sequence made only of sequences is marked **Master**. Tick **Hide** on a mini sequence to keep it off the list; it still plays inside the others and from the Stream Deck.

+ Add sponsor ads adds a step that plays the next one to five sponsor ads in rotation (Chapter 7).

WHEN A SEQUENCE STARTS, REPEATS AND ENDS

SETTING	CHOICES	USED FOR
Starts	By hand · N minutes before the picked game's start (only if Studio was running then; opened later, play it by hand) · at the breaks you tick (QT, HT, 3QT) · at full time, a few seconds after the siren	Run-up 15 minutes out; the breaks and full time from the scoreboard's clock (the PlayHQ scorer's or yours).
Repeat	Play once · until stopped · until play resumes · until N minutes before the start	A break sequence loops until the clock starts again, finishing the step on screen.
Then	Go to a scene · turn off what it turned on · leave everything · play another sequence (one for each output it plays on)	Breaks go back to Live game; Run-up hands over to Walk out.
Music	None · a song or playlist from a step, from a start mark, with a fade in; ends by fading, cutting or carrying on into the next sequence	Warm-up music through the run-up that carries into the walk out.

A sequence that starts by itself waits a few seconds as **Starting**, with **Start now** and **Cancel**: press **Cancel** to hold the break back for a shot after the siren and start it by hand when the ball is dead, or **Stop** (or a scene) to come back from a break early. **Automatic sequences on** above the cards is the one switch for all of that, and it is remembered. Only one sequence plays at a time on each output; going to a scene, playing another sequence or hiding everything stops it without its Then. A sequence only ever turns off sources it turned on itself.

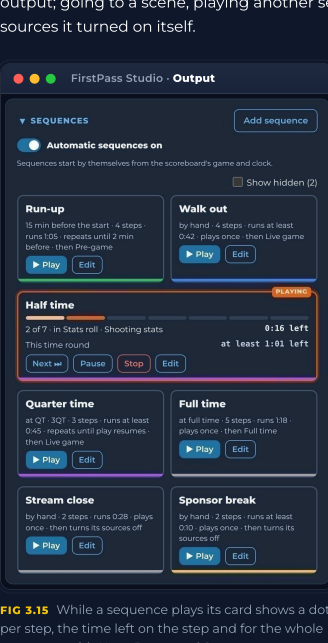


FIG 3.15 While a sequence plays its card shows a dot per step, the time left on the step and for the whole sequence, with Next, Pause and Stop.

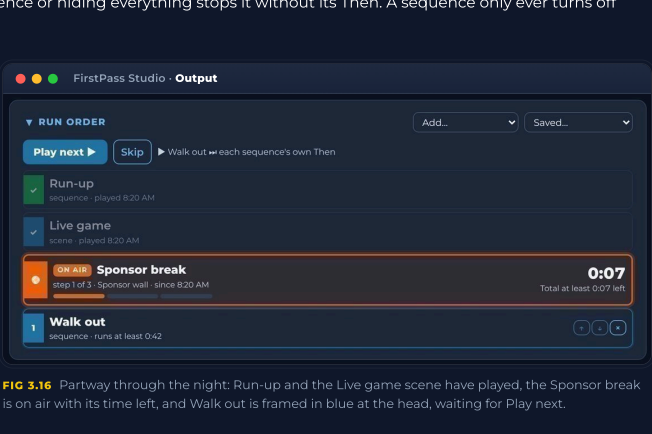

RUN ORDER
Play next ▶
Skip
▶ Walk out ⏭ each sequence's own Then
Run-up
sequence · played 8:20 AM
Live game
scene · played 8:20 AM
ON AIR
Sponsor break
step 1 of 3 · Sponsor wall · since 8:20 AM
0:07
Total at least 0:07 left
Walk out
sequence · runs at least 0:42

FIG 3.16 Partway through the night: Run-up and the Live game scene have played, the Sponsor break is on air with its time left, and Walk out is framed in blue at the head, waiting for Play next.

Sequences belong to the competition in use and change with **Use this one**; the sources and scenes are shared. The break lengths on the setup page (quarter time, half time, three quarter time) let the editor say how many times a break sequence fits in each break, and time Commentator stats' **Back in** countdown.

THE RUN ORDER

Above Scenes and Sequences sits the **Run order**: tonight's plan as a numbered list of sequences and scenes, in the order they play. It reads like a playlist: the last two things it has played sit at the top, greyed with a tick and the time they played (any before them fold behind **played earlier tonight**), then what's on air now in orange with **ON AIR**: the sequence playing, with its step, a dot per step, the step's time left in large figures and the whole sequence's under it, or the scene that's on when no sequence plays. Under that come the numbered rows still to play. **Add...** puts a scene or sequence at the end (Run-up, Walk out, Live, Quarter time, Live, Half time...), rows drag or move with their arrows, and **×** takes one out. **Play next** plays the head now, a sequence or a scene, and **Skip** drops it without playing it (stepping through a sequence is the Sequences panel's own Next). A sequence that ends looks at the run order first and plays what is queued after it; its own **Then** runs only when nothing is queued, so Then is the fallback and a sequence built for one night still works alone on another. Whatever is at the head leaves the list as it starts, however it was started, and moves up to the played rows (a skipped or removed row never played, so it doesn't). On Output 1, a sequence that starts by itself from the game is put at the head with its few seconds' countdown rather than fired blind: **Cancel** takes it out again and **Play next** plays it when the ball is dead. Playing a scene or sequence by hand leaves the list alone unless it was the head, and nothing in the list starts by itself. The status board's **Run order** line and the bar's **Run order** chip say what Play next will do: ▶ the head, ⏭ what follows it. The note under **Play next** says the same. The run order is Output 1's, the production feed, so it shows Output 1's list whichever output tab is picked; the other outputs have none, and a sequence that starts by itself there just starts. A sequence that plays on more than one output says so on its row, "also on Commentary · then Commentary loop": **Play next** starts it on all of them, as its card does, and after it each other output does its own **Then**. When the run order moves on by itself, another output joins the next sequence only if nothing is playing there; one that was playing the sequence that just ended does its own Then.

Saved... keeps tonight's order under a name for next week, loads one in place of what is queued, renames or deletes one, or clears the list (the played rows too; what's on air carries on). A game in the Production Planner can have a saved order that loads as the game comes up on the scoreboard. The played rows are for tonight: a relaunch starts them again. The run order and the saved orders live with the competition's sequences, so a relaunch picks the night up where it was. On the Stream Deck, the plugin's **Run order** key plays what's next, skips it, loads a saved order or clears the list, and names what plays next so you can see what a press will do; without the plugin, `/deck/runorder/next` plays what's next and `/deck/runorder/skip` drops it. The Sequence key's own Next still ends the step on screen.

3.8 Music, playlists and mics

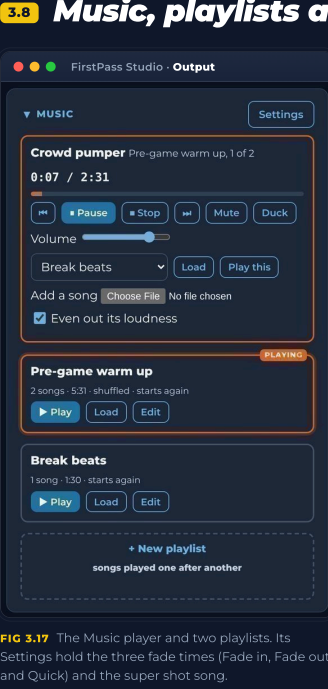
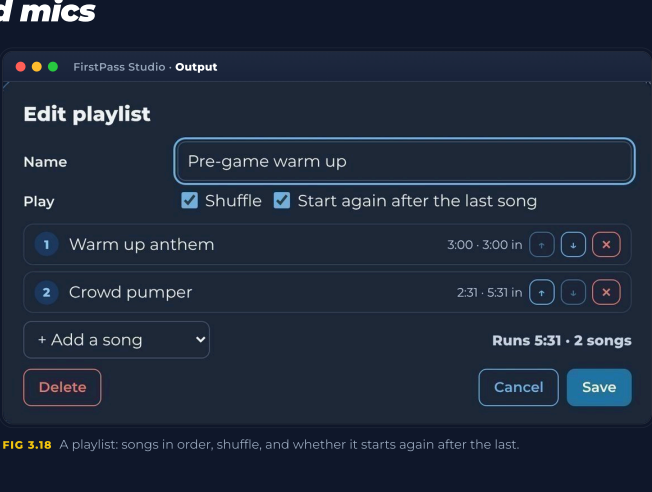

FirstPass Studio · Output
MUSIC
Settings
Crowd pumper Pre-game warm up, 1 of 2
0:07 / 2:31
Pause
Stop
Mute
Duck
Volume
Break beats
Load
Play this
Add a song
Choose File No file chosen
Even out its loudness
PLAYING
Pre-game warm up
2 songs · 5:31 · shuffled · starts again
Play
Load
Edit
Break beats
1 song · 1:30 · starts again
+ New playlist
songs played one after another

FIG 3.17 The Music player and two playlists. Its Settings hold the three fade times (Fade in, Fade out and Quick) and the super shot song.



Edit playlist

Name: Pre-game warm up

Play: ☑ Shuffle ☑ Start again after the last song

1 Warm up anthem 3:00 · 3:00 in

2 Crowd pumper 2:31 · 5:31 in

+ Add a song | Runs 5:31 · 2 songs

Delete | Cancel | Save

FIG 3.18 A playlist: songs in order, shuffle, and whether it starts again after the last.

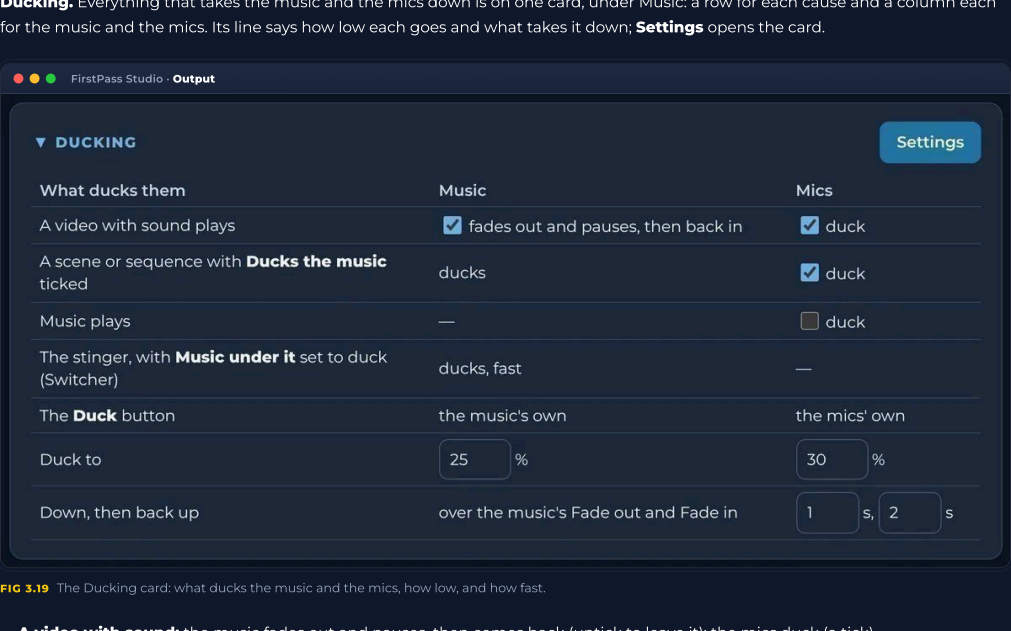
The Music player plays songs from the media folder (MP3, M4A, AAC, WAV or AIFF; use music you're licensed to stream), one song or a playlist, shuffled or not. Add a song with **Even out its loudness** ticked and it is levelled to match the others. Music plays where Output 1's **Sound goes to** says, never in the Scene View.

- **Play** fades in; **Pause** and **Stop** fade out (slower by default). **Duck** lowers it and keeps it running; **Mute** keeps it running silently.
- While a video with its sound on plays, the music fades out and pauses, then fades back in from where it was, both over the **Quick** time (untick it under **Ducking** to leave the music playing). A themed intro or outro does the same. A video without its Sound ticked leaves the music alone.
- A sequence can bring its own music, and a sequence inside it plays its own while it runs. Going to a scene or another sequence fades a sequence's music out over its Music ends time; going to a sequence with music of its own, the new music fades in as the old fades out.
- The **super shot song** cuts in when the super shot period starts in a super shot division, loops, and fades out when the period ends.

Mics. Wireless effects mics on court and crowd mics go out with the music: plug their receivers into the Mac (directly or through an audio interface) and add each one under **Mics**, with its input and channel (two receivers on one stereo input are two mics, left and right). The mics are Output 1's, like the music: heard while Output 1 is on, never in the Scene View, so a speaker near a mic can't feed back. The first time, the Mac asks whether FirstPass Studio may use the microphone.

- A mic can be plugged into the switcher instead (an ATEM Mini's Mic 1 or Mic 2): the switcher mixes it and Studio moves its fader for the same level, mutes and ducks. Move the fader on the switcher and that becomes the mic's level; Studio never moves it back. Its meter shows the sound coming into the switcher's input, with a mark where its fader is.
- Each mic has a level, a mute, a **Low cut** for wind and rumble and a meter; **Level** and **Mute all** cover every mic, and a limiter stops a shout clipping the stream.
- A mic that hears nothing for 10 seconds while it's listened to (a flat battery, a receiver turned off, a pulled cable) says **Silent for 12 s · check its receiver and battery** in amber on its card, and the status board's Mics line says **Silent**. It clears as soon as sound comes in. A muted mic is never silent, and nor is one nobody is listening to (Output 1 off, or a switcher that sends no levels).
- **A phone as a backup commentator mic.** When a commentator's headset fails, their phone can stand in. Press **+ Phone mic** in Mics and scan the QR code with the phone, on the same Wi-Fi; the first time, the phone asks the Mac (either Allow works). On the phone's **Commentator mic** page, pick the mic and press **Start**: it comes in as a mic named after the phone, marked **phone (backup)**, with the same level, mute, low cut, ducking and Silent warning as the others, mixed into Output 1. A **wired or USB-C** mic plugged into the phone sounds best; a Bluetooth headset's mic sounds like a phone call and arrives about a fifth of a second late. The page keeps the screen awake, says **On air** while Studio hears it (or **Muted in Studio**), and if the connection drops it beeps in the commentator's ears and starts again by itself, while the mic's card and the status board's Mics line say **Dropped**. **Hear Studio** on the phone plays Output 1's sound back in the commentator's headphones: the music, videos and page sounds and the other mics (their co-commentator), never their own voice, with an Off; the phone's own volume buttons set how loud. It's a moment behind the stream (more on Bluetooth headphones), and needs headphones: on the phone's speaker it would come back in through the mic. It needs the Secure phone link turned on (a phone lets a page use its mic only over https), and Output 1 on. Keep the phone on the page with Do Not Disturb on: switching apps or a call stops its mic.
- The mics **duck** while a video with its sound plays and during a scene that ducks the music, and while music plays if you tick it (it starts unticked); each is a tick box under **Ducking**. **Duck** ducks them by hand. A mic marked **Never ducks (an external mixer does it)** stays up: the mixer does the fading.
- **Music for the commentators:** in Output 1's **Sound goes to**, **Add a destination** and pick the commentators' mixer: its sound plays there too, while the SDI or the Mac's sound output carries on as before. Only the mics you tick for that destination are in it, so tick the crowd mic and leave the commentators' own unticked and they won't hear themselves back late. On the mixer (not in Studio), keep what comes in from the Mac out of its audio out back to the Mac (its mix minus), or the music comes back through their mics and goes out twice.
- Stream Deck: `/deck/mics/mute/toggle`, `/deck/mics/duck/toggle`, `/deck/mics/up`, `/deck/mics/down`, and one mic with `/deck/mic/court-mic/mute/toggle`.

Ducking. Everything that takes the music and the mics down is on one card, under Music: a row for each cause and a column each for the music and the mics. Its line says how low each goes and what takes it down; **Settings** opens the card.



DUCKING Settings

What ducks them	Music	Mics
A video with sound plays	☑ fades out and pauses, then back in	☑ duck
A scene or sequence with **Ducks the music** ticked	ducks	☑ duck
Music plays	—	☐ duck
The stinger, with **Music under it** set to duck (Switcher)	ducks, fast	—
The **Duck** button	the music's own	the mics' own
Duck to	25 %	30 %
Down, then back up	over the music's Fade out and Fade in	1 s, 2 s

FIG 3.19 The Ducking card: what ducks the music and the mics, how low, and how fast.

- **A video with sound:** the music fades out and pauses, then comes back (untick to leave it); the mics duck (a tick).
- **A scene or sequence with Ducks the music ticked:** the music ducks; the mics duck too (a tick).
- **Music playing:** the mics duck under it if ticked (off to start with).
- **The stinger**, with its **Music under it** set to duck on the Switcher card, ducks the music fast. The **Duck** buttons duck the music or the mics by hand.
- **Duck to:** one level for the music (25% to start) and one for the mics (30%). The music goes down over its **Fade out** time and back up over its **Fade in**; the mics have their own down and up times (1 and 2 seconds).

3.9 Video output

Settings over the picture opens the **Output settings**, which send the scene to your switcher. Pick one of three routes, then press **Output on** in the row over the picture. While the output is on its settings are locked; turn it off to change them. `/deck/output/on`, `off` and `toggle` do the same from a Stream Deck, and the outputs go dark when Studio goes off air.

ROUTE	WHAT IT NEEDS	WHAT THE SWITCHER GETS
Display	One or two extra displays on the Mac (HDMI into the switcher), set to exactly 1920 × 1080, not HiDPI or scaled	**Chroma**: the scene over a solid colour on one display (green by default, blue or any colour). **Fill and key**: fill on one display, key (white where the scene shows) on the other. A plain M1 drives one extra display; Pro and Max chips drive more.
Blackmagic	An UltraStudio or DeckLink with Blackmagic Desktop Video installed, and the device free	**Fill and key** through the device's external keyer (fill on one SDI out, key on the other), or fill only, at the switcher's 1080p format (1080p50 is picked first). The fill is premultiplied: turn the switcher's Pre Multiplied Key on for clean edges, and leave Straight off (it leaves a light fringe round logos and text). **Sound goes to** the SDI with the fill to start (48 kHz stereo, in time with the picture): turn that switcher input's sound on. The Mac's sound output can be picked instead, or added as well.
NDI®	NDI's free runtime for Mac, and a wired network	Fill and key as two sources, chroma as one, or one source with transparency for receivers that key on NDI's alpha. They appear as this Mac's name with the output's name in brackets (or a name of your own).

HD OR 4K

The **HD** and **4K** buttons over the picture set the picture for every output at once. In 4K each NDI and Blackmagic output sends 2160p at the frame rate you picked for it (2160p50 for 1080p50). Nothing else changes: scenes, sources, Stream Deck keys, links, NDI names and the switcher key stay as they are, and pressing HD puts everything back exactly as it was.

- **Sharp where Studio draws.** The graphics keep their 1920 × 1080 layout and are drawn at twice the density, so text, shapes, logos drawn by Studio and the animations' sparks and confetti are sharp. Pictures and videos are only as sharp as their files: a 1080p video or a small logo looks softer.
- **Only for 4K gear.** An HD switcher can't take 2160p. A Blackmagic device without the 2160p format (an HD UltraStudio) stays at 1080p and its card says so; a display is left alone, since a 4K screen set to look like 1920 × 1080 already draws at 4K.
- **Pick a 4K format directly.** In 4K a Blackmagic output's video format list also has the device's own 4K formats, such as 2160p50. In HD they're hidden, and the output sends the 1080p format at the same rate.
- **Any graphic's link works at 4K too.** Opened in a browser source or window set to 3840 × 2160, a graphic draws itself at twice the size, with nothing added to its link.
- **Check the Mac keeps up.** A 4K output is about four times the work of an HD one; the performance line under the output says whether it's keeping up.

MORE THAN ONE OUTPUT

One output is all most nights need. To give a camera its own graphics, press **Add an output** beside Output on (up to four). Each output is its own picture with its own route and its own on/off, so a second output can feed an upstream keyer tied to a camera and its graphics cut with that camera, while Output 1 stays the downstream key over everything. Name an output in its card ("Wide camera", say) and that name is used everywhere, including its Stream Deck links.

- **Sources are shared, on/off is per output.** The Sources list follows the output tab picked above the Scene View: each source's On / Off is for that output, in its colour (orange, pink, blue, green for Outputs 1 to 4), and shows or hides it there only. **Hide all** clears that output too. Pick another tab to set another output. A new output starts with everything off.
- **Every output plays its own video sound; the music and the mics are Output 1's.** Each output plays the sound of the videos and animations on it. The music (the player's, a sequence's and the super shot song) and the Mac's mics go out on Output 1 only, so a second output never takes the production's sound, and a sequence played on another output alone plays without its music. A commentators' feed is a destination added to Output 1's **Sound goes to**, with the mics ticked for it.
- **Scenes and sequences are shared; each output plays its own.** Pick an output's tab over the Scene View and clicking a scene or sequence plays it on that output (the sections say which). **Hide all** clears that output only (or the one the Sources list's **Show** is set to), and says which, so clearing a monitor never blanks the stream. Each output has its own current scene and plays one sequence at a time, so the same scene or sequence can be on two outputs at once; coloured dots on scene cards, and on sequence cards each output's tick with how the sequence is there (Playing, Paused, Starting or Stopped (it ended there, played out or cut short); a hollow dot and "off" while that output isn't sending), show where each one is on. A sequence's **Plays on** ticks (on its card and in its editor) pick the outputs it plays on together: started by itself, or played on any of them, it starts on all of them at once, and each output has its own **Then** for when it ends there, so a quarter-time sequence can go back to the live scene on Output 1 while the commentary monitor goes on to its stats loop. Played on an output it isn't ticked for, it plays there alone (`/deck/output/2/sequence/<name>/only` does that on a ticked one). Stop on a card stops it on that output only. Remove every output it plays on and the sequence stops starting by itself until you tick another, so it never lands on air by surprise. A Sponsor ads step off Output 1 plays its ads in that output's own ad sources. `/deck/scene/<name>` and `/deck/sequence/<name>` play on Output 1; `/deck/output/2/scene/<name>`, `/deck/output/2/sequence/<name>`, `/deck/output/2/scene/next` and `/deck/output/2/sequence/stop` (and the rest) play on Output 2.
- **Sponsors count on Output 1, and on outputs you tick.** An extra output's card has **Counts for sponsors**, off to start. Tick it only when everything on that output is seen on the stream, like a second downstream key. Leave it off for a camera's upstream key or the commentary monitor: Studio can't tell when they're on air, and the report shouldn't claim time nobody saw. Ads there show whose turn it is without using it up, so the order on air stays fair.
- **The Scene View has a tab per output**, coloured to match, so you can check what each one is sending. Switching tabs is seamless: the picture carries straight on, as the new output's view takes over once it has loaded.
- **Routes can't be shared.** Two outputs can't use the same display, the Blackmagic device or the same NDI name; the earlier output keeps it and the later one says so on its card. Two HDMI displays give either two chroma outputs or one fill and key output. Several NDI outputs can run at once, each with its own name, which suits feeding a switcher's inputs over the network.
- **Stream Deck.** `/deck/output/2/toggle` (or the output's name) turns Output 2 on or off, `/deck/output/all/off` turns them all off, and `/deck/output/2/layer/<source>/toggle` shows or hides a source on Output 2. Source settings lists these links for each output.

EACH OUTPUT COSTS Every extra output runs its pages once more on the Mac. Try the number of outputs you plan to use on your own Mac before a stream night: while an output is on, its card shows its CPU, memory and frame rate (amber when it can't keep up), with Studio's total under the status, and the log notes them once a minute.

A COMMENTARY MONITOR

Commentators can have graphics of their own: a monitor at the commentary desk with the live stats they talk to, never seen on the stream. The **Commentator stats** page is made for it: one full screen with both team lists on the left (each team's time since its last goal, and each athlete's games played this season), the game stats big in the middle (each team's logo with its ladder position, record and form under it, head to head under the names, then shooting %, quarters won, best quarter, super shots, runs, biggest lead and lead changes), down the right the game clock with a pill under it for the commentary team (**Starts in 4:32** to the scheduled start, **Back in 2:15 · Quarter time** at a break, counted from the clock reaching 0:00 by the competition's break minutes, then **Due back now** in pink until the clock runs again, and during play the super shot window), who is on the mic (from the Production Planner) and up next, then a list of the latest moments (lead changes, comebacks, super shots, hot streaks), newest at the top, each with its quarter, game time and how long ago it happened (it lists every moment a goal makes, even the ones the stream skips, and opened part way through a game it starts with that game's moments so far), and across the bottom the last 10 goals, the current run in big numbers and the score worm. Its stats refresh themselves every 15 seconds and when the score changes (untick **Auto-refresh** in its options, or pick another frequency), and the clock, moments and goals follow the game as it happens. Other graphics can go on the commentary feed too: the last 10 goals and the scoreboard follow the game live, and the rest show what they read each time they come on.

- **On the same Mac.** Add an output for the commentary monitor (a display, or NDI to a screen on the network) and turn sources on for it with its pills or its Stream Deck keys, such as `/deck/output/commentary/layer/ladder/toggle`. Play scenes and sequences on it from its Scene View tab or its own Stream Deck links while the stream carries on as normal.
- **On a second Mac.** Run Studio on another Mac with the same competition and game, and Follow PlayHQ on. It reads PlayHQ's live scoring for itself, so it needs nothing from the stream Mac. Its Output 1 goes to the commentary monitor, and everything on it is for the commentators.

On the commentary output (or the second Mac), make a **commentaryLoop** sequence that steps through what commentators find useful (game stats, shooting stats, the last 10 goals, the ladder) with **Repeat** set to until stopped, and give the commentators a Stream Deck of their own:

- A key per graphic they want on demand, each going to a scene with that graphic (on the same Mac, `/deck/output/commentary/scene/ladder`). Going to a scene stops the loop on that output only, so their pick stays up and the stream is untouched.
- A **Loop** key to go back to looping when they're done: `/deck/output/commentary/sequence/commentaryloop` on the same Mac, or `/deck/sequence/commentaryloop` on a second Mac. With the FirstPass Studio plugin, use Scene and Sequence keys with **Output** set to the commentary output; they light pink while it's on.

Each output's **Sound goes to** lists where its sound plays: the Mac's sound output to start, or on a Blackmagic output the SDI with the fill; add the Mac's sound output as well, or any of the Mac's sound outputs such as the commentators' mixer. For a screen with no speakers, such as a billboard over NDI, pick **No sound** (last in the list): the output plays no sound anywhere, its card says **Sound: none** and the status board says **No Audio** for it. On Output 1 that silences the music and the mics too, as they go out through Output 1 only. Each output's card says it in a line (Sound: SDI + Mixer (crowd mic); Sound: Mac, not over NDI, since NDI carries no sound), and the status board's **Sound out** rows say it for every output, with what each device plays at and how loud. The **Scene View** on the Output page shows the scene as it is right now, on a checkerboard so see-through parts show; **Pop out** opens it in its own window to move to another screen, with a floating list of On/Off buttons. The Scene View never plays sound: an output's sound plays where its Sound goes to says, on or off air, so nothing plays twice and nothing plays from a page that happens to be open.

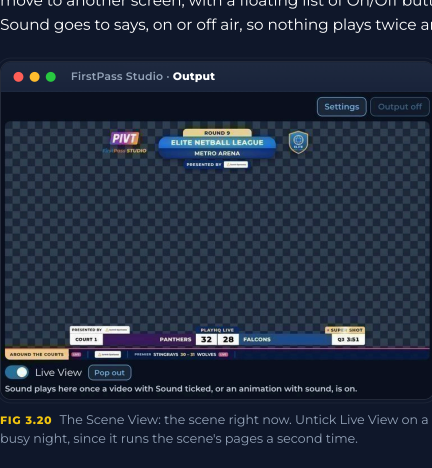

FirstPass Studio · Output
Settings
Output off
PIVT
ELITE NETBALL LEAGUE
METRO ARENA
PANTHERS 32 28 FALCONS
Live View
Pop out
Sound plays here once a video with Sound ticked, or an animation with sound, is on.

FIG 3.20 The Scene View: the scene right now. Untick Live View on a busy night, since it runs the scene's pages a second time.

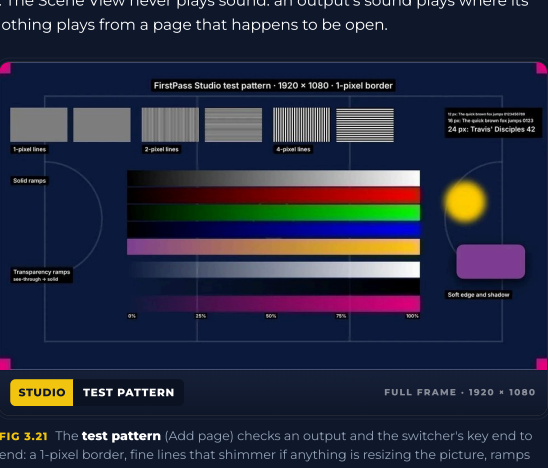

FirstPass Studio test pattern · 1920 × 1080 · 1-pixel border
STUDIO TEST PATTERN
FULL FRAME · 1920 × 1080

FIG 3.21 The **test pattern** (Add page) checks an output and the switcher's key end to end: a 1-pixel border, fine lines that shimmer if anything is resizing the picture, ramps for banding and soft shapes for the key. Not for air.

CHROMA KEY SAFE A chroma key cuts out one colour, so anything half see-through fringes. With a chroma output picked, **Chroma key safe** is on: every graphic goes out with solid panels, no glows or soft shadows, and fades become cuts. Pick another output and everything is back as it was. The Output page also lists any competition or club colour close to the key colour.

THE SWITCHER

Studio can also drive a Blackmagic ATEM switcher over the network, so the camera cuts can be pressed from the Output page or a Stream Deck beside the graphics. Open the **Switcher** card on the Output page, press **Set up** and **Look for switchers**: Studio finds the ATEMs on the network the way ATEM Software Control does (macOS asks the first time to let it use the local network) and remembers the one you pick by name. Typing its address works too.

- **Each output's key, checked.** In each output's setup, **Into the switcher as** says which switcher key its graphics come into (usually downstream key 1 for Output 1, upstream keys 2 to 4 for the others, an upstream key for chroma) and which inputs its fill and key are plugged into. Studio reads that key's settings and lists anything that doesn't suit the output, such as the wrong fill source, Pre Multiplied Key off, or Straight ticked on a Blackmagic output. **Set up this key for me** fixes just those, after asking, and leaves the key on or off air as it was.
- **Graphics that go with a camera.** An output on an upstream key can tick the cameras it **goes with**, say a bench graphic with the bench camera. Studio's cuts, autos and stingers then bring its key on with those cameras and take it off with the others, on the same frame. With **Keys follow the camera** on, a cut on the switcher's own panel brings the key a frame or two later. A key turned off by hand stays off until the camera changes. The output's card says **On air with Bench** or **Waiting for Bench**.
- **Inputs in tally colours.** Each input shows by its name from ATEM Software Control, red on program and green on preview. Click one to put it on preview, then **Cut** or **Auto**. Fade to black and each upstream and downstream key are a press away too.
- **Only what your switcher has.** An ATEM Mini or Mini Pro has 4 inputs, one upstream key and one downstream key, so fill and key into the downstream key leaves two cameras; a Mini Extreme has 8 inputs, four upstream keys and two downstream keys. Studio offers what it finds, and says so when a key or link asks for something the switcher doesn't have.
- **Your hands stay on the switcher.** Its own buttons, ATEM Software Control and other controllers keep working, and the card follows them. Studio sends nothing by itself, not even on connecting.
- **If it drops off the network, or is off when Studio connects,** the card turns red within about five seconds and Studio keeps trying. Presses meanwhile are refused rather than saved up, so a cut never lands late.
- **Stinger cuts.** Add the Stinger page as a source and the card's **Stinger** button plays it and cuts the switcher underneath, right in the middle of the bands. It plays on the output on a downstream key, which is Output 1 unless another output that's switched on is. If the cut shows before the bands do (an output's picture can reach the switcher a few frames late, more over NDI), set the **line-up** in frames and press **Test the line-up** before you go live. Each output keeps its own line-up. **Music under it**, beside them, says what the music does as the stinger plays: carries on, ducks for the scene you cut to, or ducks while it plays and comes back up.
- **Camera cuts in sequences and scenes.** A sequence step's 🎥 button, or a scene's 🎥 **Switcher** in its settings (the pencil), adds switcher actions: cut, auto or stinger to a camera, put one on preview, a key on or off, fade to black, or run a macro, as the step starts or a set number of seconds in. So the walk-out sequence can cut to the bench camera as the line-ups come up and bring the scoreboard key back as it ends, with nobody on the switcher. They happen only on Output 1, and if the switcher isn't answering at that moment the action is skipped and logged rather than done late. You can always cut by hand on the switcher itself.
- **Stream Deck.** The plugin's **Switcher** key puts a camera on preview or program, plays a stinger, cuts, autos, turns a key on or off or runs a macro, and lights like the switcher's own buttons: red on program, green on preview. While the switcher isn't answering it goes dark with a yellow dot. Without the plugin, the links are `/deck/switcher/cut`, `auto` and `ftb`, `/deck/switcher/preview/2` (or `preview/Camera%202`), `/deck/switcher/usk/1/toggle`, `/deck/switcher/dsk/1/auto` and `/deck/switcher/macro/1`.

3.10 A night, start to finish

With the sample league's scenes and sequences, a solo operator's evening looks like this. Everything in the right-hand column happens by itself.

Arrive

Open the app. Studio goes on air, picks tonight's first game on Court 1 with Follow PlayHQ on. Turn the video output on and check the test pattern on the switcher.

15 min out

Run-up starts by itself: the Pre-game scene, then the sponsor wall, match up and ladder in turn over the warm-up playlist, repeating until 2 minutes before the start.

2 min out

Run-up hands over to **Walk out**: The Clash smashes the two logos together, the position match ups run through GS to GK, the countdown holds until the clock starts, then the Live game scene comes up and the music fades.

Q1 to Q4

The scoreboard follows the PlayHQ scorer's score, quarter and clock. Super shots, hot streaks and moments play by themselves. The super shot tab and song mark the last five minutes of each quarter. The operator watches the stream and, if they like, keys up a ladder lower third or a lower third for a commentator.

Quarter time

Quarter time starts a few seconds after the siren: shooting stats, the last 10 goals and the ladder, then two sponsor ads and the sponsor card, over the break music, round and round until play resumes, then back to Live game.

Half time

Half time does the same with Up next and the ladder lower third added, and the house ad if no sponsor has one due.

Full time

Full time starts after the final siren: the winner celebration, the full time card, game stats, the score worm, a sponsor break and Up next (which becomes the end-of-night card after the last game).

Next game

Automatic selection moves the board to the next game on Court 1 as its electronic scoring opens, with the quarter back to Q1 and a full clock. Run-up starts again 15 minutes out.

Close

Press **Stream close**: the Clear scene, then the outro with the night's teams. Turn the output off. The sponsor report has the night's ready to send.

3.11 The Production Planner

The **Production Planner** tab plans the season ahead: which court Studio streams each game night, and who is on the mic for every game. Studio streams one court a night; a second court is a second Mac running its own Studio, with its own plan.

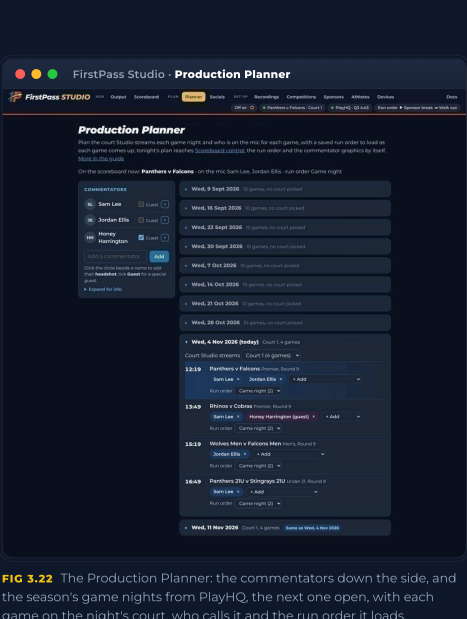


FIG 3.22 The Production Planner: the commentators down the side, and the season's game nights from PlayHQ, the next one open, with each game on the night's court, who calls it and the run order it loads.

COMMENTATORS

Add the people who call your games to the list on the left. Tick **Guest** for a special guest. The list belongs to the competition; as you type a name, people from your other competitions are suggested, so someone who calls in two leagues is added in a tap. Click the circle beside a name, or drop a photo on the row, to add their **headshot** for the Socials tab's Commentary team picture, the fixture board and the commentators' lower thirds; the small × on it takes it off again.

A NIGHT

Every game night in the season is listed, read from PlayHQ's fixture, by date rather than round. Open one, pick the **Court Studio streams**, and its games on that court appear in time order. Tap **+ Add** on a game to put up to three people on the mic; tap × to take someone off. Everything saves as you go. A night you haven't planned takes the plan of the night before it (the same court, and the same people and run order game by game), marked **Same as** that night, so a season with a regular team is planned once and only the nights that differ need changing. **Same as the night before** drops a night's own plan again.

A RUN ORDER FOR EACH GAME

Each game's **Run order** list holds the run orders saved on the Output page (Saved...). Pick one and it loads into the run order as that game comes up on the scoreboard, whether Studio picked the game or you did, in place of whatever is queued. If that clears anything, the run order panel says so (**Cleared the 2 items queued: the Production Planner loaded "Big game" for Stormz v Titans**) until you press **OK**. What's on air carries on, and it loads once as the game arrives: restarting Studio mid-game doesn't load it again. A game set to **None** leaves the run order alone, and the run order's own buttons work as ever. A run order that gets renamed stays picked; one that gets deleted is flagged on its games.

WHEN THE FIXTURE CHANGES

PlayHQ's fixture changes through a season: a game moves court for availability, or is refixtured to another date. The commentators belong to the time on the court, because that is where the desk and cameras are set up, so they stay put and the game is flagged **Fixture changed** with what was there before. A game that has gone from the court keeps its row, so its commentators can be moved to wherever it went.

ON THE NIGHT

When the night arrives, the plan's court goes to Scoreboard control, which follows it game by game as usual. The **Commentators** graphic, with no names typed in its options, names whoever is planned for the game on the scoreboard, and says **Special guest commentators** when everyone on the mic is a guest. Its **Lower thirds** look puts up a person lower third for each of them from one source, side by side, with a **Special guest** tab over a guest and, if you like, their headshots. **Commentator stats** shows them under the clock, and the **Fixture board** full page puts the whole night's fixture on a screen in the competition room or backstage, with a camera on the streamed court's games and who is on the mic for each: their headshots (or a silhouette) with their names, under an **On the mic** tab beside the teams. Manual control always wins: if you change the court or pick a game by hand on Scoreboard control, the Planner says so at the top, so the commentary team can be moved to match. Swapping someone in the Planner takes effect the next time the graphic comes on.

3.12 Socials

The **Socials** tab turns a game night into pictures for Instagram and the league's other channels, from the same fixture the stream uses, so the posts and the broadcast always agree. Pick the **Game night** (by date, so every division playing that night is in it), **Upcoming** or **Results** (a night with results opens on Results), and the sizes: **Square** 1080 × 1080, **Portrait** 1080 × 1350 (the most room in the feed) and **Story** 1080 × 1920. Every picture shows as a preview in each size, in the competition's colours, font and logos. Click a preview to see it as big as the window allows; ← and → step through the others, and **Esc** or a click closes it.

Show narrows the list to the pictures you want, like a scene does on the Output page. **All** lists every picture. **New picture set** makes a set from a name and a tick for each kind of picture (Match up, Form guide, Ladder and so on); it starts with every kind on the list you're looking at ticked. Pick a set and only those kinds are listed, on both Upcoming and Results, and only those are saved; each button shows how many pictures it lists this night. A set picks kinds, not games, so the same set works every week. ✎ on a set changes its name or ticks, or deletes it. Your sets, and the one picked, are remembered for the competition. When more than one division plays that night, **Division** on the same row lists only that division's pictures (its match ups, ladder, top shooters, season draws and so on) along with the ones for the whole night, like the Game night fixture, Results and Byes.

UPCOMING PICTURES

The **Game night fixture** lists every game of the night on every court, by time. A **Match up** for each game puts the two logos side by side with the date, time, court and venue, one for each club to share. A **Form guide** for each game shows both teams' last five results (oldest first, with the scores) and how their meetings this season went. Each division gets a **Round preview**: its games with each team's ladder place and points, the game of the round (the two best placed teams; **Top of the table** when it's 1st v 2nd) outlined, and any team a win could lift into the finals places. **Byes** lists the teams not playing that night, division by division. **Milestones coming up** names athletes whose next game would be their 50th, 100th or another 50; Studio counts this season's PlayHQ line ups plus any games typed in on the Athletes tab (**Games**), so type in earlier seasons' games for it to know. In the last three rounds before a division's finals, its **Finals race** shows each team as in the finals, out, or still a chance, from the ladder and the games left. In the last four rounds, and on finals nights, **Countdown** shows the rounds left before the finals or the days to the grand final. On a division's first night, each of its teams gets a **Season draw** with every game of its season. **On the stream** shows the court Studio streams, with each game's division and who calls it, from the Production Planner. **Commentary team** is a picture for each game someone calls, with that game's commentators side by side with their headshots and the game's time and teams once underneath, to share before the livestream. Someone who calls more than one game is on each of their games' pictures, so the social team can post them one at a time or grouped as they like. Long names shrink together so the row stays balanced, breaking only between words or after a hyphen; anyone without a headshot shows their initials, and a guest gets a **Special guest** pill. The **Polaroid** is a tilted white print for a match photo: drop a photo on it, drag the preview to frame it and set the zoom. With no photo, its window is see-through, so the social team can put their own photo under it in Canva or a phone app.

RESULTS PICTURES

After the night, **Results** shows every finished game's final score, division by division, with the winner in the highlight colour. Each game gets a **Full time** card, with the logos and score big and who won by how much, and a **Polaroid** with the score as its caption and the division and round under it. **Round in review** picks out the night's biggest win, closest game and upsets (a team at least half the ladder, or 4 places, below the team it beat). Each division that played gets its **Ladder**, with the finals line, labelled with the round it is after (that night's, when it is that division's latest night with results; else the latest round played), and on that night each position since before it when **Highlight movers** is ticked. Each division also gets its own **Top shooters**, naming its best six that night from each game's line ups (with super shots only where that division plays them), and its **Tournament top shooters**: the best eight over every finished game so far, after this night's round. It's called **Season top shooters** when the fixture runs 15 days or more; pick either name under Game day on Competitions setup to override. On a carnival day with several rounds, set **As at** to an earlier round to count only the games up to it. Tick **Highlight movers** and each shooter shows the places they climbed or fell since the round before, even when both were played that day, with NEW for anyone new to the top eight; the tick is remembered for the competition. **Sharpshooter** ranks each division's most accurate shooters that night (10 attempts or more), **Super shot leaders** (in divisions with super shots) the most super shots that night and over the season, **Team super shots** (in the same divisions) every team's super shots that night and over the season, and **On fire** the shooters whose goals a game over their last three games rose most. **Night in numbers** adds the night up: games, points, super shots, close games, the biggest win, the highest score, the shooting % and the upsets (or, on a night without one, the average winning margin), and, when Studio has every game's whole story, the lead changes, the most goals in a row and the runs of 5 goals. A club with more than one team playing gets a **Club round up** of its teams' results (clubs come from Competitions setup). **On a roll** names the teams unbeaten all season or on three wins or more in a row. Pick a team for its **Team season card**: its wins and losses, ladder place, percentage, biggest win and top shooter so far. On a division's last night, **Season wrap** names its champions (once known), top shooter, sharpest shooter, most super shots and biggest win of the season. A game appears once PlayHQ has its final score; until a night has one, each Results picture says so.

Some pictures are picked by hand on their card. **Player of the match** (one per game) takes a name from either team's line up, with their photo when they approved its use and their shooting that game; type a medal's name, such as an Anzac medal for a special round, and it becomes the title. **Team of the round** (one per division) takes an athlete for each position from that night's line ups. On both Upcoming and Results, **Athlete spotlight** shows one athlete from the Athletes tab (their photo, teams, positions, games played and goals this season) and **Sponsor spotlight** one sponsor big with a thank you, and once the night has been streamed, how long they were on screen. **Thanks for watching** closes the night with the next game night and the sponsors.

For each game Studio streamed, or followed in the background (the night's other games), Results adds a **Game story** (the score quarter by quarter, quarters won, the best quarter, the longest run, the biggest lead and the lead changes) and a **Score worm** (the margin over the game). Studio keeps each game's goals and quarter scores on this Mac as it follows it, so they're there after the scoreboard moves to the next game; a game picked up part way says Studio didn't see every goal. When a second court is streamed by another Mac running Studio, each Mac copies the other's finished games, so the Socials tab on either one has every court's pictures, even on a Mac that was off air for part of the night: while both are on the same network they find each other by themselves and copy every few minutes, and with iCloud Drive on (same Apple account) each finished game also goes into **iCloud Drive/FirstPass Studio/game-stories**, for nights the Macs never saw each other (a game Studio was quit part way through goes there too, half an hour later, at Studio's next look). On a night with copies, Results says how many games came from which Mac and which way. Copying only reads from the other Mac; nothing there changes. When both Macs have the same game, the better one is kept: the one with fewer gaps, then the streamed one over one only followed (a game scored by hand on the stream has goals PlayHQ never had). Where one Mac has a gap (its internet dropped, or it picked the game up part way) and the other Mac saw those goals one by one, the gap is filled with them, and the game's quarters, runs, worm and moments are worked out again; nothing more is asked of PlayHQ, and Results says how many games had gaps filled from which Mac. This happens during the game too: a Mac turned on part way through a game, or whose internet dropped for a while, asks the other Macs on the network for that game so far (every 30 seconds while it has a gap) and fills it there and then, so its worm, runs, streaks and Commentator stats have the whole game, then carries on following the game by itself; the moments it fills in aren't played on air. A game also gets **Moments of the game**: the lead changes, comebacks, runs and super shots Studio saw, in order. Every finished game, streamed or not, gets **Game stats** (team against team from PlayHQ's team counts: goals, super shots, attempts and accuracy, plus quarters won, the longest run, the 3 and 5 goal runs and the biggest lead when Studio streamed or followed it) and a **Shooting duel** (each team's shooters). From the games Studio streamed or followed, the night also gets **Runs of the night** (each game's longest run of goals), **Super shot time** (the points each team scored in each quarter's super shot period, where the clock ran) and **Comeback of the night** (the winner who came back from furthest behind, with the game's worm). Upcoming adds a **Team sheet** for each game (both squads with games played) and, once two teams have met this season, a **Head to head**. **Watch it back** turns the replay's link into a QR code: paste the video's link (it starts with your channel's, from Live viewers on Competitions setup, when there is one).

Finals get their own pictures. After a division's last home and away round, Results adds **Finals bound**, the teams that made the finals with what each one's ladder place earned (a double chance, a week off, an elimination final), and the **Finals draw**, the finals week by week. On each finals night the draw is there again, under Upcoming to post before the games and under Results with the winners filled in, and the Match up, Full time and Results pictures name each final, such as Major semi final or Grand final. Once the grand final has a winner, Results adds **Champions**: the winners' logo, their name and the grand final's score. Call them **Premiers** instead under Finals on Competitions setup. A division gets these when it has a finals format on Competitions setup.

SPONSORS, THE LOGOS AND SAVING

Each picture has a **Sponsor strip** tick, which puts your naming, elite and major sponsors along the bottom and is remembered for next week. Under **Options**, **FirstPass Studio mark** (bottom right) and **PIVT logo** (bottom left) turn those on or off for every picture. **Template** sets the look of every picture: **Default** is the stream's own, and Dynamic, Junior, Retro, Professional, Thunder, Gradient, High contrast, Neon, Poster, Splash, Game Day, Hero and Final each bring their own shapes and lettering, and three netball sets draw on the game itself: Court (the court's lines and circles), Bib (a bib with its position letters) and Goal (the ring, net and post), and five sporting sets bring the kit around it: Bottle, Cones, Sneakers, Stadium and Locker room. Star (☆) the templates you like and tick **Favourites only** to list just those, and the one in use; it's remembered for the competition. Any template works with any colours, and every title stays on one line beside the competition logo, shrinking a little where it would otherwise wrap. A short ladder or a couple of games grow to fill the picture, and a Story keeps its words and logos out of the top and bottom, where Instagram puts its own bar and reply box. A long team name shrinks to stay on one line in its row. Athletes' photos come from the Athletes page, and only for athletes ticked **Approved for photo use**. Top shooters shows each shooter's photo automatically. On a Match up, Full time or Polaroid card, pick an athlete by name, narrowing the list by division and team. Hero and Final stand the picked athletes tall, in front of the title; a cut-out PNG with a transparent background looks best. **Colours** gives every picture a different look: **Competition colours** is the stream's own, and the others swap the competition's main, highlight and accent colours around (Accent first, Highlight ground, Accent ground), or go darker (Night), paler (Day) or down to two colours. **New colour set** makes your own from three colours, for a finals night or a sponsor round. A set marked **low contrast** has titles that may be hard to read. Your choice is remembered for the competition. Untick anything you don't want (**Select all** and **Select none** tick or untick every picture at once), then **Save pictures**: they are saved as PNGs in a folder for the night in **Pictures › FirstPass Studio**, which opens in Finder. **Share...** saves them too and opens the Mac's Share menu, to AirDrop them to your phone or send them by Messages or Mail. Tick **As a zip** to also save them as one .zip beside the folder; Share then sends that one file, which is easier to email or upload. Tick **As a carousel** to post them together as one Instagram carousel: the ticked pictures line up above Save in the order they'll post, and you drag one (or use ‹ ›) to move it. A carousel is one size and up to 20 pictures; each is marked 1/6, 2/6 and so on in its top right corner, with an arrow to swipe on (the first says Swipe), and they're saved numbered in that order in a folder of their own. A picture that can't be made, such as Milestones on a night with none in the line ups, is skipped and the rest are still saved; the line beside Save says which were skipped and why, in amber. In a carousel the rest are numbered again without a gap (1/5, 2/5...).

From a phone. The Socials tab works on a phone (or another computer) paired with **Allow everything**. On the Mac, press **Open on your phone** at the top of the Socials tab and scan the QR code with the phone's camera, on the same Wi-Fi as the Mac. The first time, the phone asks the Mac to pair: type a name, press **Ask the Mac** and choose **Allow everything** on the Mac, and the phone goes on to the Socials tab. After that the QR code (or the Studio link with `/socials.html` on the end) opens it straight away; add it to the Home Screen to open it like an app. Pick the pictures as on the Mac, then press **Save to this device**, and the phone shows them: press and hold one and choose **Add to Photos**, or tap **Download** under it. With **As a zip** ticked there's a button for the zip too. Each picture's caption shows above them, with **Copy**. The Mac app has to be running, since it draws them; nothing is left on the Mac, and the pictures stay ready for the phone for an hour. **Share...** isn't offered on a phone, because a phone's browser only shares files from a secure (https) link.

Studio on a phone. On a phone, Studio's bar has just the three phone pages: **Socials**, **Remote** (The Remote on a phone) and **Commentator** (a phone as a backup commentator mic, in Mics). Studio's own link opens Socials there. **Open on your phone** on the Mac has a QR code for each of the three. The Mac's other pages still open on a phone at their own links.

CAPTIONS

Each picture has a **Caption** box under it, for the words that go with the post. Open it, pick a style (**Hype**, **Professional**, **Emotional**, **Neutral** or **Fun**), tick **Short** for one or two sentences, and press **Write caption**. An AI writes it from the same facts the picture shows, and only those: the scores, teams and stats on the picture, never a miss or a low number, in Australian English. Your hashtags go on the end. Read it before you post it. The text is yours to change, or type your own without the AI at all. **Again** writes another, **Undo** brings back what you had, and **Copy** copies it to paste into the post. The next picture starts in the style you picked last. Under the box is who wrote it and about what it cost.

Save pictures also saves each ticked picture's caption, under the picture's name, in **Captions.txt** in the folder with them (and in the zip). On a phone, the captions come back with the pictures, each with **Copy**. The captions are kept until you pick another game night.

Setting it up. The captions are written by Claude, ChatGPT or Gemini, with your club's own API key, so the club pays the AI company for what it uses: usually a fraction of a cent a caption. On the Mac, open the **Captions** card at the top of the Socials tab:

1. **Written by:** pick Claude (Anthropic), ChatGPT (OpenAI) or Gemini (Google).
2. **API key:** press **Get a key**, make a key on the company's site (it asks you to add a payment method), paste it in and press **Save key**. Studio checks the key with the company first, keeps it in the Mac's Keychain and only ever shows its last four characters. **Remove key** takes it off this Mac.
3. **Model:** the models your key can use, each with its price. The smaller, cheaper models (Claude Haiku, GPT mini, Gemini Flash-Lite) write captions well.
4. **Hashtags:** your hashtags and @handles, which go on the end of every caption as typed.

The card shows the model's price and what this month's captions cost, with a link to the company's billing page, which has the real figure. When Studio doesn't know a model's price, the card says so in amber and the captions show the words used instead of a cost. Studio writes one caption at a time, and at most 30 in 10 minutes. The key and the AI settings can only be changed on the Mac; a phone paired with **Allow everything** can write captions with them.

3.13 Recordings

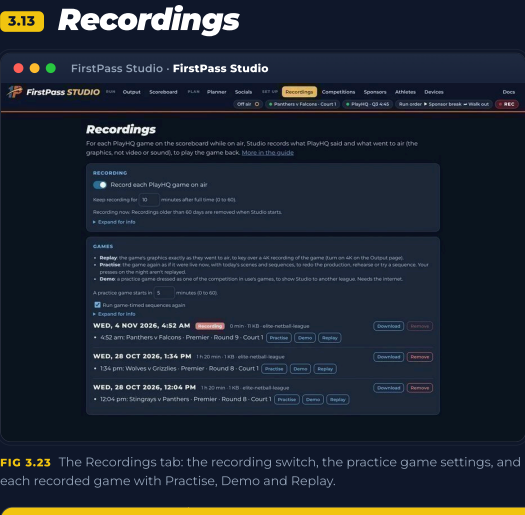


FIG 3.23 The Recordings tab: the recording switch, the practice game settings, and each recorded game with Practise, Demo and Replay.

While Studio is on air, it records each PlayHQ game on the scoreboard. A recording holds what PlayHQ said (score, clock, quarter, line ups) and what went to air (the scene, the scoreboard and what each graphic showed), with the time of each. It has no video or sound, so a game is about a megabyte. Recording starts when the game is due on the scoreboard (10 minutes before the start, as set by automatic game selection) and stops 10 minutes after full time. It also stops when the scoreboard moves to the next game or Studio goes off air. Recording is on unless you untick **Record each PlayHQ game on air** on the **Recordings** tab, or press **Stop** beside Record on the Output page's status board. Recordings stay on this Mac for 60 days and hold nothing secret.

The Recordings tab lists each recorded game with three buttons. Each plays the game back for a different job:

	REPLAY	PRACTISE	DEMO
Plays	The game's graphics exactly as they went to air	The game again, as if it were live now	The game as one of another league's
Scenes, sequences and settings	The night's, as they were	Today's: change them first to fix the night	Today's, with the other league's teams, athletes, logos and colours
Sequences	Only what went to air	Start by themselves again from the game	As Practise
Your presses on the night	On screen as they went to air	Not replayed: you produce it again	Not replayed
Needs the internet	No	No	Yes
Use it to	Put 4K graphics on a game recorded in 4K, for YouTube	Redo a night's production, rehearse, or try a new sequence	Show Studio to another league

Whichever one runs, a red strip on every Studio window says so, with **Stop**. Its controls are at the top of the Recordings tab, and the Stream Deck plugin's **Replay** key works them too. Nothing in it counts for your sponsors and it isn't recorded as a game of its own. **Stop** puts the live scoreboard back exactly as it was (a Replay never touches the live scene; a practice game plays on it, as a game night does).

PUT 4K GRAPHICS ON A RECORDED GAME

01 **On the night:** stream at 1080p as usual and record the camera in 4K, on a HyperDeck if you can, with its timecode set to **Time of day**. Studio records the game by itself.

02 **Afterwards:** on the Recordings tab, press **Replay** beside the game. Studio puts that game's graphics back on every output and waits, paused, at the start of the game.

03 On the Output page, turn on **4K**, so the outputs send the graphics at 2160p. Key them over the 4K video in a 4K switcher or recorder. Or add the output's **Output canvas** link (Devices tab, Links) to an editing or streaming program as a web page source at 3840 × 2160.

04 **Line the graphics up with the video.** If the video is on a HyperDeck, set **Follow** to **A HyperDeck's timecode** and type its IP address. The graphics then follow the HyperDeck as you play, stop and scrub it. Another player can do the same over SDI into the UltraStudio (**Timecode into the UltraStudio (SDI)**). With neither, pause the video on a moment you can see, such as the first centre pass. Pick the same moment in the list, press **Jump there**, then press Play on both together.

05 If the graphics are a little early or late, **Nudge** them a frame or a second. With **Follow**, Studio remembers the sync for that recording.

06 Record the result. Studio adds no video or sound: use the night's programme recording for the sound. Press **Stop** when you're done.

Replay shows the night's data in **today's** version of each graphic, so a graphic fixed since the night shows the fix. A graphic you have removed from the Output page since still plays, because the recording keeps its link. A video or image you've deleted from the media folder comes up blank, because the recording only keeps its name.

REDO A NIGHT'S PRODUCTION

If something went wrong on the night, such as a sequence at the wrong time, a scene that needed changing or a graphic that needed fixing, fix it first. Then run the game again with **Practise**:

01 Fix the scenes, sequences or graphics on the Output page.

02 On the Recordings tab, leave **Run game-timed sequences again** ticked. Set **A practice game starts in** to at least the minutes of your earliest pre-game sequence (a Run-up 15 minutes out needs 15). A pre-game sequence due before the practice starts is skipped, as on a night Studio was opened late.

03 Press **Practise**. The game is due that many minutes from now, and Studio runs as it does on a game night. The scoreboard follows the recorded score and clock, and sequences start by themselves from the game, at the head of Output 1's run order.

04 Produce it as you would live: anything you did by hand on the night, do again. **Jump to** goes to the start, a quarter or full time, and **Pause** stops the clock.

Practise replays only what PlayHQ said, so a game scored by hand rather than live on PlayHQ has no score or clock in a practice. A practice game doesn't follow a player's timecode: to lay it over a video, start them together and use **Pause** and **Nudge** (see below). Practise is also the way to rehearse before the season or try a new sequence.

GRAPHICS ON ANOTHER COURT'S GAME

The night's other games Studio follows in the background (The night's other games) are recorded too, each on its own from Studio's first reading of it until 10 minutes after full time, named **(followed)** and marked **followed, not streamed** on the Recordings tab. Nothing of them went to air, so they have no Replay: play one with **Practise**, and Studio runs it on the scoreboard as if it were live, so you can produce it over somebody's video of that game, from a phone or a camera:

01 Press **Practise** beside the game, then the quarter the video is on (**Q1**, **Q2**...) and **Pause**. Each quarter's button is the moment PlayHQ's scorer started that quarter's clock.

02 Cue the video on that quarter's first centre pass and pause it.

03 Press Play on both together (the Stream Deck's practice **Play** key helps).

04 Watch for a goal: if the score changes before or after the ball goes through, **Nudge** the game a second at a time until they match.

Its recording has what Studio asked PlayHQ that night (the fixture, the ladders, the live scoring); anything else a graphic needs, such as the line ups, is read from PlayHQ when you play it, so it needs the internet, and the line ups show the final numbers.

DEMO GAMES

A demo game is a practice game dressed as another league's game, for showing them Studio in their own colours. Set up their competition and switch to it, then press **Demo** beside one of your recorded games. Pick one of their divisions and a finished game. Studio plays your recording as that game: their teams, logos, colours and athletes, with your recording's score, clock and moments. Athletes are matched by position, so your GS's goals become their GS's. The **Athletes** table shows who each of your athletes is shown as, and you can change any of them. **Swap home and away** lets either team win. **Save demo** keeps the whole setup by name, so on the day it's one click under **Saved demos**. Their competition is read from PlayHQ, so a demo needs the internet.

CHAPTER 4 OF 11

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

- 1 Two people, not one
- 2 Switching to manual
- 3 The score
- 4 The clock and the quarter
- 5 Breaks and celebrations by hand
- 6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

Production manager

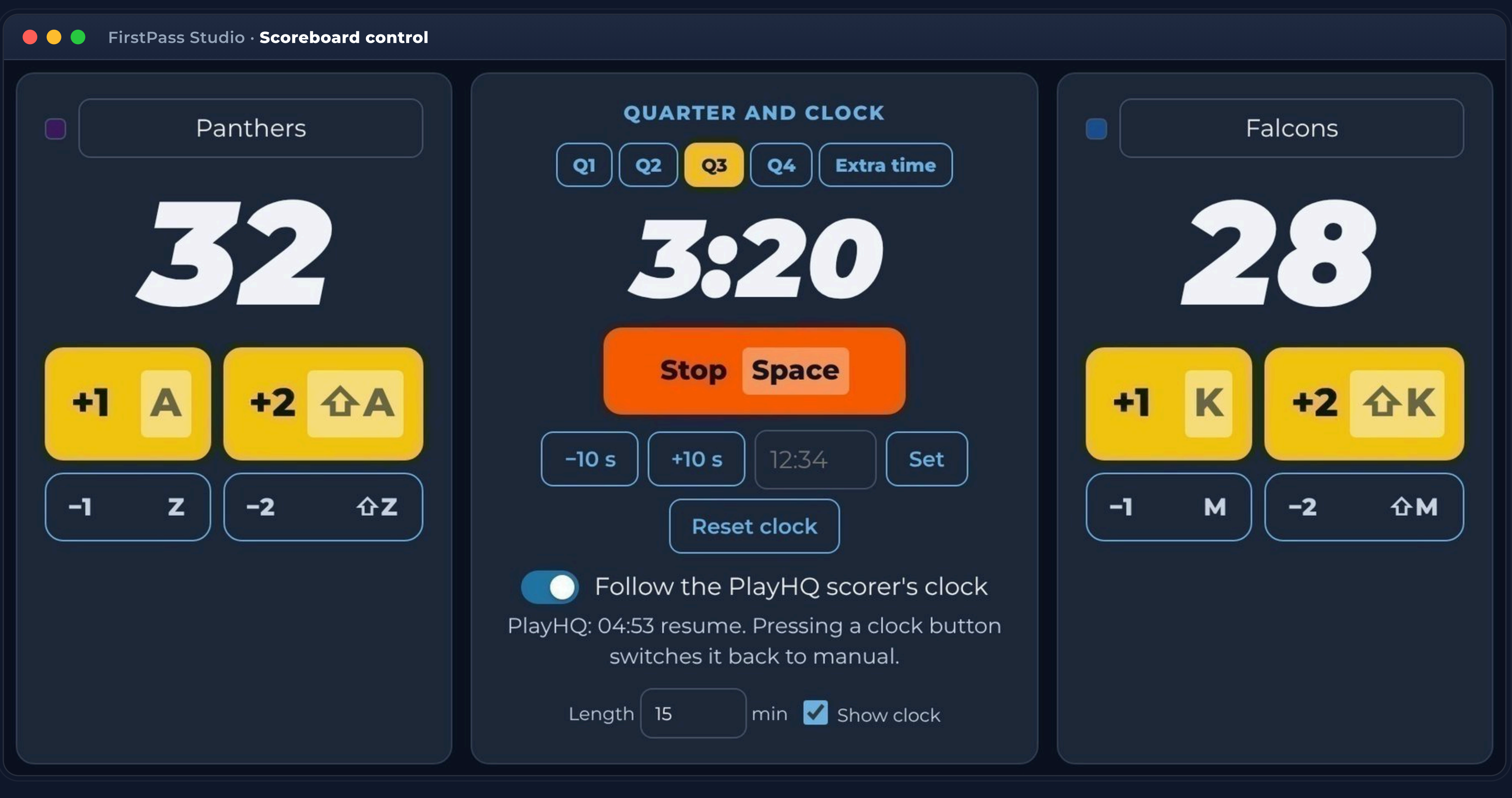
- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> **If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.**
>
> CHAPTER 4 · RUNNING IT BY HAND

4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again. If PlayHQ is still behind your score when you press it (the scorebench hasn't caught up yet), the board keeps your score, quarter and clock until PlayHQ catches up or has something new from the scorer, then follows it; the PlayHQ chip reads **your score held** meanwhile. So you can press Follow PlayHQ as soon as the scorebench is back without losing the goals you scored by hand.



Follow the PlayHQ scorer's clock

PlayHQ: 04:53 resume. Pressing a clock button switches it back to manual.

Length 15 min ☑ Show clock

FIG 4.1 Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

If the scorebench drops off line instead (its wifi, say), PlayHQ still answers but its score stops moving. Studio spots this two ways: PlayHQ says the game is under way while its live feed says it isn't being scored (or it's five minutes past the start), or nothing new comes from the scorer for four minutes while its clock runs. Then the bar's PlayHQ chip turns purple and reads **PlayHQ not live**, and the status board's PlayHQ line reads **Not live**; hover over either for why. Fix the scorebench if you can, or score by hand until it's back and press Follow PlayHQ: your score stays on the board until PlayHQ catches up. When PlayHQ catches up with several goals at once, the score jumps straight to it, the goals that came together show as a gap on the last 10 goals and the score worm (their order and times aren't known), and no super shot or moment plays for them. If it catches up before the next quarter starts, as it does when the scorebench comes back at quarter time, that quarter's score still counts on the game stats.

4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

4.4 The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.

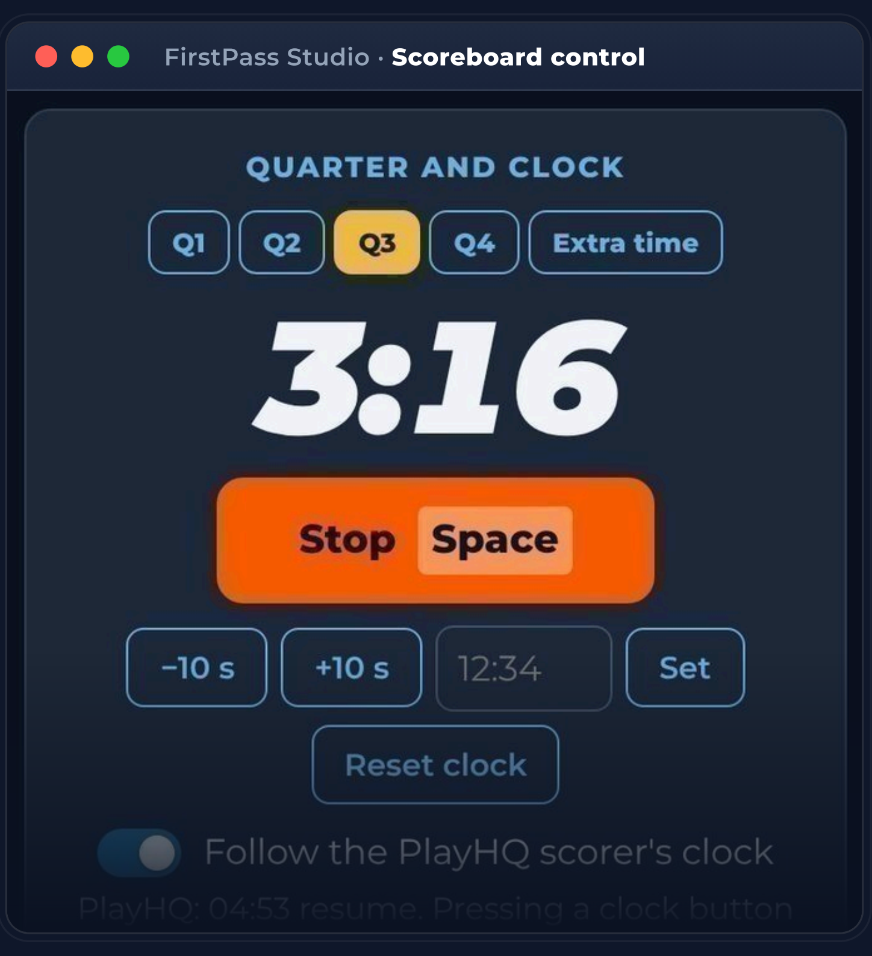


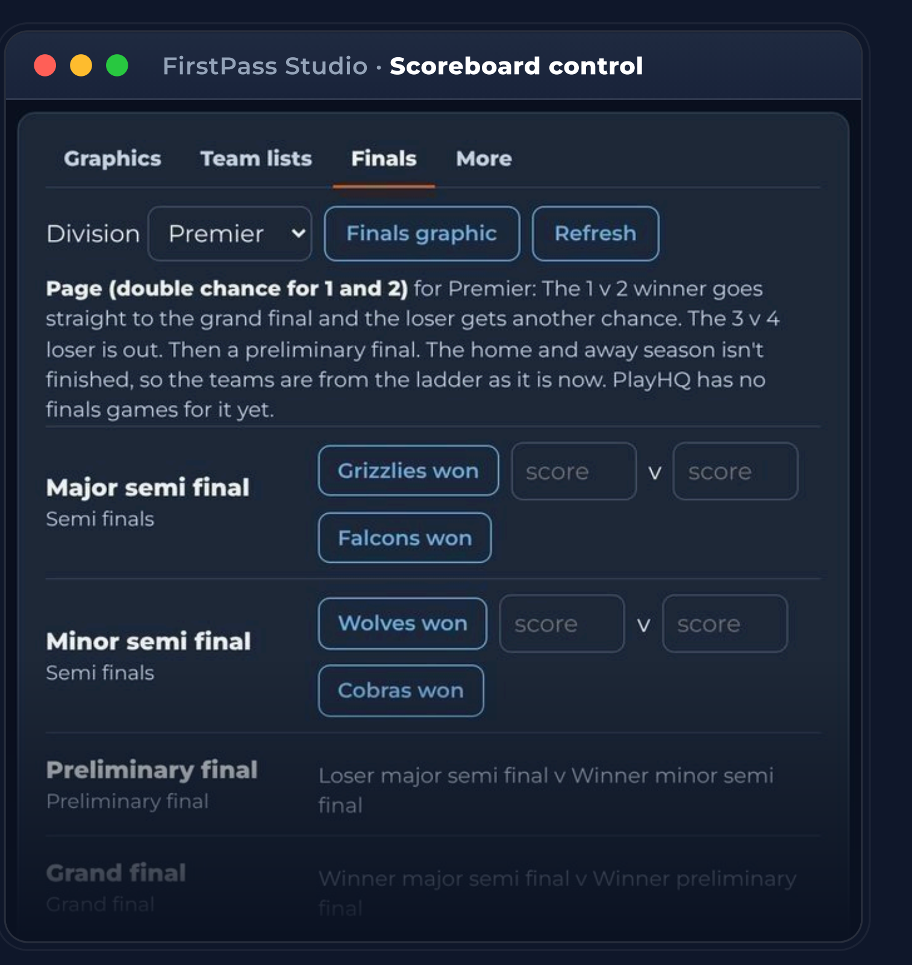
FIG 4.2 Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

4.6 Finals



Page (double chance for 1 and 2) for Premier: The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. The home and away season isn't finished, so the teams are from the ladder as it is now. PlayHQ has no finals games for it yet.

Major semi final Semi finals

Minor semi final Semi finals

Preliminary final Preliminary final — Loser major semi final v Winner minor semi final

Grand final Grand final — Winner major semi final v Winner preliminary final

FIG 4.3 Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 5 OF 11

The graphics

Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its address; the options named here are the ones worth knowing.

1 Always on · 2 Before the game · 3 During play · 4 After the game · 5 The competition · 6 The breaks · 7 Checking the outputs

All of them follow the game night, game and court picked on Scoreboard control, take the competition's colours, font and logos from its setup, and fit long team names. Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits. Every page, with its link, its options and a Preview button, is also listed under **Links** on the Devices tab.

5.1 Always on

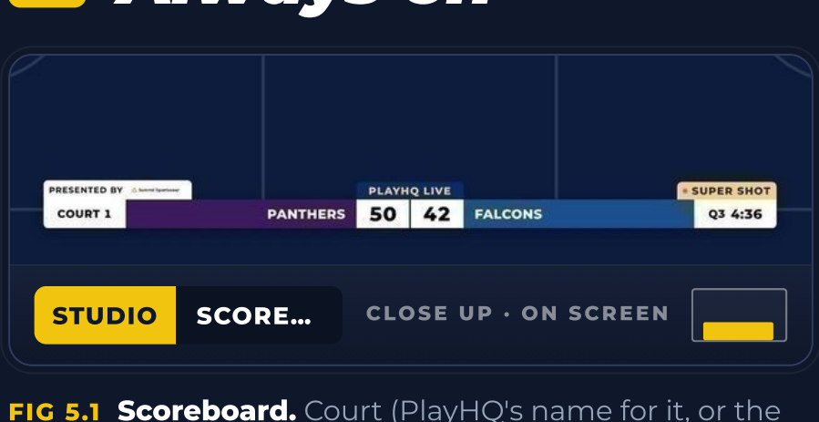


FIG 5.1 Scoreboard. Court (PlayHQ's name for it, or the one typed on Scoreboard control), both teams with their logos and club colours, the score, the quarter and the clock, centred at the bottom above the ticker. Options: position (top centre, bottom left and so on), super shot tab on or off, and the naming-rights sponsor's tab.

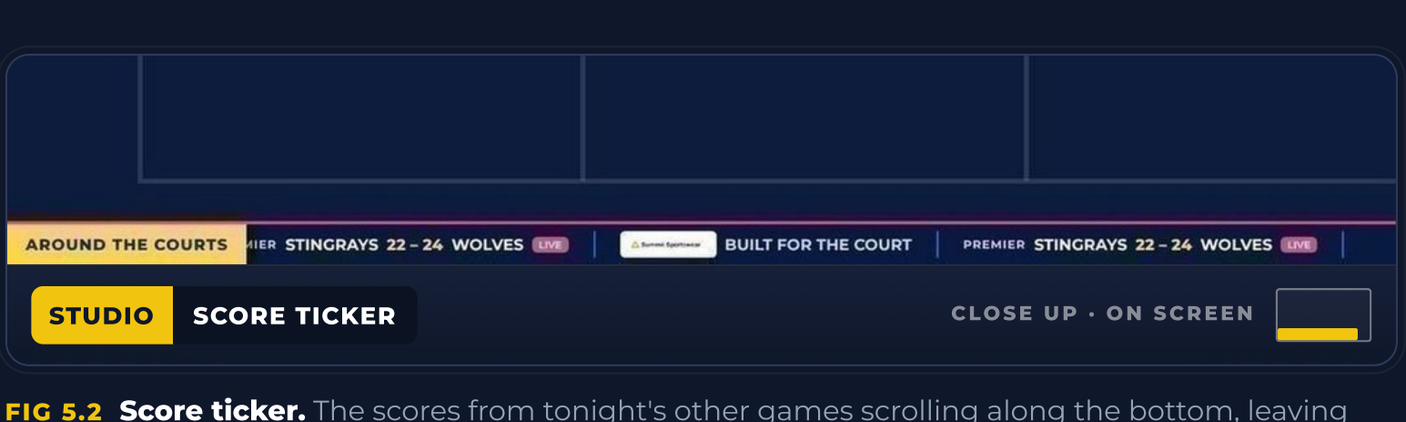


FIG 5.2 Score ticker. The scores from tonight's other games scrolling along the bottom, leaving out the game on the board, updated every 30 seconds. Speed from the control page or a Stream Deck key; a Partner sponsor's logo takes a slot in the line.

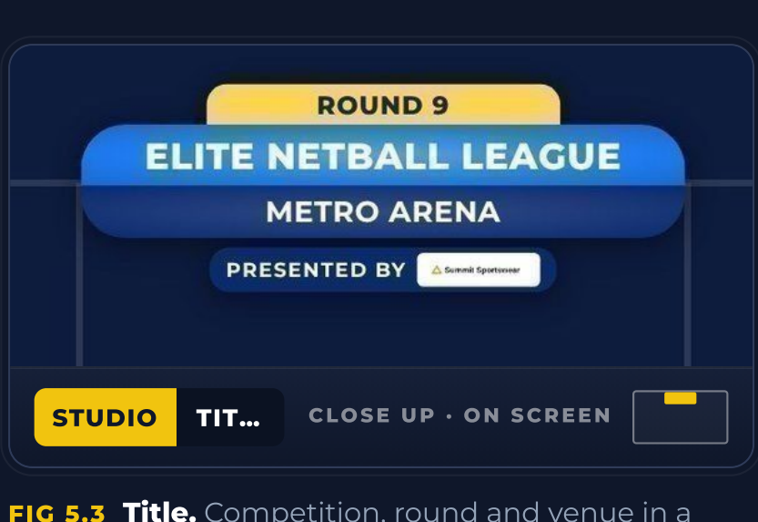


FIG 5.3 Title. Competition, round and venue in a banner at the top. Options: an emoji either side, the competition logo above it, top or bottom, and the naming-rights sponsor under it.

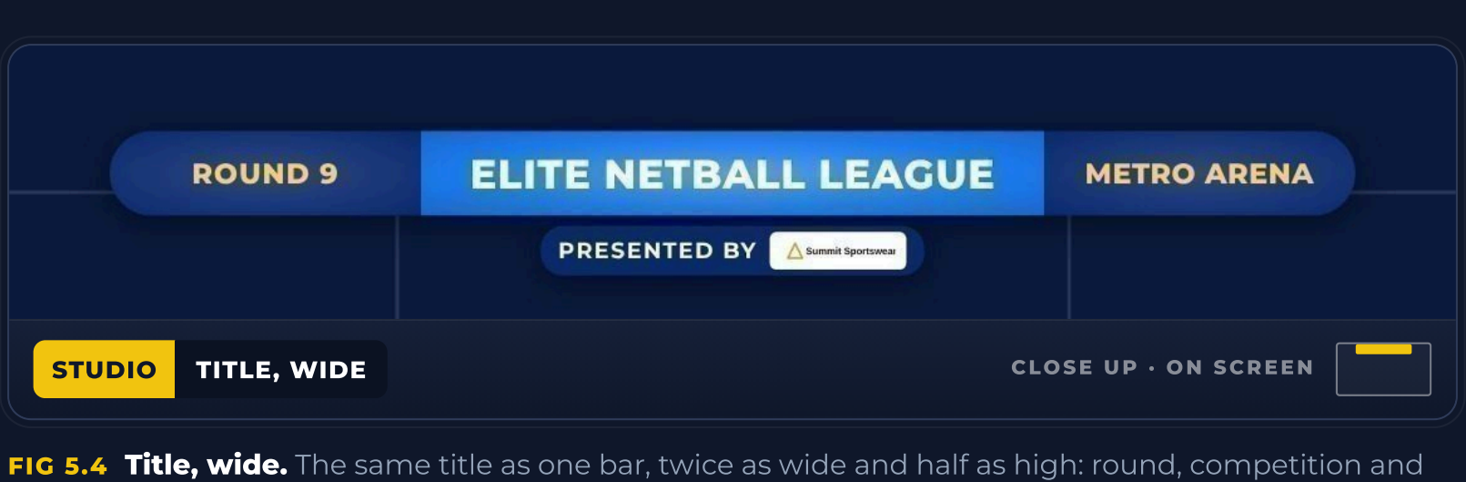


FIG 5.4 Title, wide. The same title as one bar, twice as wide and half as high: round, competition and venue side by side, the name in the middle. Its ends are dark with yellow words, or both yellow, whichever suits the competition's colours. With it, put the commentators box under the title, since the wide bar reaches its corner spot.

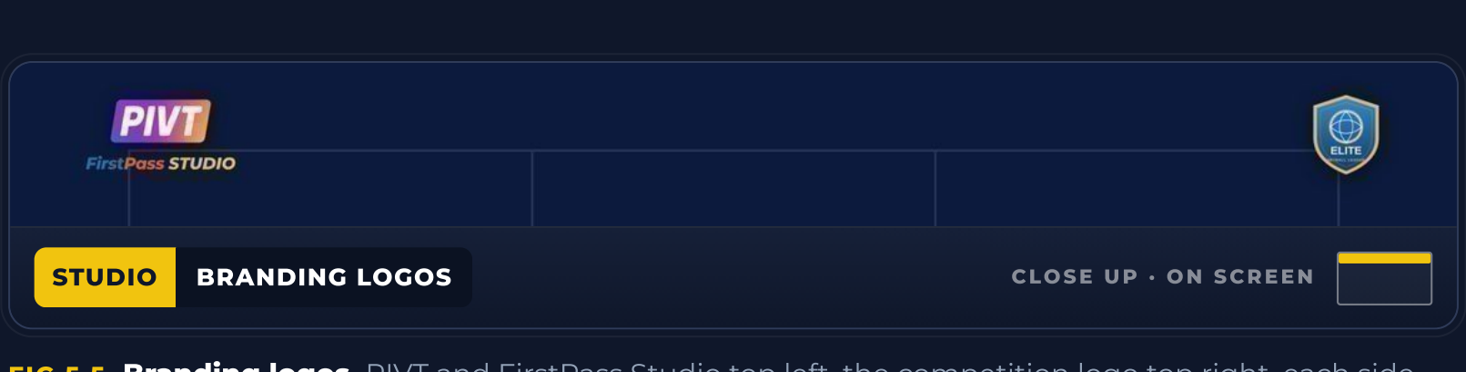


FIG 5.5 Branding logos. PIVT and FirstPass Studio top left, the competition logo top right, each side 100px in from its edge of the frame so the corners match. Each can be hidden; the gap from the edges (or from the centre instead), height and size adjust the placement.

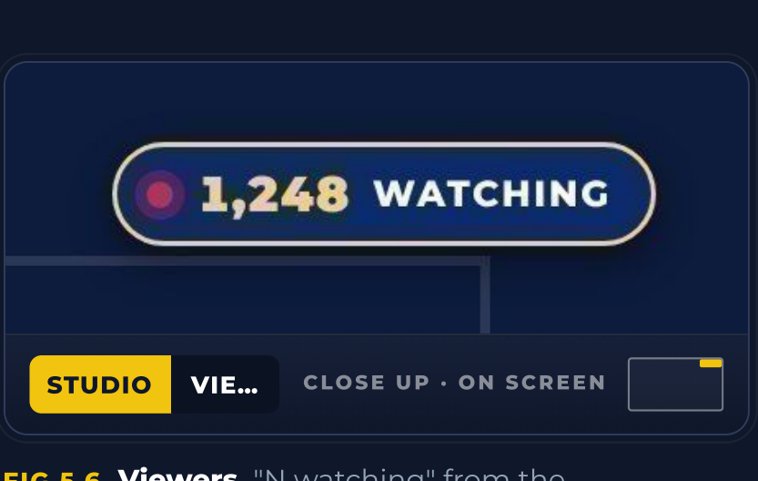


FIG 5.6 Viewers. "N watching" from the competition's YouTube and Facebook streams, top right, shown only once the count passes the competition's minimum. **Shows: peak** makes it a card for the end of the night.

5.2 Before the game

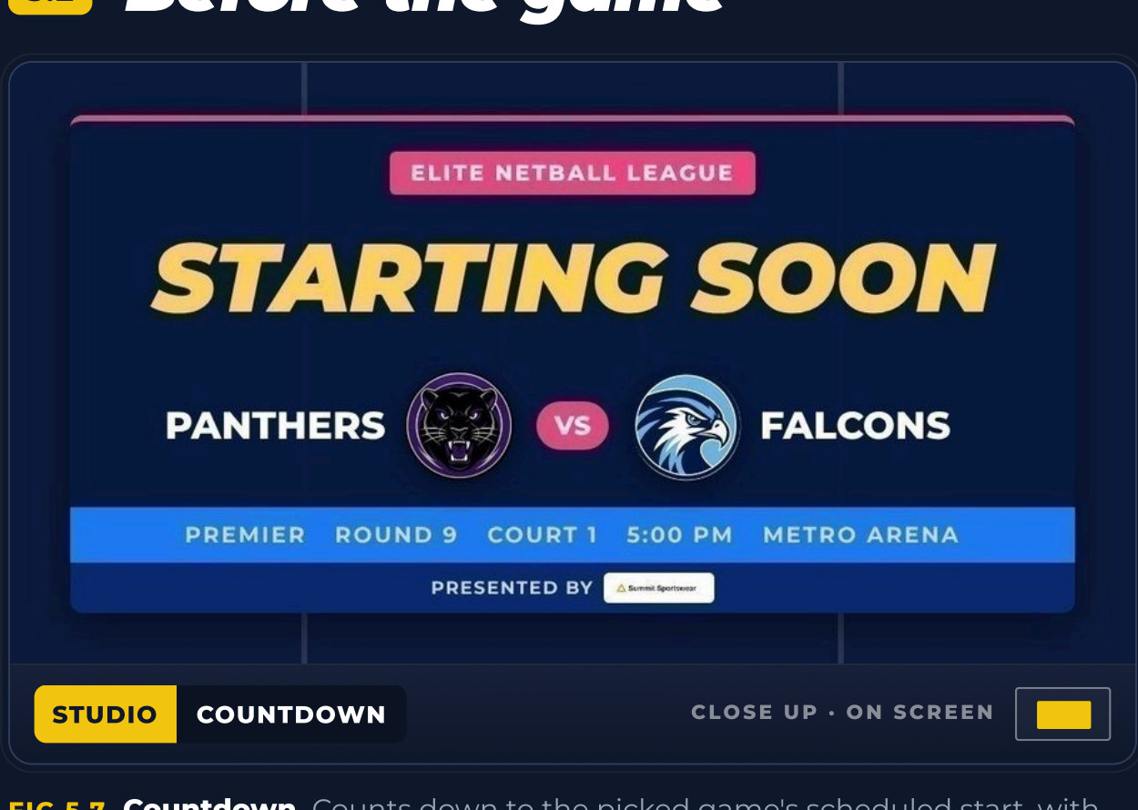


FIG 5.7 Countdown. Counts down to the picked game's scheduled start, with the two teams. Options: heading, the words at zero, a fade over the last seconds, a sponsor, a **Mini** size (a small timer like the title banner, which can sit where the title does) and a position on a 3 x 3 grid (top, middle or bottom; left, middle or right).

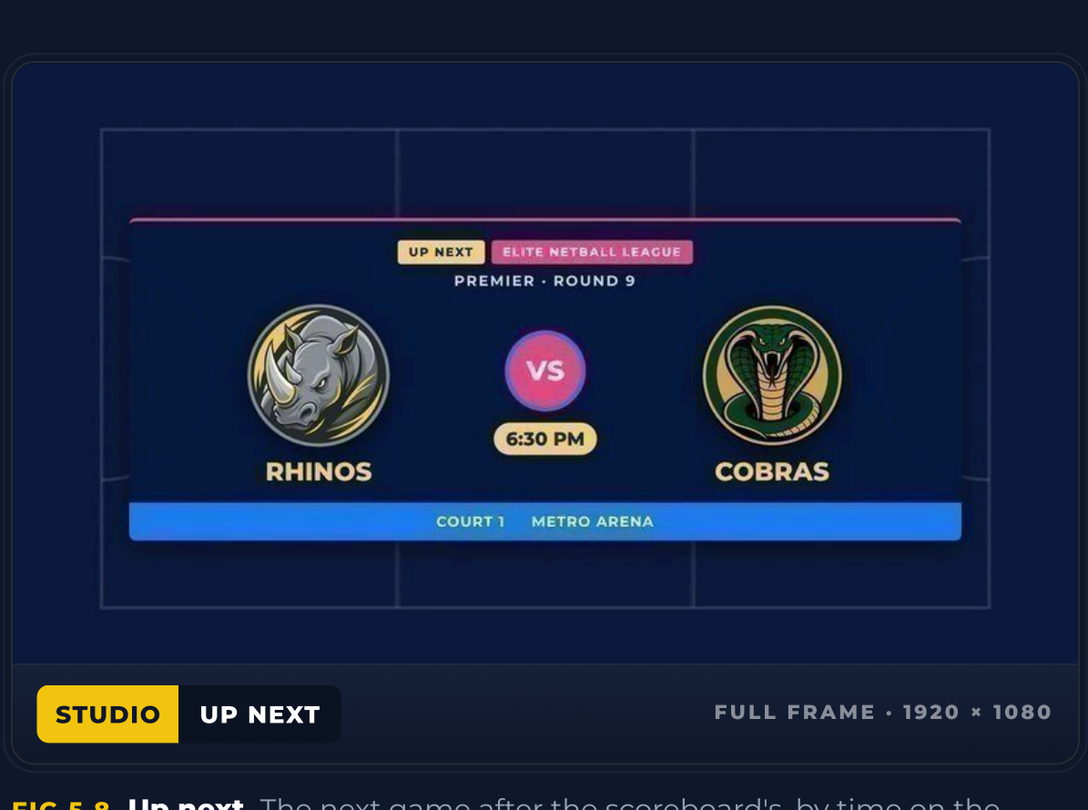


FIG 5.8 Up next. The next game after the scoreboard's, by time on the same court that day (or any court), whatever round or division it is in, with its time, teams and logos. After the last game it becomes the end-of-night card.

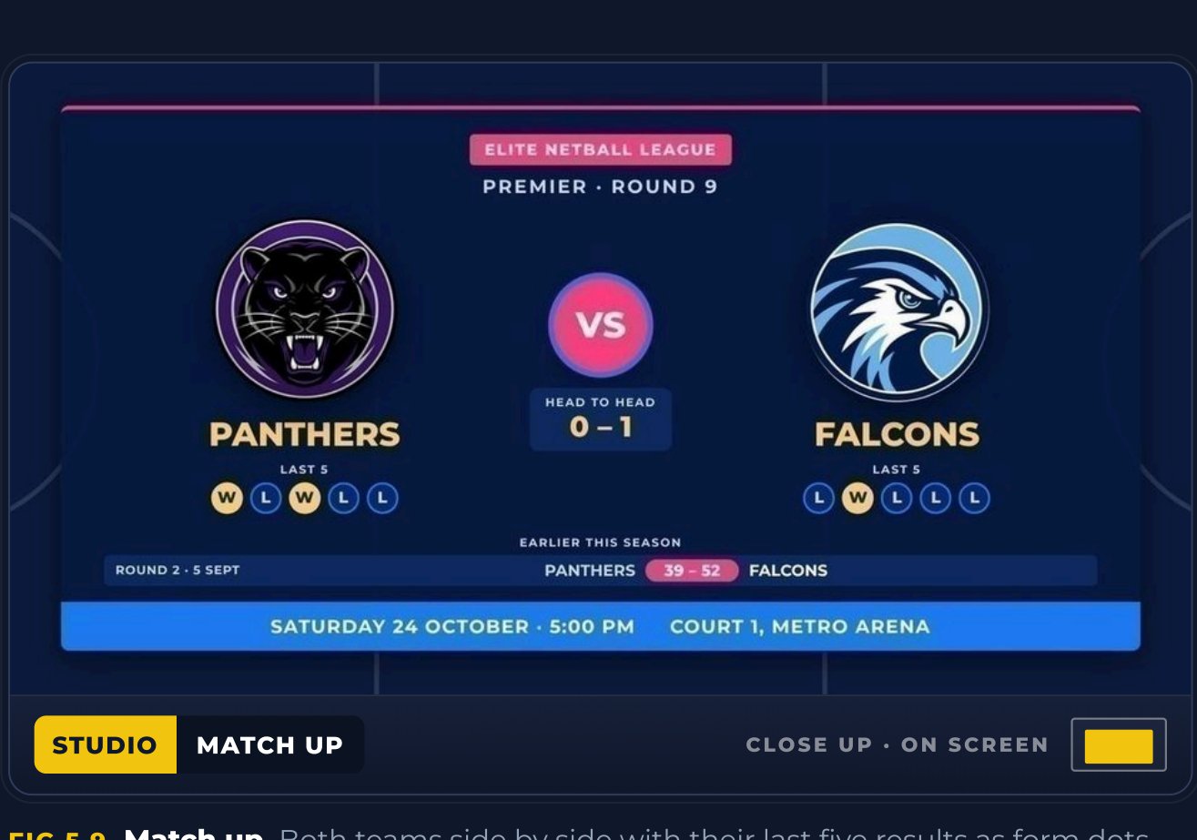


FIG 5.9 Match up. Both teams side by side with their last five results as form dots and this season's head to head. **Form** and **Head to head** can each be hidden.

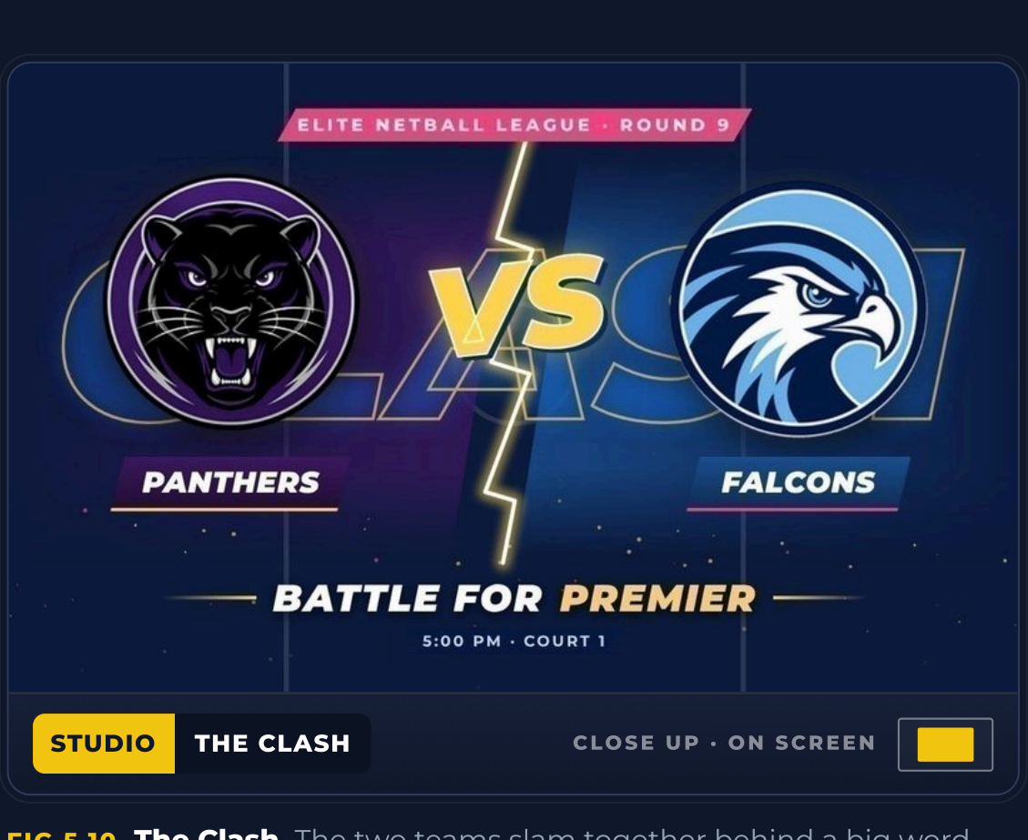


FIG 5.10 The Clash. The two teams slam together behind a big word (IMPACT, CLASH, SHOWDOWN, your own) in their club colours. Options: the word, the bottom line, animate away after N seconds.

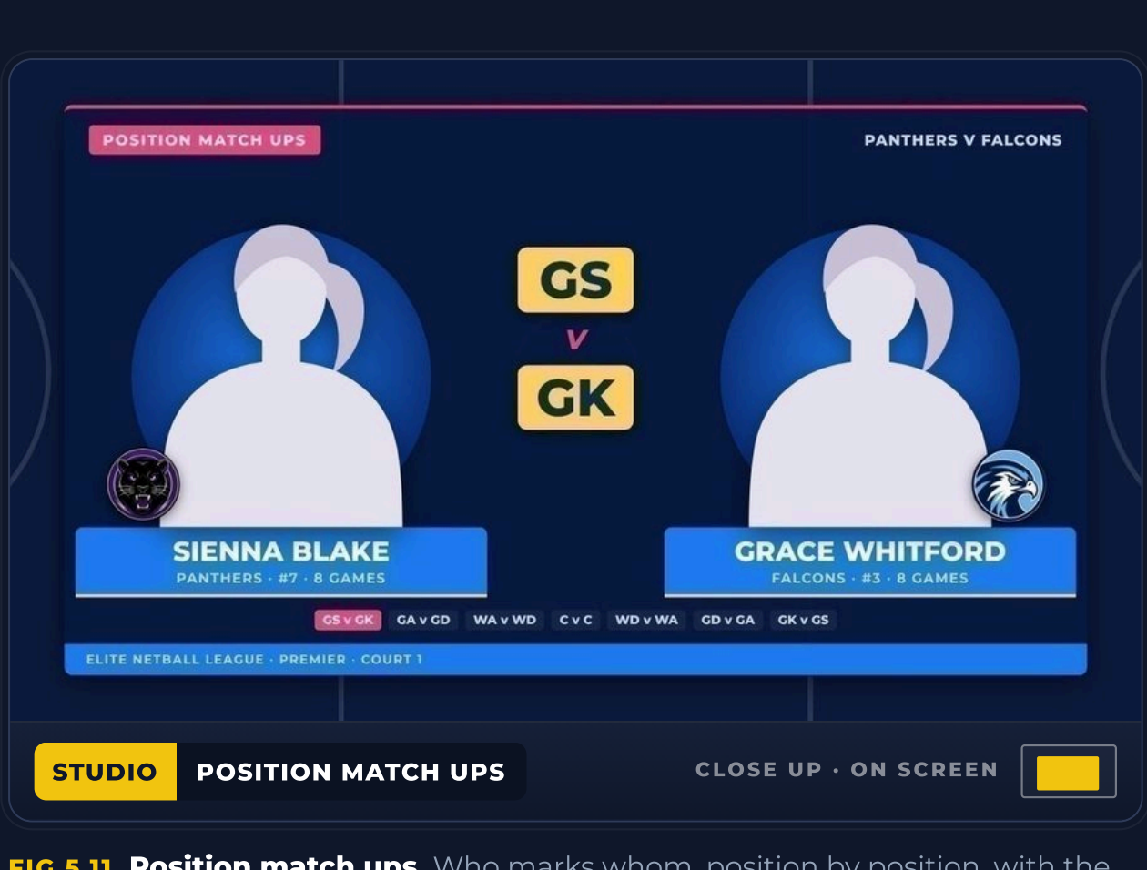


FIG 5.11 Position match ups. Who marks whom, position by position, with the athletes' photos from the Athletes page or a silhouette. Options: seconds per pairing, loop or stop, start with GS or GK, one pairing only, animate away at the end.

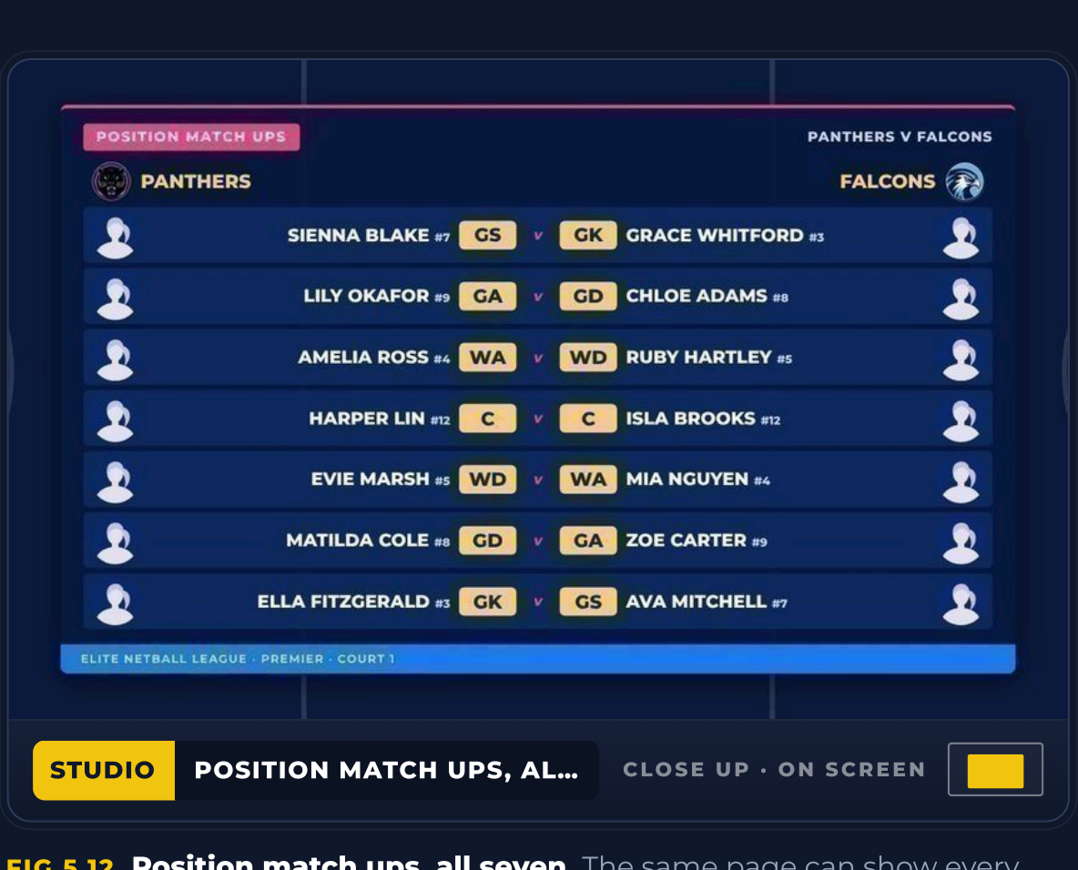

POSITION MATCH UPS
PANTHERS V FALCONS
PANTHERS
FALCONS
SIENNA BLAKE GS v GK GRACE WHITFORD
LILY OKAFOR GA v GD CHLOE ADAMS
AMELIA ROSS WA v WD RUBY HARTLEY
HARPER LIN C v C ISLA BROOKS
EVIE MARSH WD v WA MIA NGUYEN
MATILDA COLE GD v GA ZOE CARTER
ELLA FITZGERALD GK v GS AVA MITCHELL
ELITE NETBALL LEAGUE · PREMIER · COURT 1
STUDIO POSITION MATCH UPS, AL…
CLOSE UP · ON SCREEN

FIG 5.12 Position match ups, all seven. The same page can show every pairing at once as one card.

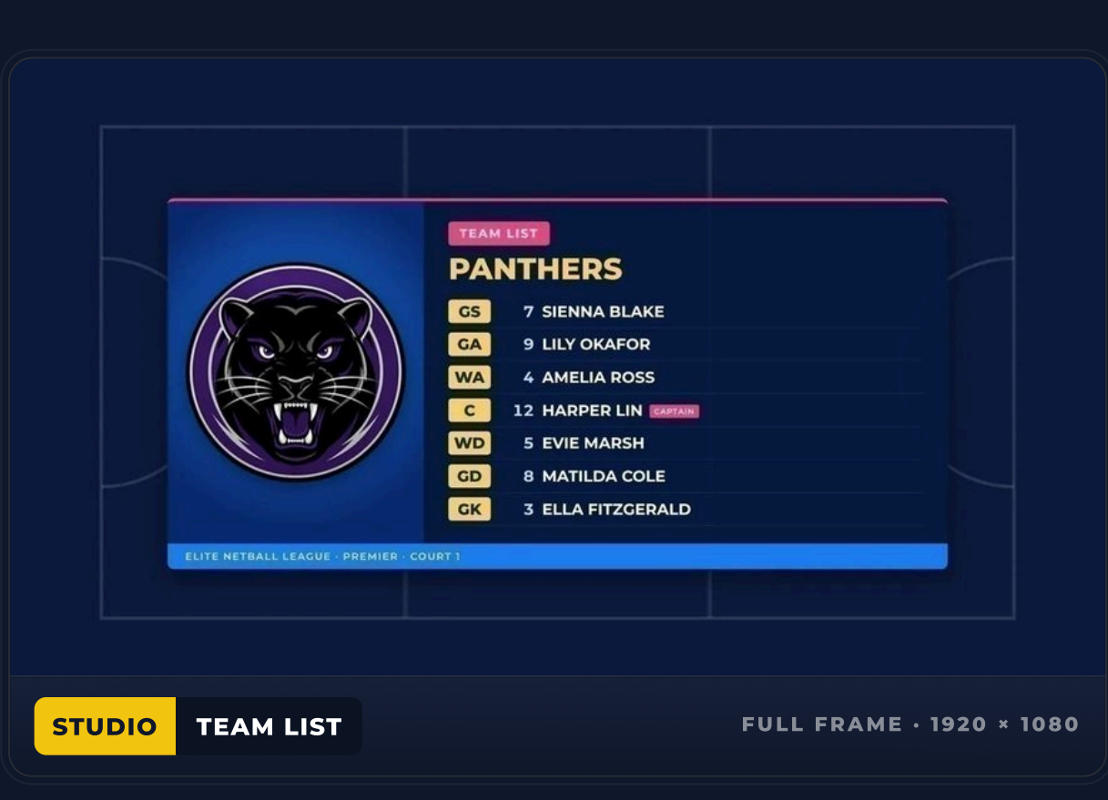


FIG 5.13 Team list. One team's line-up from PlayHQ with positions, captain and bench, warmed up before the game so it shows at once. Add one source per side.

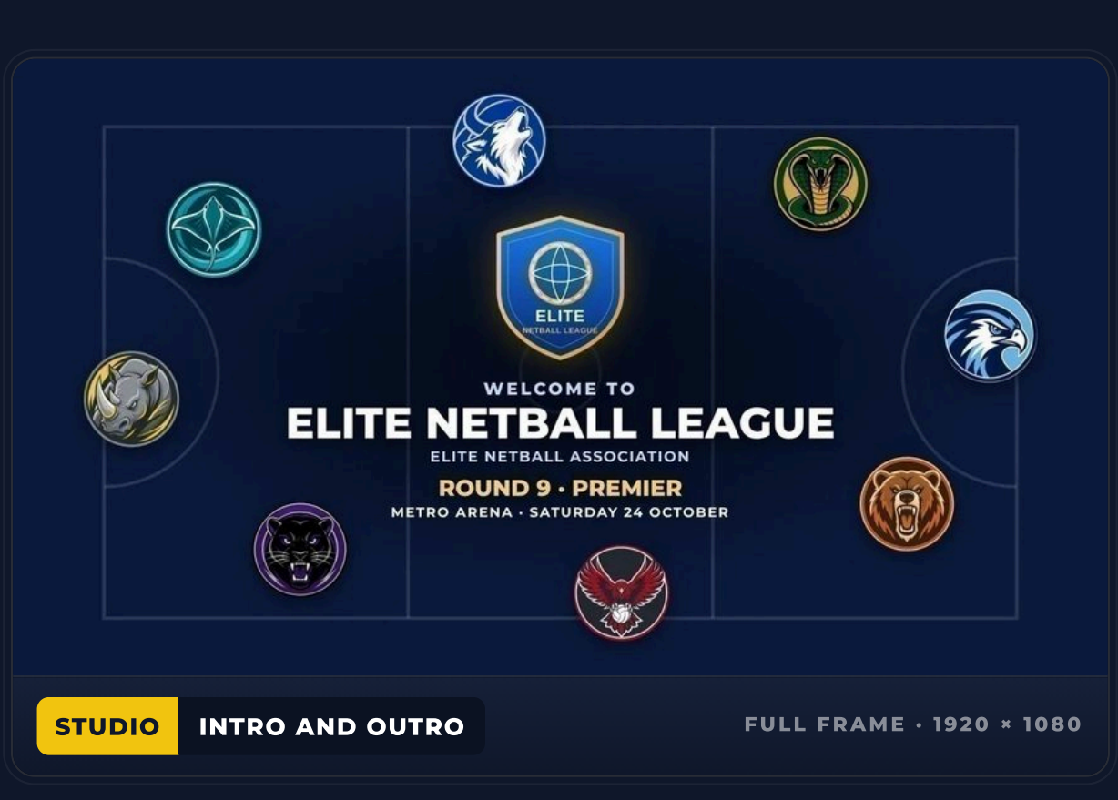


FIG 5.14 Intro and outro. Six full-screen styles, about 8 seconds, with the round's teams and a yellow message line. Options: style, teams from a division or round, time on screen, a theme song with its timing and volume, sound effects, a background.

5.3 During play

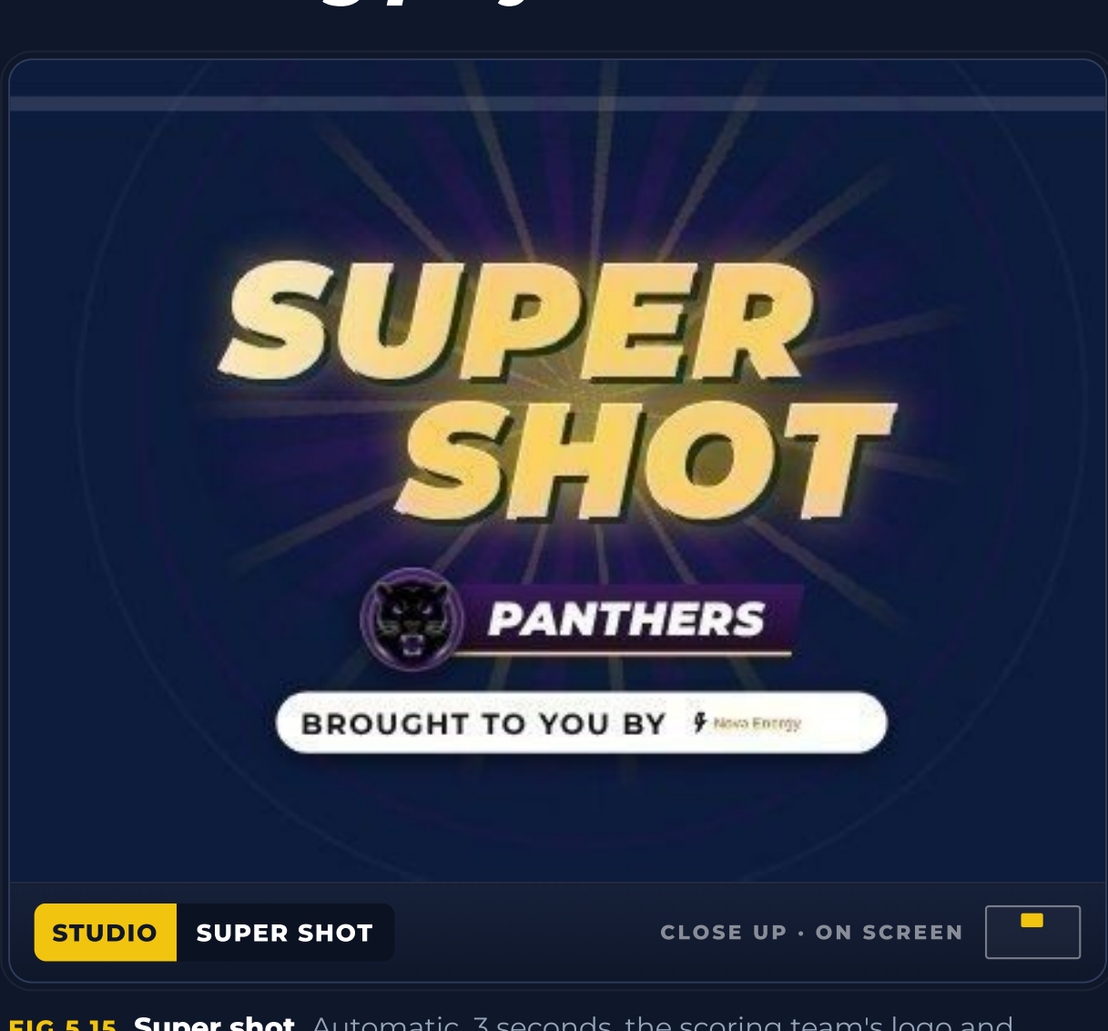


FIG 5.15 Super shot. Automatic, 3 seconds, the scoring team's logo and colour; the sponsored-moment tag names who brought it. **Plays: home / away / no team** makes a manual one.

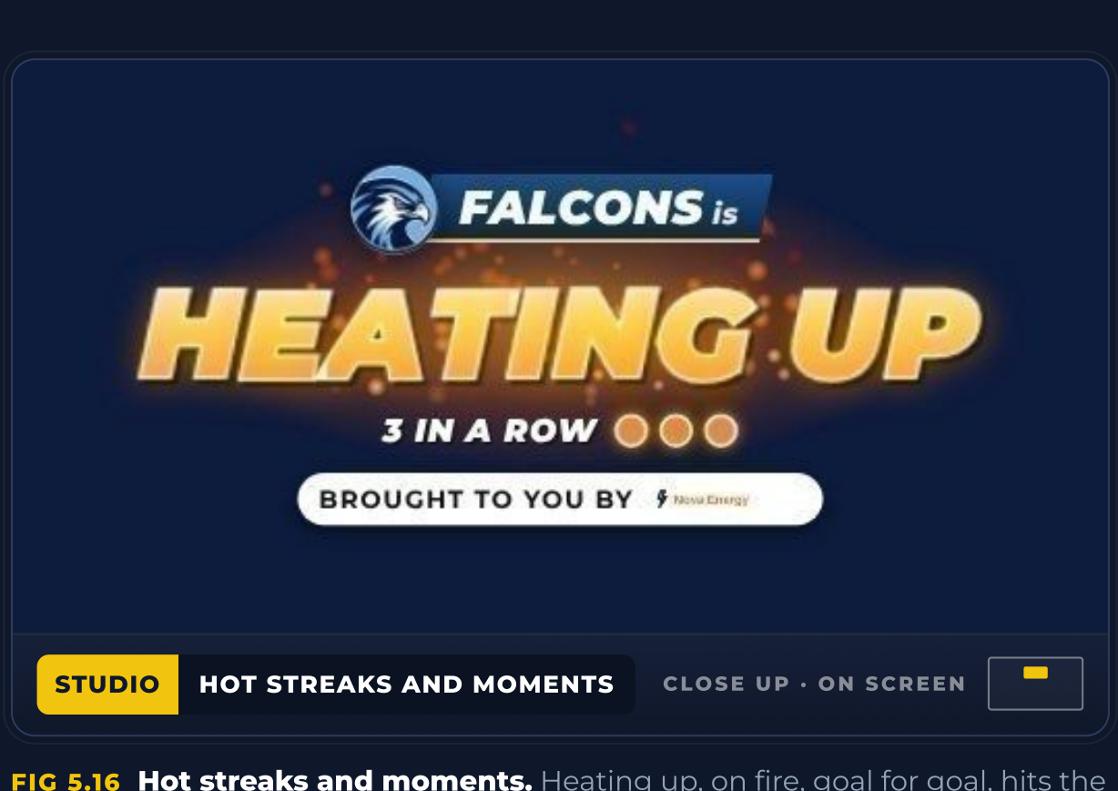


FIG 5.16 Hot streaks and moments. Heating up, on fire, goal for goal, hits the front, all square, coming back, on the siren, back to back. Automatic from the goal order, or **Plays** picks one to play once.

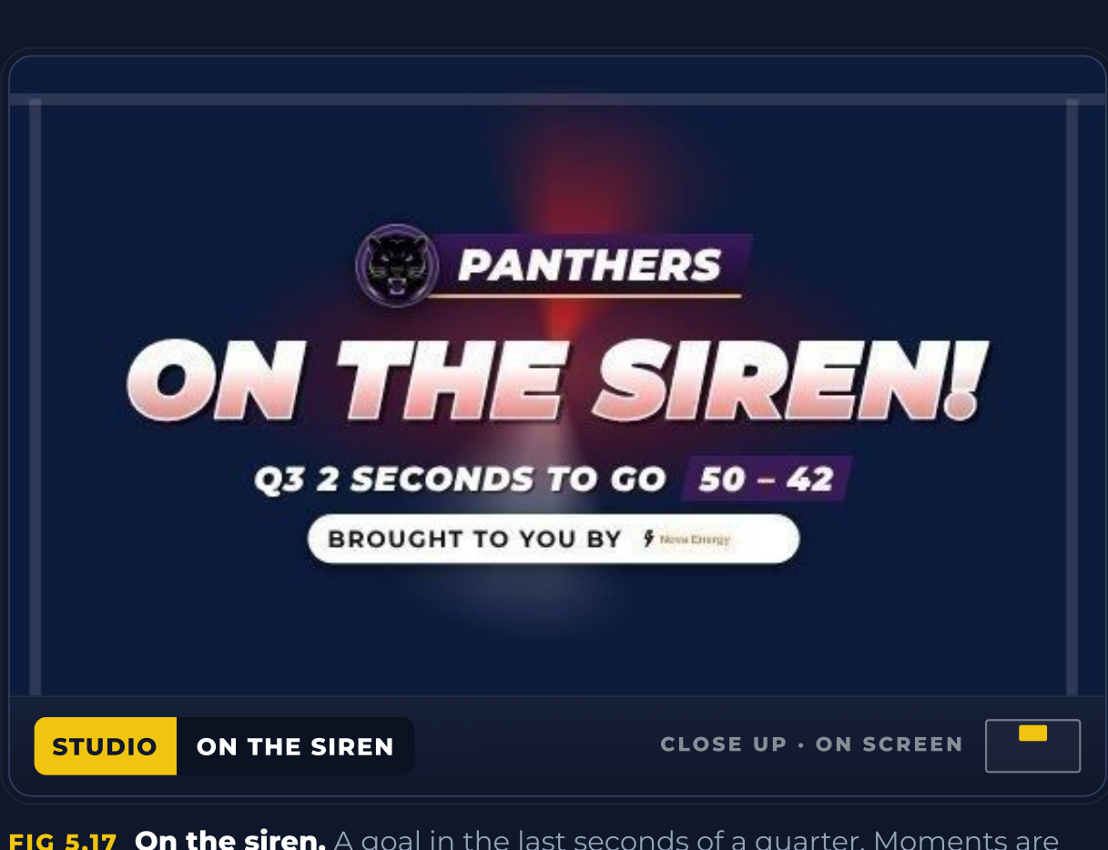


FIG 5.17 On the siren. A goal in the last seconds of a quarter. Moments are the only animations that need nothing from you; they also take the sponsor's "brought to you by" tag.

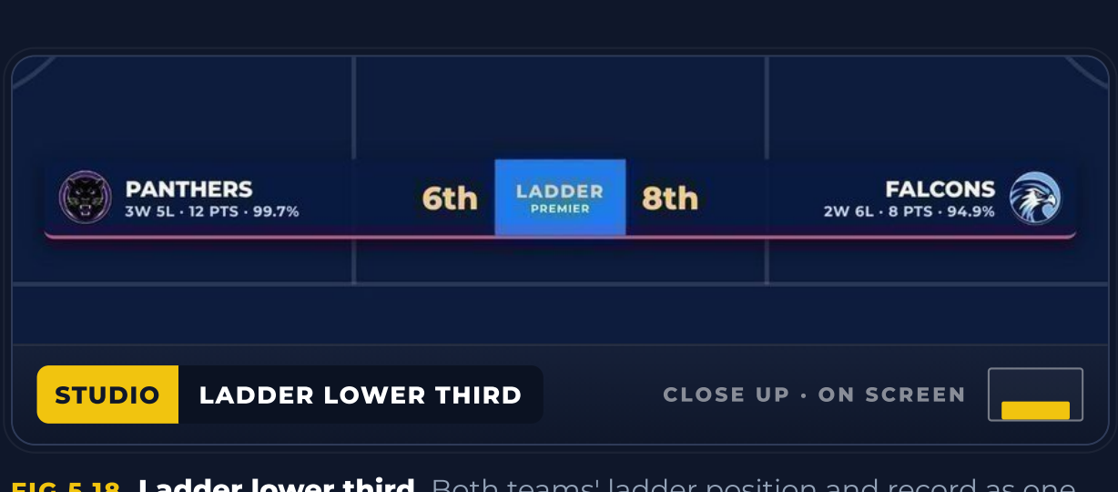


FIG 5.18 Ladder lower third. Both teams' ladder position and record as one strip right above the scoreboard, the same width as it, for a glance during play without covering the game.

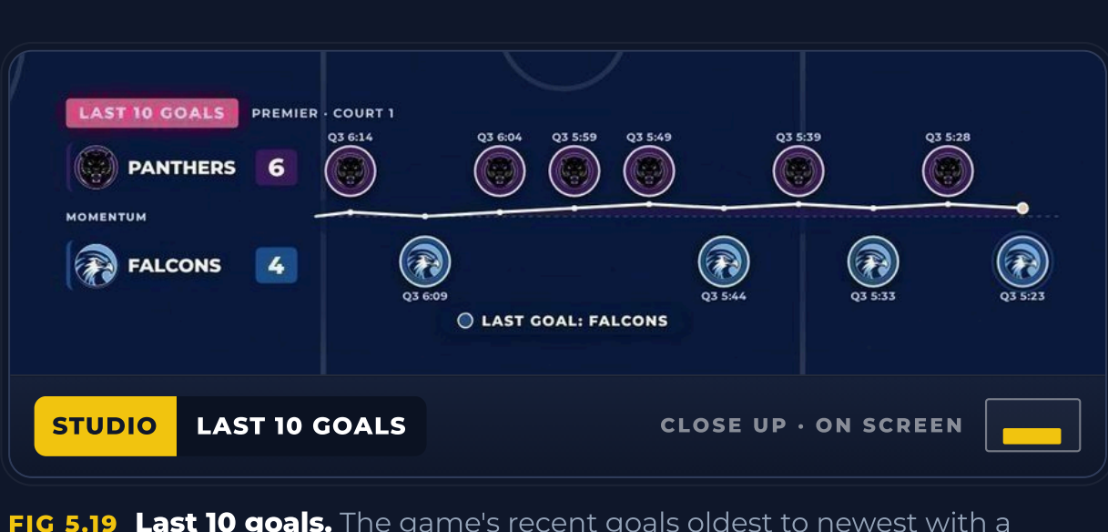
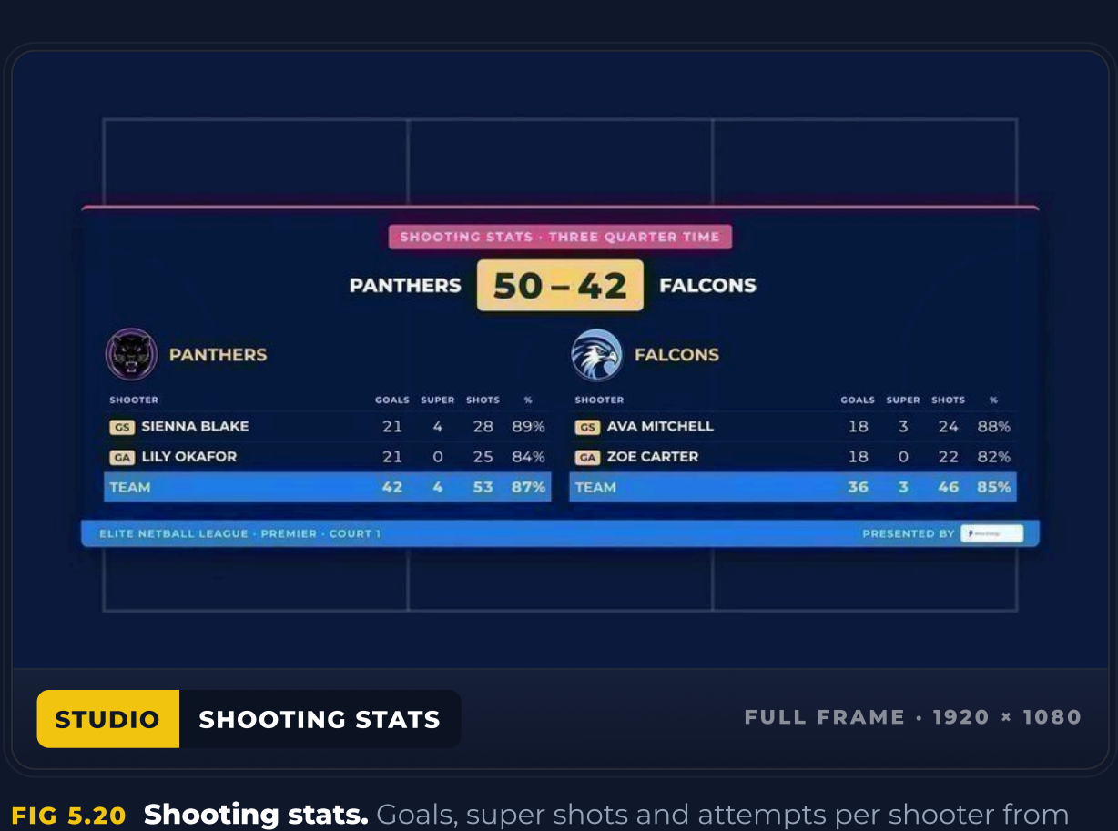


FIG 5.19 Last 10 goals. The game's recent goals oldest to newest with a momentum line and the current run, worked out by Studio from the score, updating while it stays on. Options: horizontal or vertical, how many goals, where it sits (top, middle or bottom, and left, centre or right) and heading.



SHOOTER	GOALS	SUPER	SHOTS	%
GS SIENNA BLAKE	21	4	28	89%
GA LILY OKAFOR	21	0	25	84%
TEAM	42	4	53	87%

SHOOTER	GOALS	SUPER	SHOTS	%
GS AVA MITCHELL	18	3	24	88%
GA ZOE CARTER	18	0	22	82%
TEAM	36	3	46	85%

FIG 5.20 Shooting stats. Goals, super shots and attempts per shooter from PlayHQ's game page, for a break or a quarter-time sequence. Super columns are left off for a division without super shots.

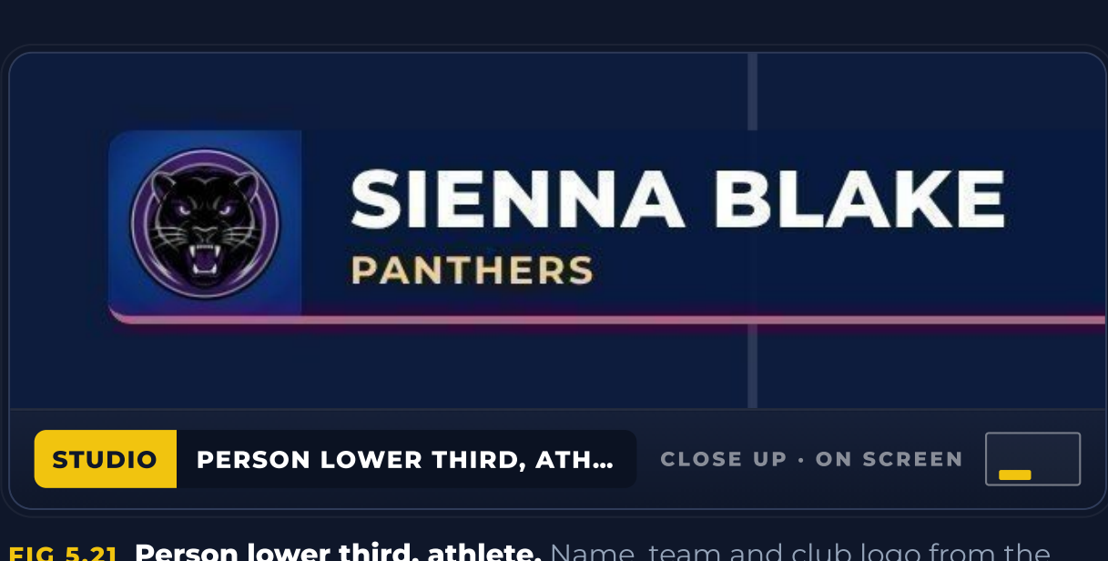


FIG 5.21 Person lower third, athlete. Name, team and club logo from the athlete register, with their photo if there is one. The Output page suggests names as you type.

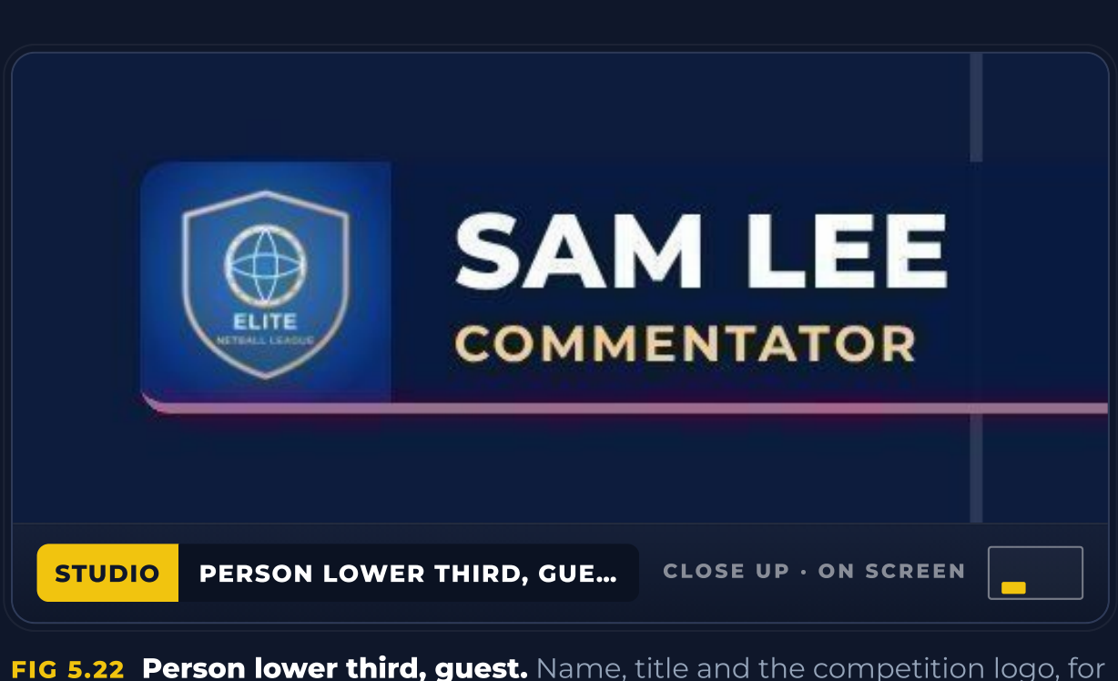


FIG 5.22 Person lower third, guest. Name, title and the competition logo, for a commentator, an official or a sponsor's guest. Options: a tab above, position left or right, animate away.

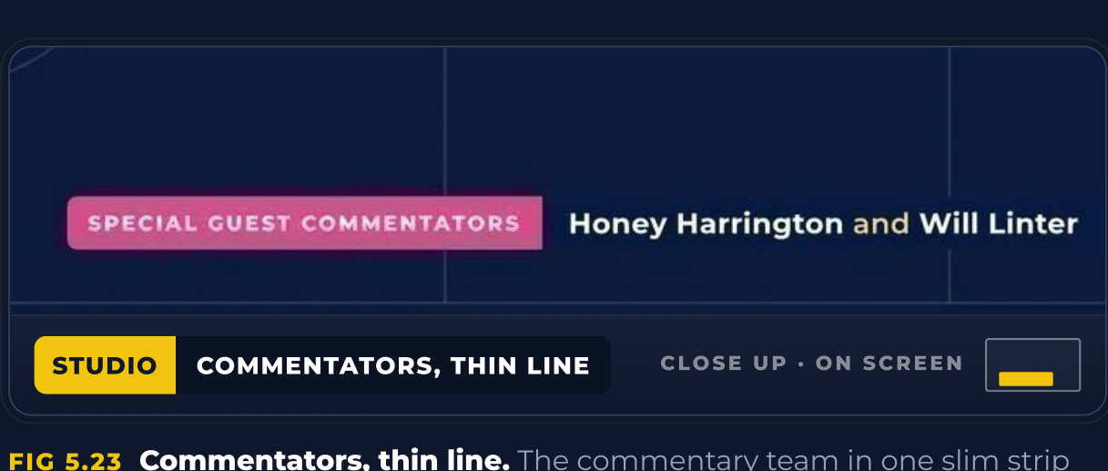

SPECIAL GUEST COMMENTATORS
Honey Harrington and Will Linter
STUDIO COMMENTATORS, THIN LINE
CLOSE UP · ON SCREEN

FIG 5.23 Commentators, thin line. The commentary team in one slim strip above the scoreboard: a pink tab, like Special guest commentators, and up to four names joined with "and". With no names typed, whoever the Production Planner has on the mic.

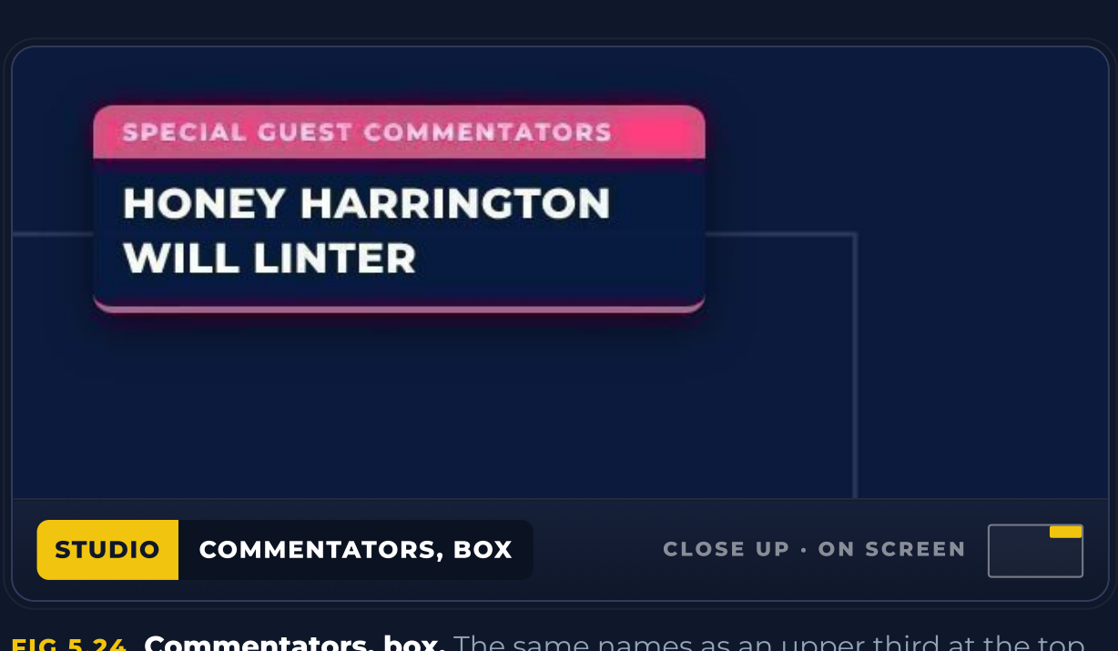


FIG 5.24 Commentators, box. The same names as an upper third at the top right, just left of the competition logo, a line per name. Options: under the title instead, the tab's words, animate away.

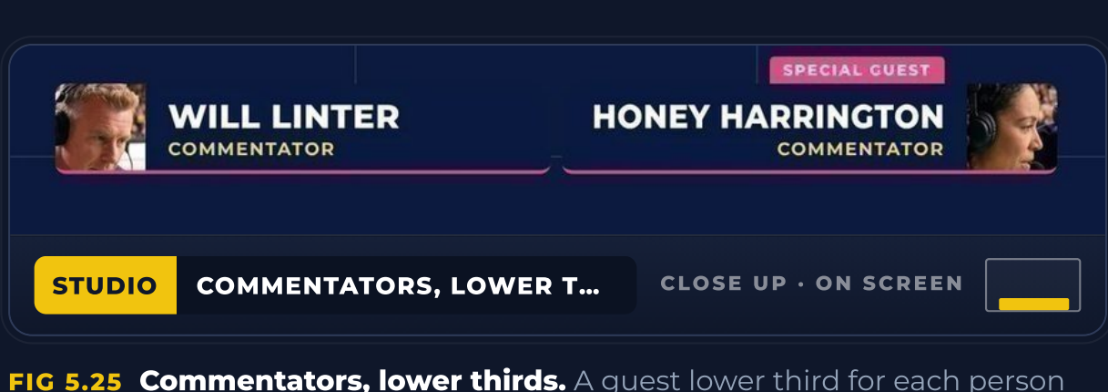


FIG 5.25 Commentators, lower thirds. A guest lower third for each person on the mic, from one source: one on the left, two left and right, three across the scoreboard's width. With no names typed, whoever the Production Planner has on, a guest with a Special guest tab. Options: each commentator's headshot in place of the logo, the line under the names, just above the scoreboard instead of at the bottom.

5.4 After the game

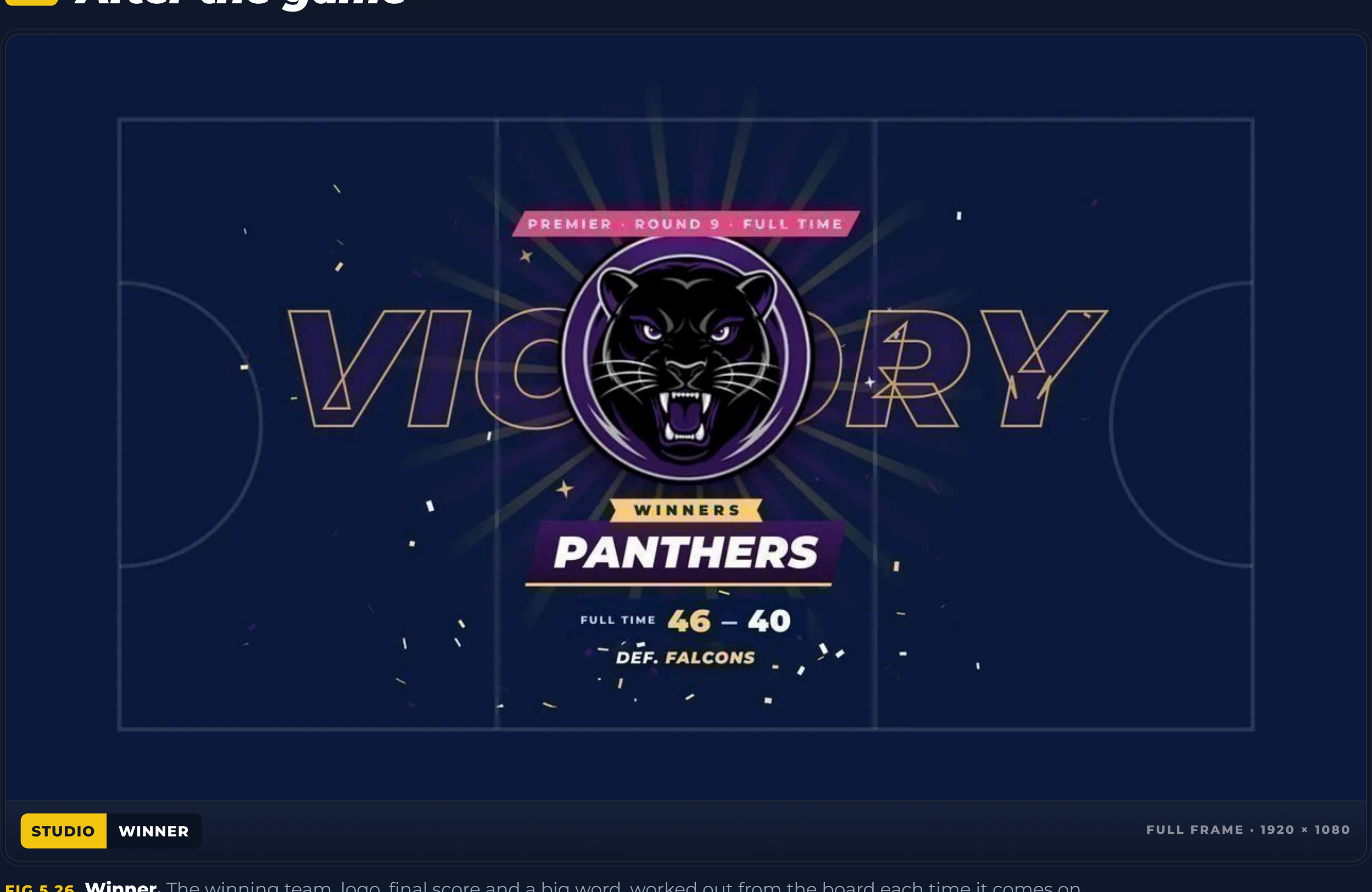


FIG 5.26 Winner. The winning team, logo, final score and a big word, worked out from the board each time it comes on. Options: force a result, the word, the line under the score, animate away.

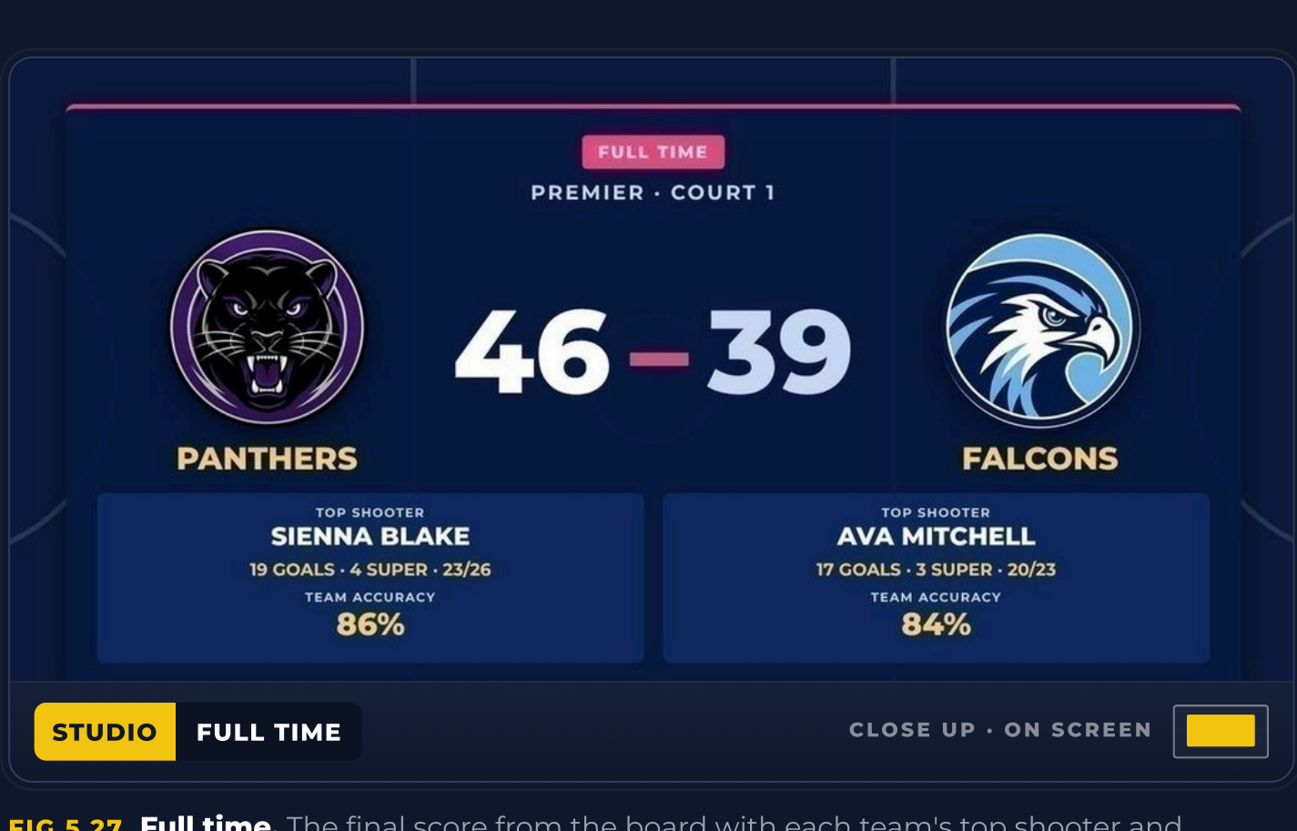
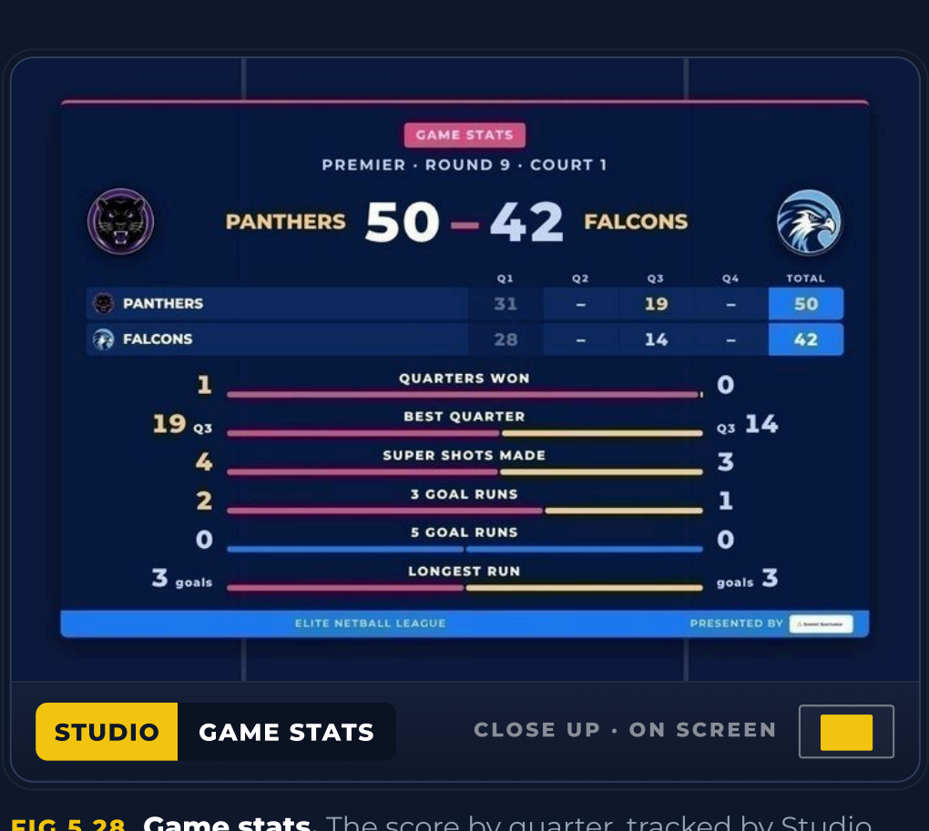


FIG 5.27 Full time. The final score from the board with each team's top shooter and shooting accuracy from PlayHQ's game stats. **Heading** makes it a half time card.



	Q1	Q2	Q3	Q4	TOTAL
PANTHERS	31	–	19	–	50
FALCONS	28	–	14	–	42

PANTHERS		FALCONS
1	QUARTERS WON	0
19 Q3	BEST QUARTER	Q3 14
4	SUPER SHOTS MADE	3
2	3 GOAL RUNS	1
0	5 GOAL RUNS	0
3 goals	LONGEST RUN	goals 3

FIG 5.28 Game stats. The score by quarter, tracked by Studio from the board, then team against team: quarters won, best quarter, super shots made, 3 and 5 goal runs and the longest run, with a sponsor's footer.

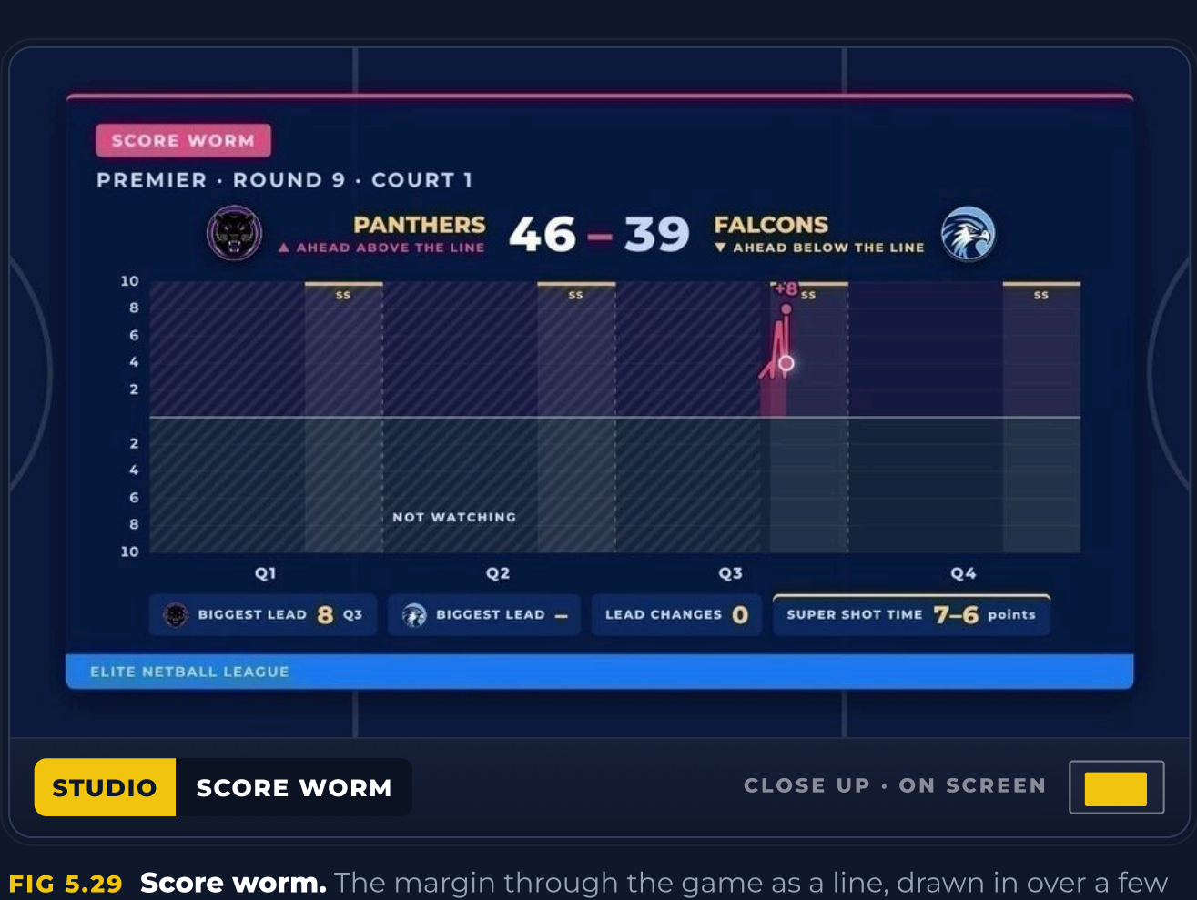


FIG 5.29 Score worm. The margin through the game as a line, drawn in over a few seconds. Options: drawing time, club colours or the competition's.

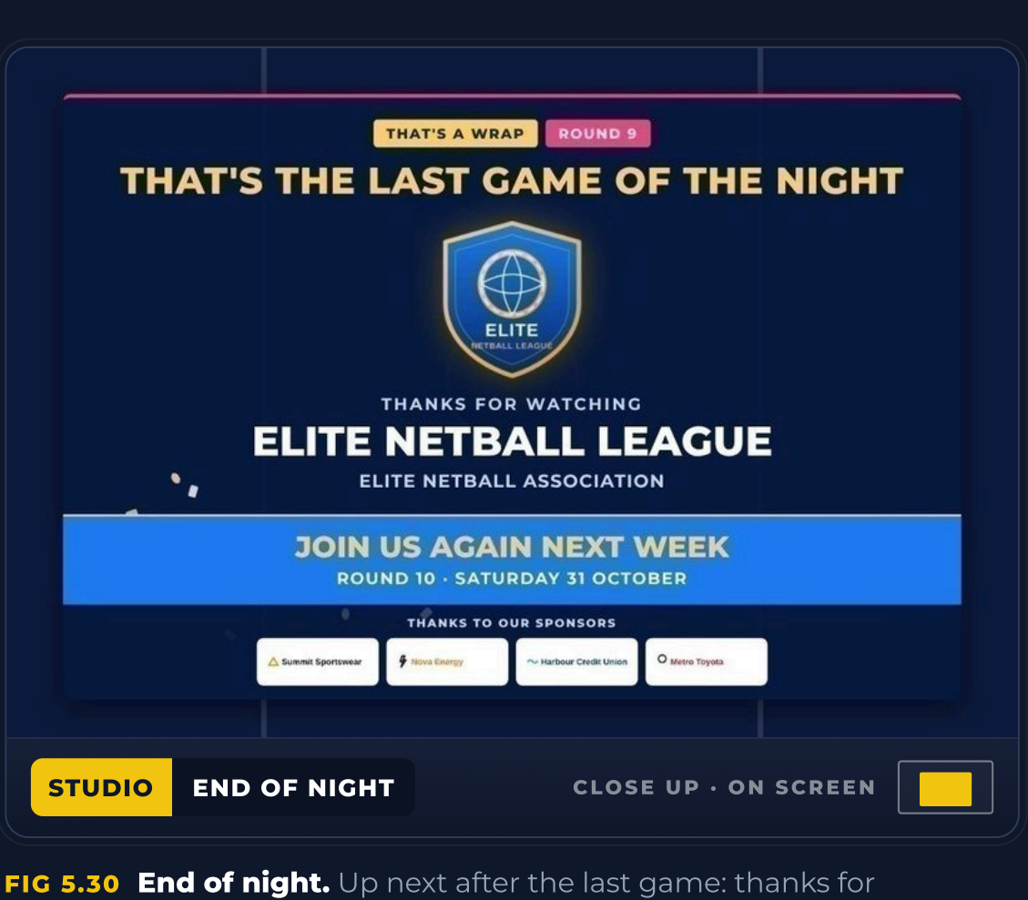
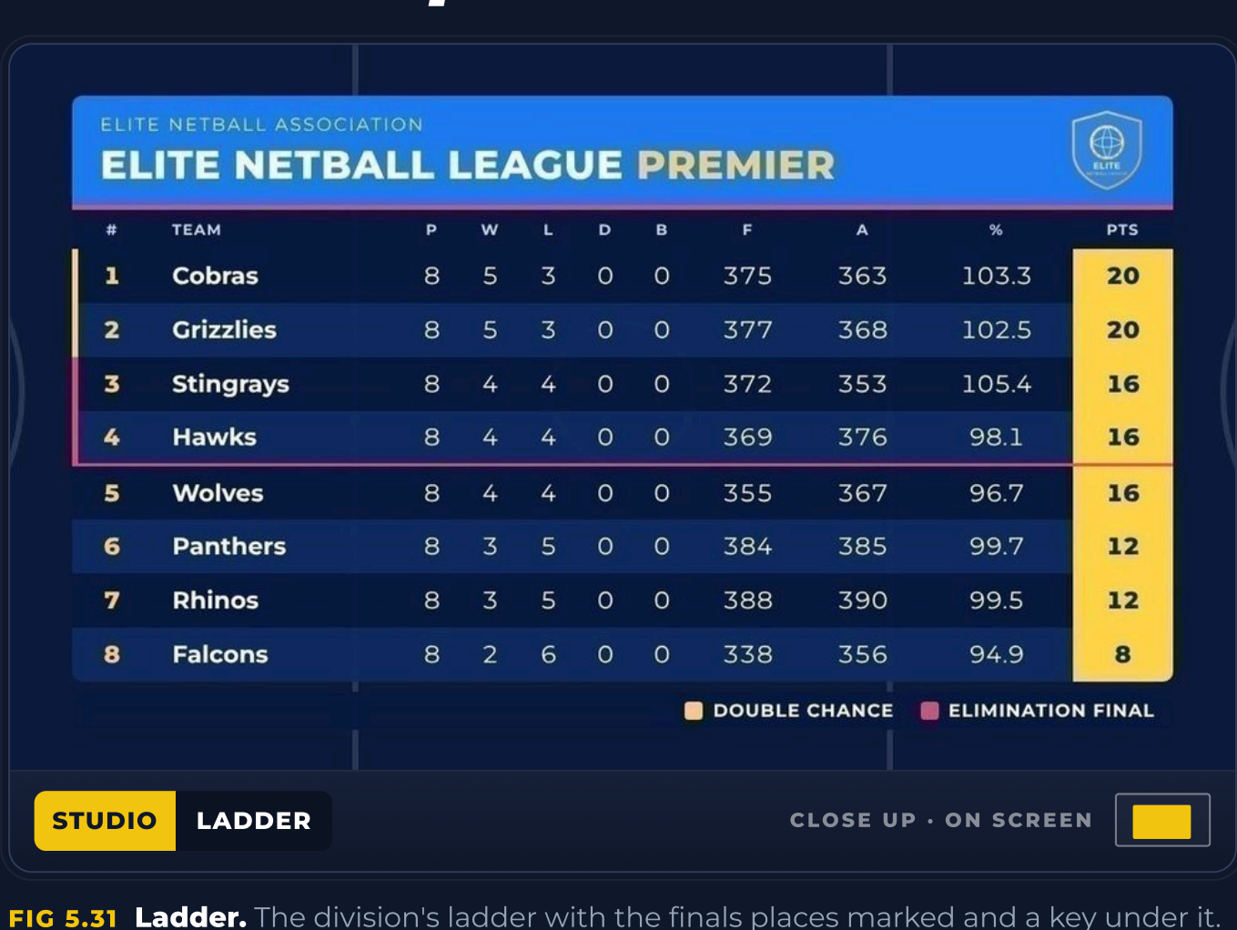


FIG 5.30 End of night. Up next after the last game: thanks for watching, next week's date or the finals, and the Supporter sponsors in a strip. Options: the invite line, sponsors on or off.

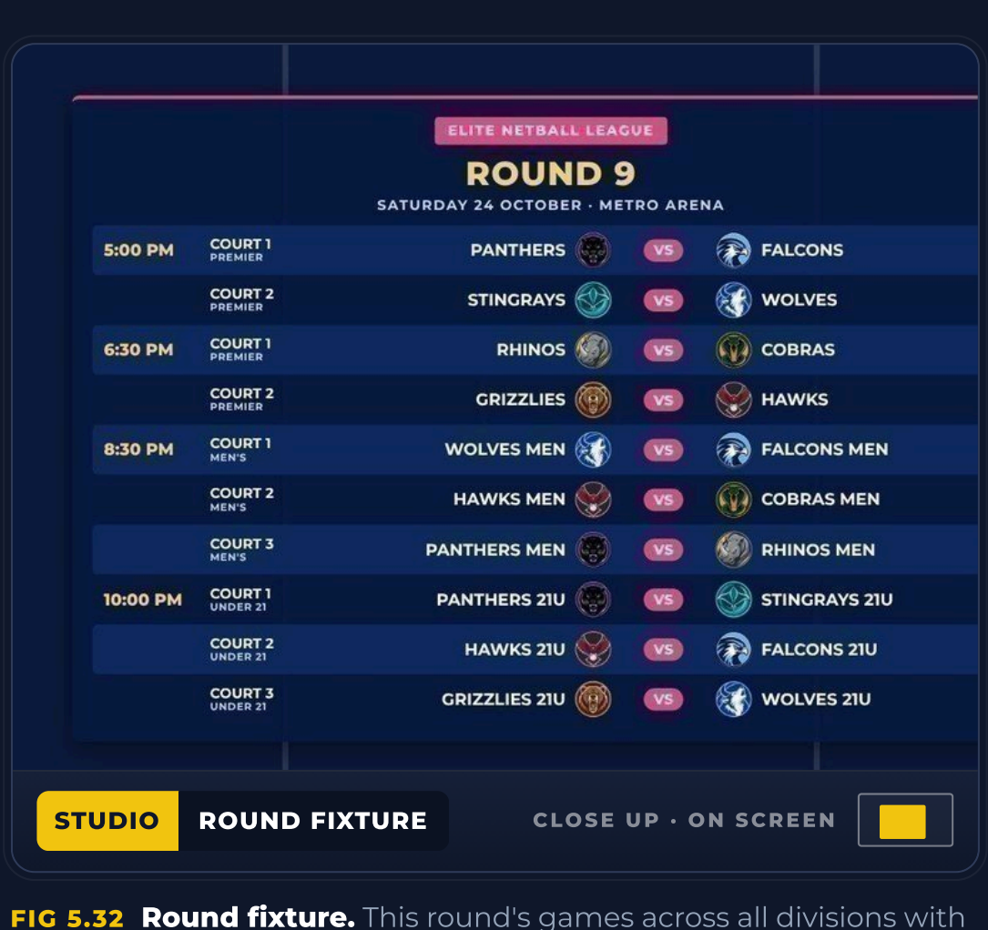
> **Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits.**
>
> CHAPTER 5 · THE GRAPHICS

5.5 The competition



#	TEAM	P	W	L	D	B	F	A	%	PTS
1	Cobras	8	5	3	0	0	375	363	103.3	20
2	Grizzlies	8	5	3	0	0	377	368	102.5	20
3	Stingrays	8	4	4	0	0	372	353	105.4	16
4	Hawks	8	4	4	0	0	369	376	98.1	16
5	Wolves	8	4	4	0	0	355	367	96.7	16
6	Panthers	8	3	5	0	0	384	385	99.7	12
7	Rhinos	8	3	5	0	0	388	390	99.5	12
8	Falcons	8	2	6	0	0	338	356	94.9	8

FIG 5.31 Ladder. The division's ladder with the finals places marked and a key under it. Options: division, rows, logo on or off, light or dark mode logo.



5:00 PM	COURT 1 PREMIER	PANTHERS	VS	FALCONS
	COURT 2 PREMIER	STINGRAYS	VS	WOLVES
6:30 PM	COURT 1 PREMIER	RHINOS	VS	COBRAS
	COURT 2 PREMIER	GRIZZLIES	VS	HAWKS
8:30 PM	COURT 1 MEN'S	WOLVES MEN	VS	FALCONS MEN
	COURT 2 MEN'S	HAWKS MEN	VS	COBRAS MEN
	COURT 3 MEN'S	PANTHERS MEN	VS	RHINOS MEN
10:00 PM	COURT 1 UNDER 21	PANTHERS 21U	VS	STINGRAYS 21U
	COURT 2 UNDER 21	HAWKS 21U	VS	FALCONS 21U
	COURT 3 UNDER 21	GRIZZLIES 21U	VS	WOLVES 21U

FIG 5.32 Round fixture. This round's games across all divisions with day, time and court. **Round** takes next, prev, +2, a number or a name like Final. When divisions are on different rounds, each game shows its own. A finals round PlayHQ has no games for yet shows that week's finals from the finals format, with the time and court TBC.

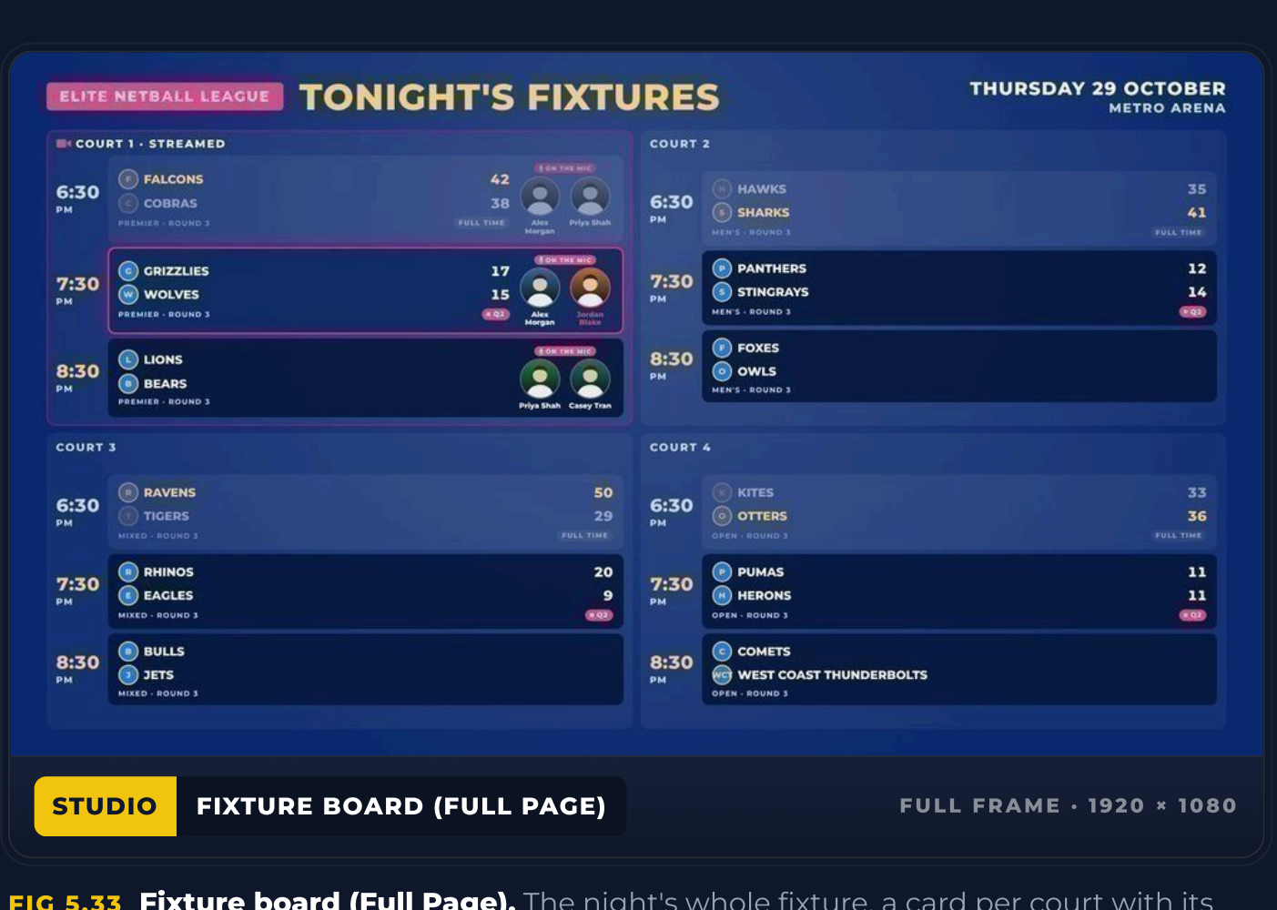


FIG 5.33 Fixture board (Full Page). The night's whole fixture, a card per court with its games by time, for a screen in the competition room or backstage, or a picture to share before the night (**Tuesday Night's Fixtures**). The streamed court has a camera and its games show the commentators from the Production Planner, **On the mic** with their headshots (a silhouette for anyone without one) where there is room, as name pills on a crowded night; games being played show their live score, and finished ones grey out with the winner in yellow. Not for the stream.

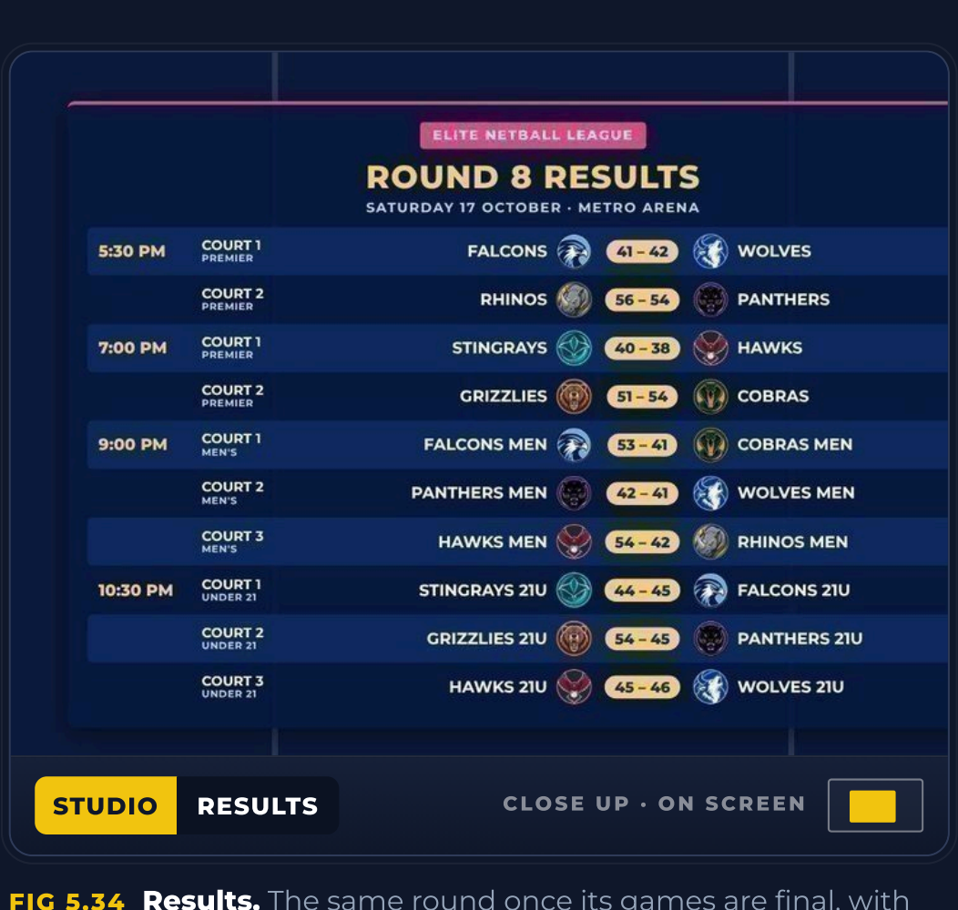
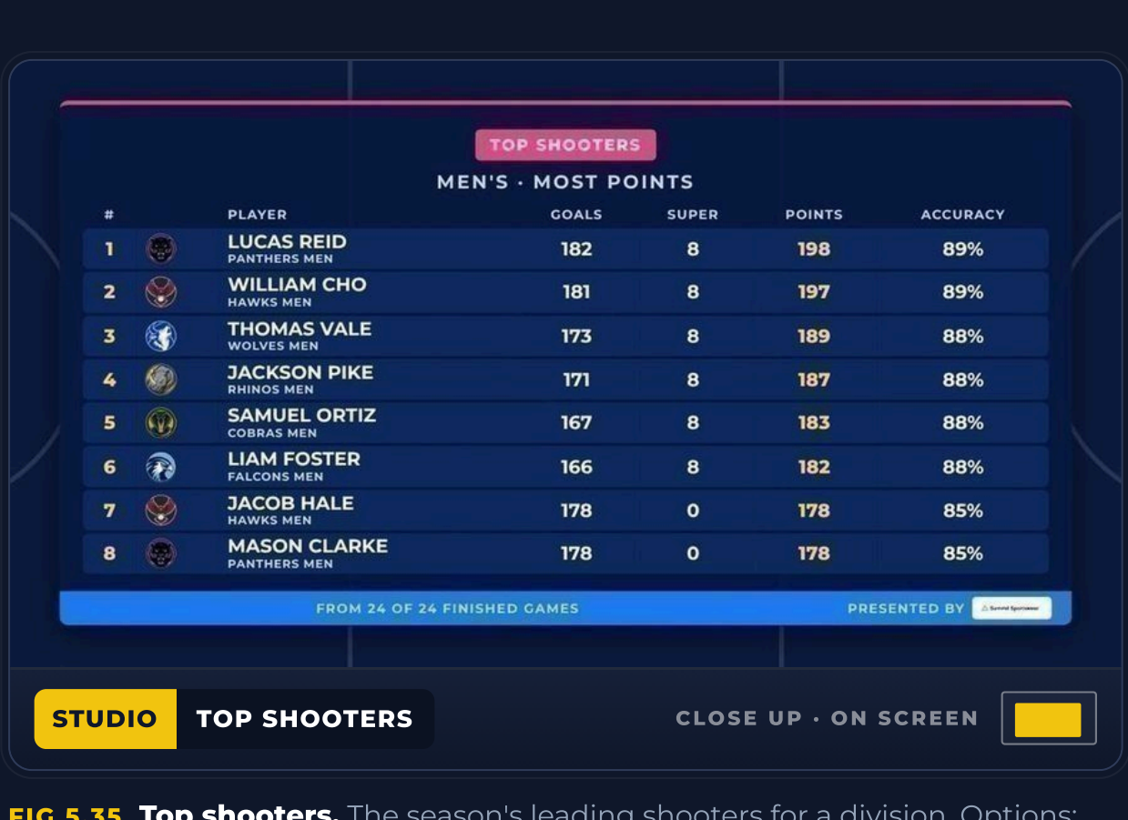


FIG 5.34 Results. The same round once its games are final, with scores. The same round option applies.



#	PLAYER	GOALS	SUPER	POINTS	ACCURACY
1	LUCAS REID PANTHERS MEN	182	8	198	89%
2	WILLIAM CHO HAWKS MEN	181	8	197	89%
3	THOMAS VALE WOLVES MEN	173	8	189	88%
4	JACKSON PIKE RHINOS MEN	171	8	187	88%
5	SAMUEL ORTIZ COBRAS MEN	167	8	183	88%
6	LIAM FOSTER FALCONS MEN	166	8	182	88%
7	JACOB HALE HAWKS MEN	178	0	178	85%
8	MASON CLARKE PANTHERS MEN	178	0	178	85%

FIG 5.35 Top shooters. The season's leading shooters for a division. Options: order by goals or average, how many shooters, a sponsor footer.

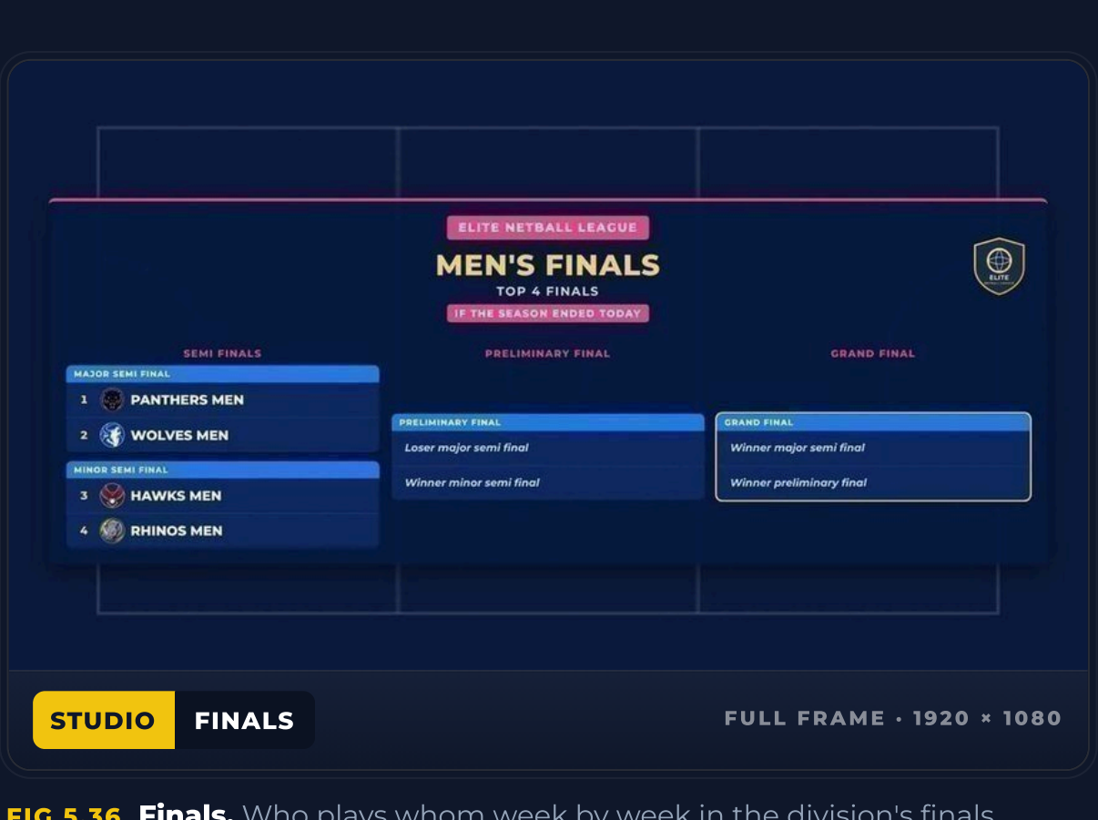


FIG 5.36 Finals. Who plays whom week by week in the division's finals format, with ladder positions and scores; "If the season ended today" until the home-and-away rounds finish. Options: one week only, the heading and the line under it.

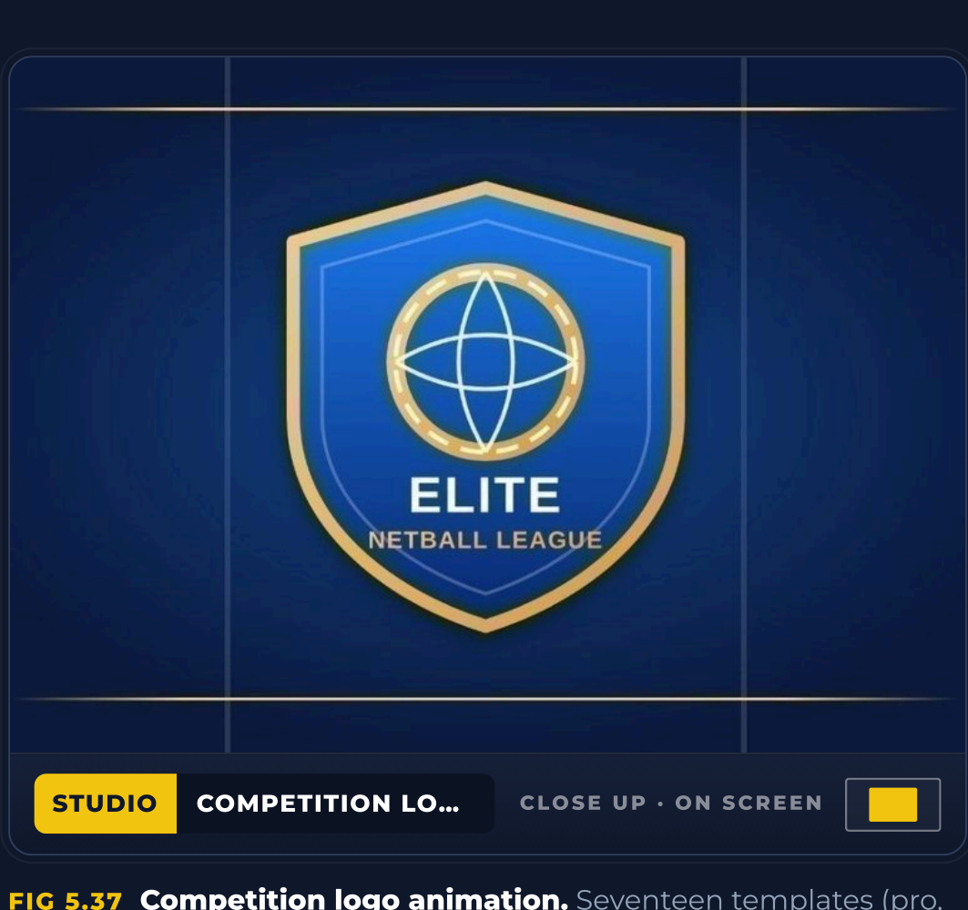


FIG 5.37 Competition logo animation. Seventeen templates (pro, coin, tiles, glitch, badge, lightning, court, bibs, net and more), about 5 seconds, with optional words and a smaller line.

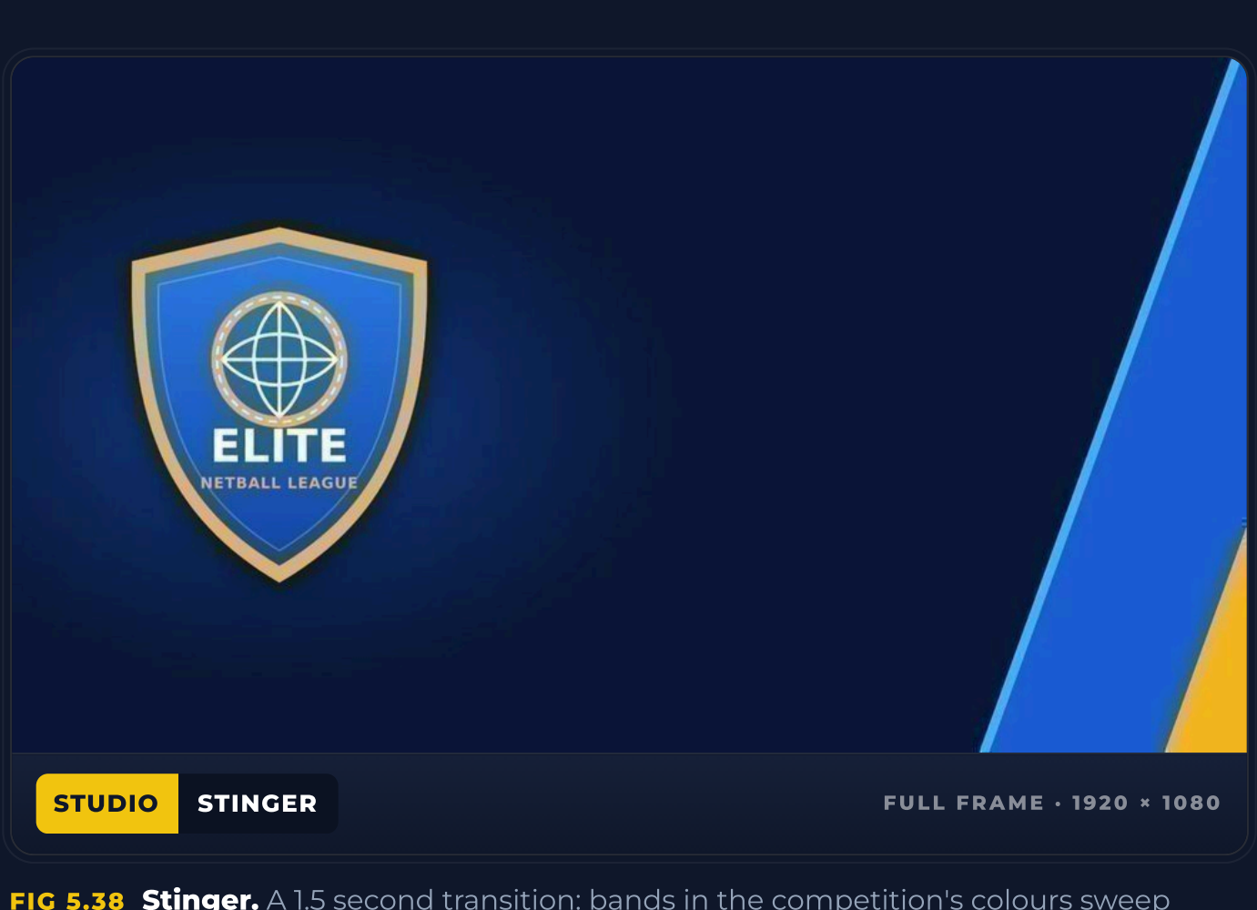


FIG 5.38 Stinger. A 1.5 second transition: bands in the competition's colours sweep across and cover the whole picture with the competition logo for half a second, so a cut can happen underneath, then carry on off the other side. Options: hold, length, logo on or off, light or dark mode logo, whoosh volume, and the music ducked for the next scene or while it plays.

5.6 The breaks

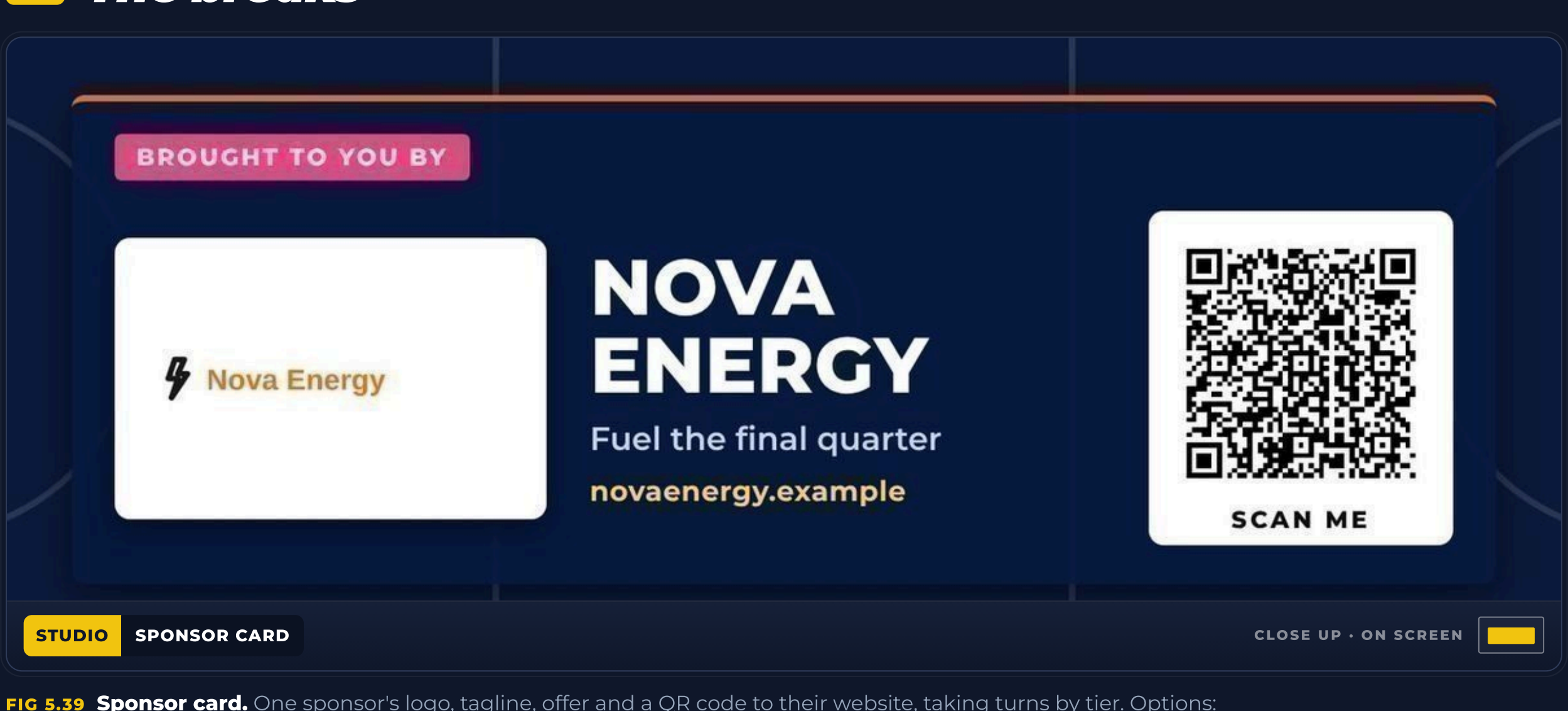


FIG 5.39 Sponsor card. One sponsor's logo, tagline, offer and a QR code to their website, taking turns by tier. Options: one sponsor always, which tiers rotate, animate away.

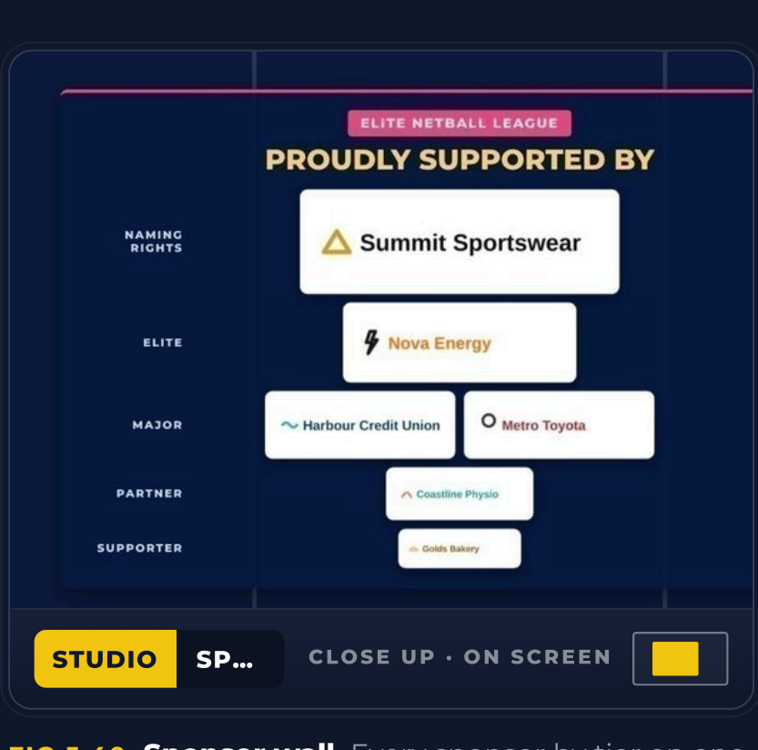


FIG 5.40 Sponsor wall. Every sponsor by tier on one card, for the run-up and the close. Options: which tiers, names under the logos.

FIG 5.41 PIVT and FirstPass Studio ad. A 10-second house ad for a break with nothing sold. A Sponsor ads step plays it when no sponsor's ad is due, if **House ad** is ticked.

5.7 Checking the outputs

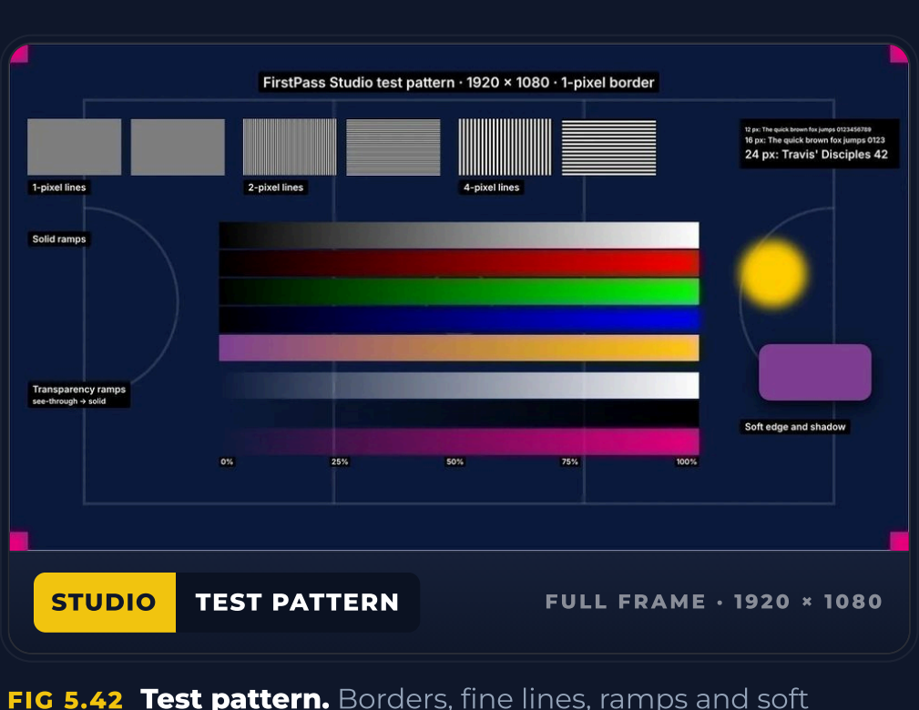


FIG 5.42 Test pattern. Borders, fine lines, ramps and soft shapes to check an output, the switcher's scaling and its key end to end before the stream. Never for air.

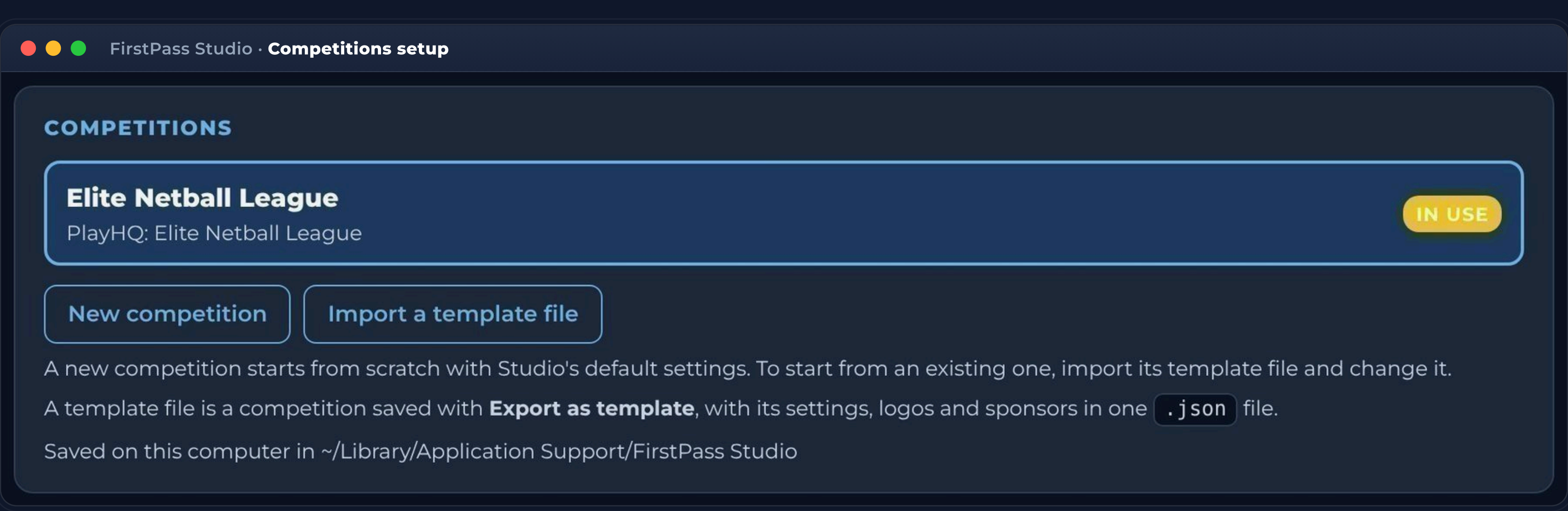
CHAPTER 6 OF 11

Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

1. PlayHQ
2. Names on the graphics
3. Game rules
4. Finals
5. Game night
6. Colours and font
7. Logos
8. Live viewers
9. Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.



COMPETITIONS

Elite Netball League
PlayHQ: Elite Netball League

A new competition starts from scratch with Studio's default settings. To start from an existing one, import its template file and change it.

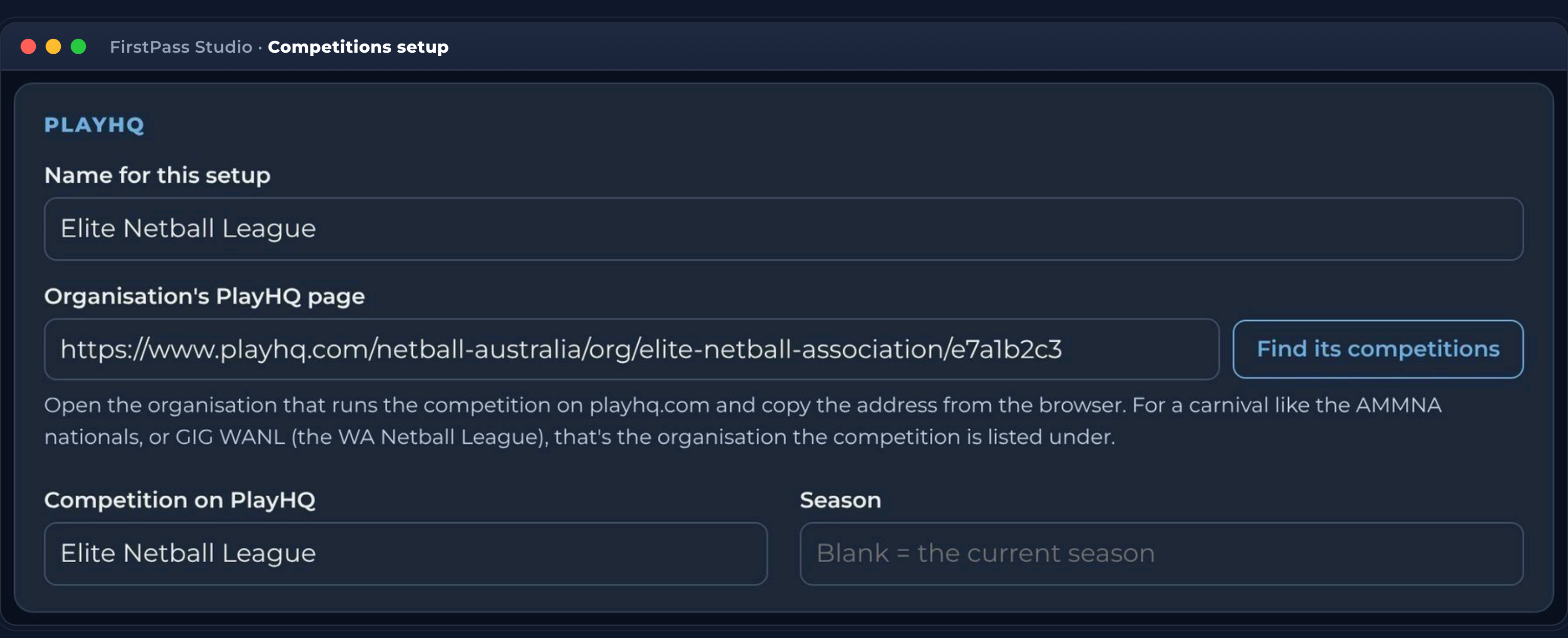
A template file is a competition saved with **Export as template**, with its settings, logos and sponsors in one `.json` file.

Saved on this computer in ~/Library/Application Support/FirstPass Studio

FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMMNA Nationals, GIG WANL and West Coast Thunder) or press New competition for Studio's defaults.

6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.



PLAYHQ

Name for this setup: Elite Netball League

Organisation's PlayHQ page: https://www.playhq.com/netball-australia/org/elite-netball-association/e7a1b2c3

Open the organisation that runs the competition on playhq.com and copy the address from the browser. For a carnival like the AMMNA nationals, or GIG WANL (the WA Netball League), that's the organisation the competition is listed under.

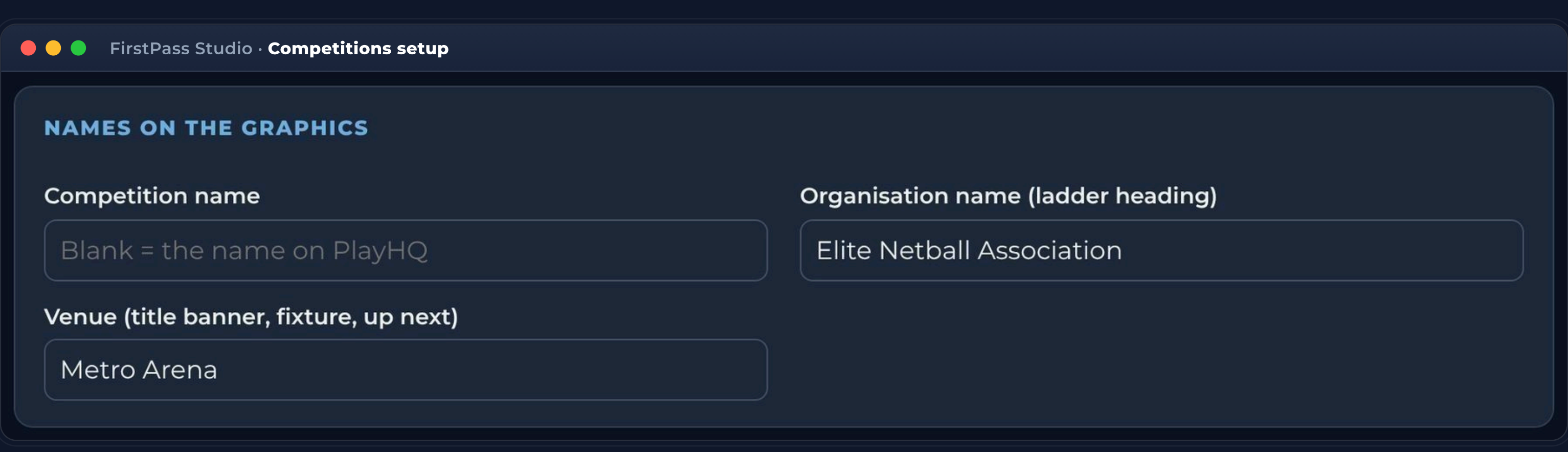
Competition on PlayHQ: Elite Netball League

Season: Blank = the current season

FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.



NAMES ON THE GRAPHICS

Competition name: Blank = the name on PlayHQ

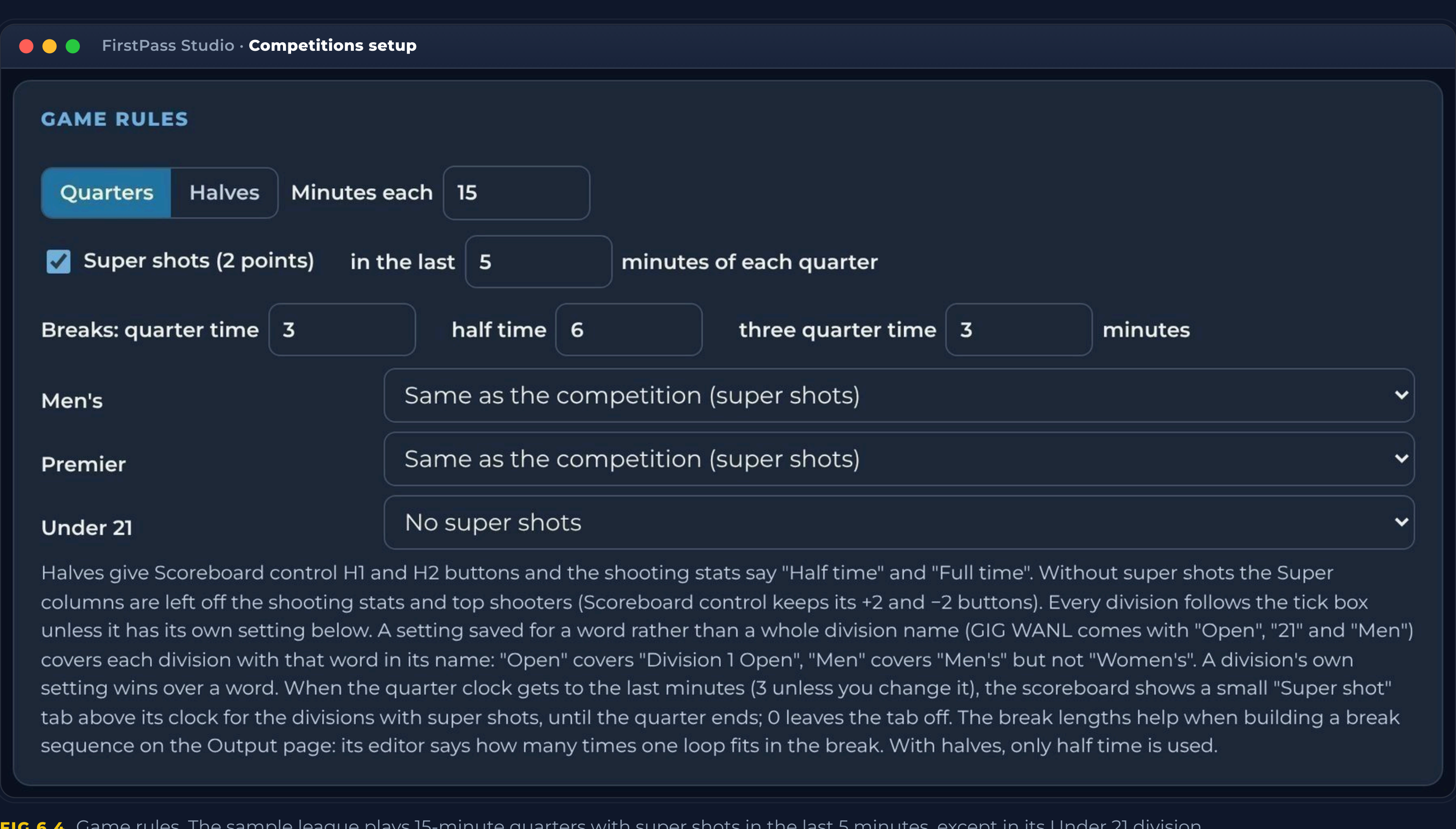
Organisation name (ladder heading): Elite Netball Association

Venue (title banner, fixture, up next): Metro Arena

FIG 6.3 Names on the graphics.

6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.



GAME RULES

● Quarters ○ Halves Minutes each: 15

☑ Super shots (2 points) in the last 5 minutes of each quarter

Breaks: quarter time 3 half time 6 three quarter time 3 minutes

Men's: Same as the competition (super shots)

Premier: Same as the competition (super shots)

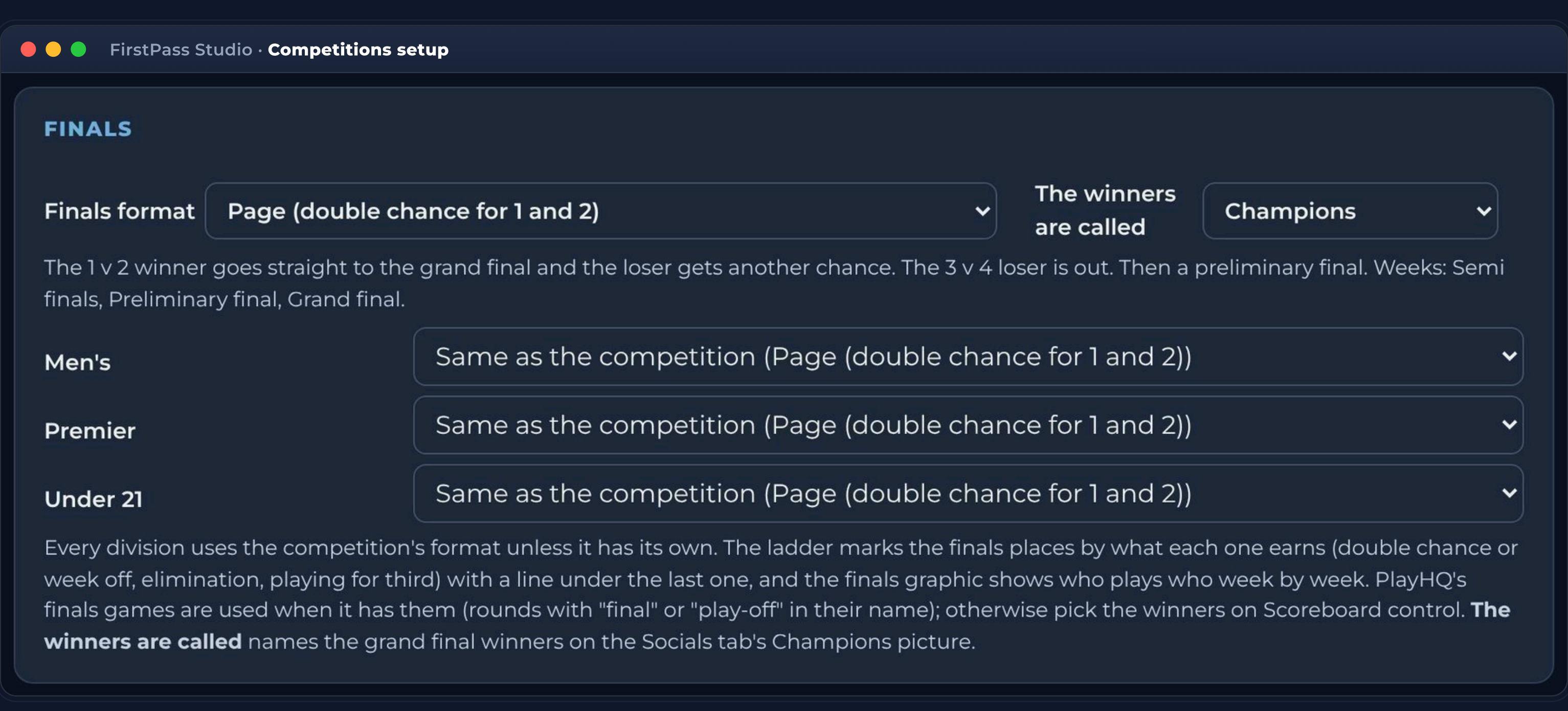
Under 21: No super shots

Halves give Scoreboard control H1 and H2 buttons and the shooting stats say "Half time" and "Full time". Without super shots the Super columns are left off the shooting stats and top shooters (Scoreboard control keeps its +2 and −2 buttons). Every division follows the tick box unless it has its own setting below. A setting saved for a word rather than a whole division name (GIG WANL comes with "Open", "21" and "Men") covers each division with that word in its name: "Open" covers "Division 1 Open", "Men" covers "Men's" but not "Women's". A division's own setting wins over a word. When the quarter clock gets to the last minutes (3 unless you change it), the scoreboard shows a small "Super shot" tab above its clock for the divisions with super shots, until the quarter ends; 0 leaves the tab off. The break lengths help when building a break sequence on the Output page: its editor says how many times one loop fits in the break. With halves, only half time is used.

FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control. **The winners are called** picks Champions or Premiers for the Socials tab's picture of the grand final winners.



FINALS

Finals format: Page (double chance for 1 and 2)

The winners are called: Champions

The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. Weeks: Semi finals, Preliminary final, Grand final.

Men's: Same as the competition (Page (double chance for 1 and 2))

Premier: Same as the competition (Page (double chance for 1 and 2))

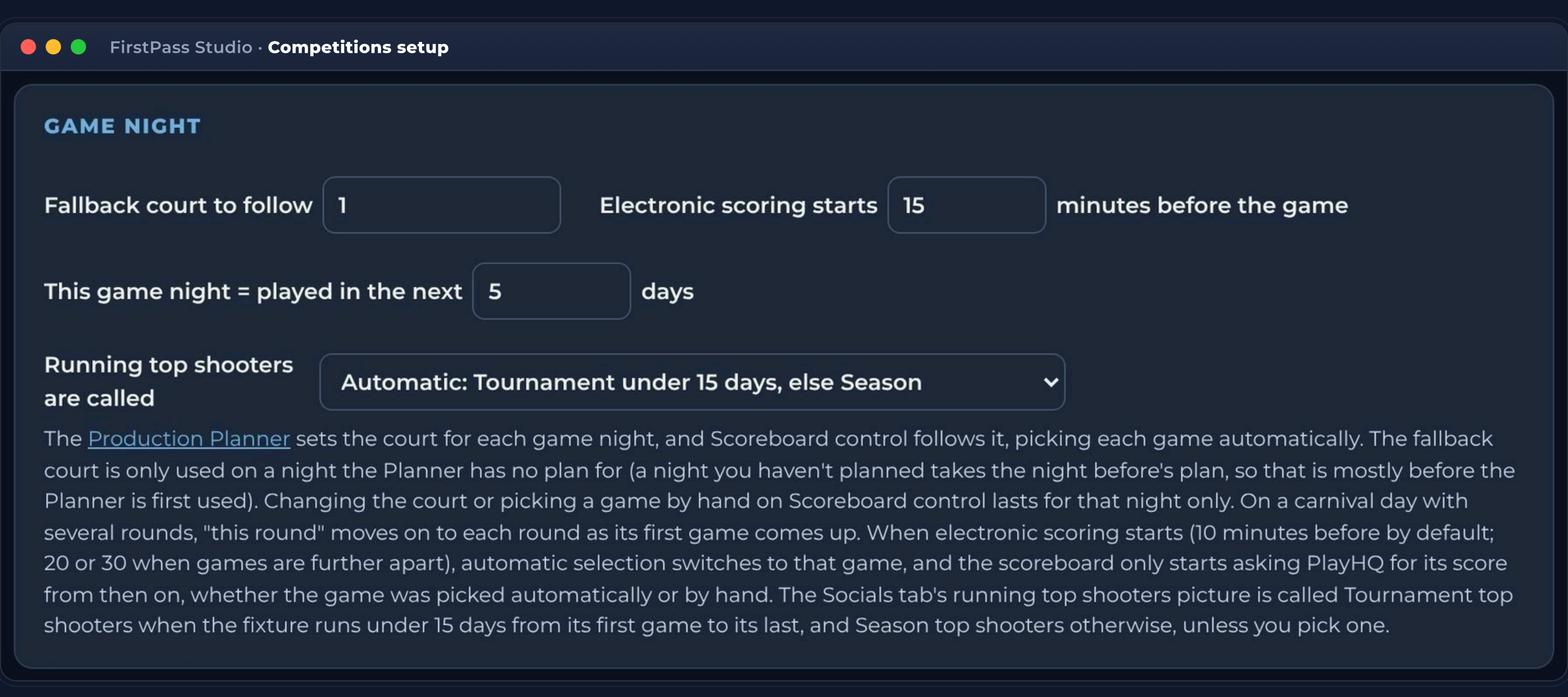
Under 21: Same as the competition (Page (double chance for 1 and 2))

Every division uses the competition's format unless it has its own. The ladder marks the finals places by what each one earns (double chance or week off, elimination, playing for third) with a line under the last one, and the finals graphic shows who plays who week by week. PlayHQ's finals games are used when it has them (rounds with "final" or "play-off" in their name); otherwise pick the winners on Scoreboard control. **The winners are called** names the grand final winners on the Socials tab's Champions picture.

FIG 6.5 Finals formats, per division where they differ.

6.5 Game night

The **Fallback court to follow**, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. The Production Planner sets each game night's court; the fallback is only used on a night with no plan, which is mostly before the Planner is first used, since a night you haven't planned takes the night before's plan. Every game night starts with the game picked automatically. Scoreboard control can change the court, pick a game by hand or pick another game night, for that night only. On a carnival day with several rounds, this round moves on as each round's first game comes up. **Running top shooters are called** sets the Socials picture's name: Automatic calls it Tournament top shooters when the fixture runs under 15 days and Season top shooters otherwise.



GAME NIGHT

Fallback court to follow: 1 Electronic scoring starts 15 minutes before the game

This game night = played in the next 5 days

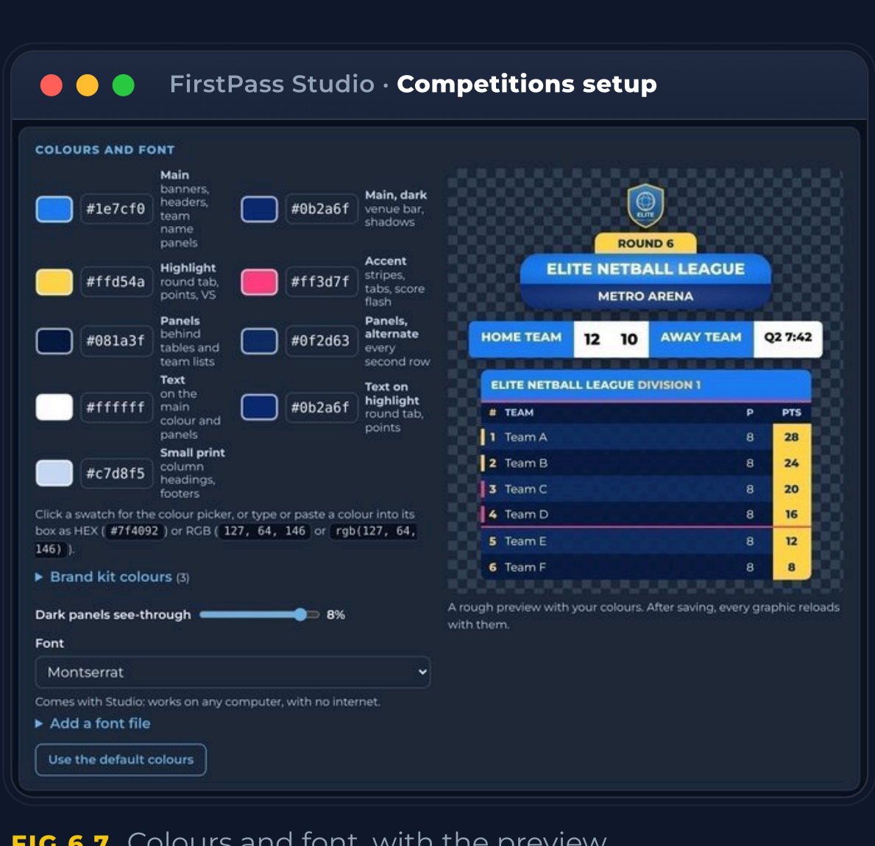
Running top shooters are called: Automatic: Tournament under 15 days, else Season

The Production Planner sets the court for each game night, and Scoreboard control follows it, picking each game automatically. The fallback court is only used on a night the Planner has no plan for (a night you haven't planned takes the night before's plan, so that is mostly before the Planner is first used). Changing the court or picking a game by hand on Scoreboard control lasts for that night only. On a carnival day with several rounds, "this round" moves on to each round as its first game comes up. When electronic scoring starts (10 minutes before by default; 20 or 30 when games are further apart), automatic selection switches to that game, and the scoreboard only starts asking PlayHQ for its score from then on, whether the game was picked automatically or by hand. The Socials tab's running top shooters picture is called Tournament top shooters when the fixture runs under 15 days from its first game to its last, and Season top shooters otherwise, unless you pick one.

FIG 6.6 Game night.

6.6 Colours and font

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.



COLOURS AND FONT

#1e7cf0 Main banners, headers, team name panels

#0b2a6f Main, dark venue bar, shadows

#ffd54a Highlight round tab, points, VS

#ff3d7f Accent stripes, tabs, score flash

#081a3f Panels behind tables and team lists

#0f2d63 Panels, alternate every second row

#ffffff Text on the main colour and panels

#0b2a6f Text on highlight round tab, points

#c7d8f5 Small print column headings, footers

Click a swatch for the colour picker, or type or paste a colour into its box as HEX (`#7f4092`) or RGB (`127, 64, 146` or `rgb(127, 64, 146)`).

▸ Brand kit colours (3)

Dark panels see-through 8%

Font: Montserrat

Comes with Studio: works on any computer, with no internet.

▸ Add a font file

ELITE NETBALL LEAGUE DIVISION 1

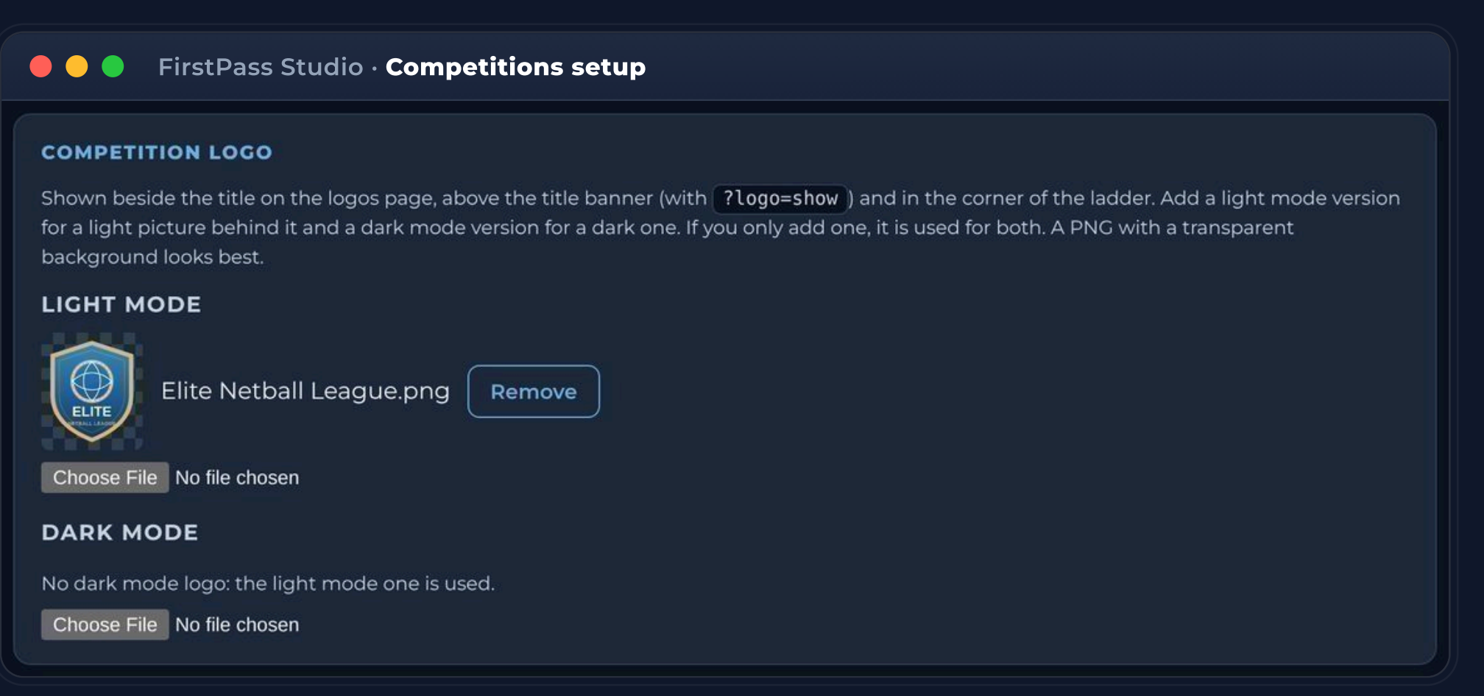
#	TEAM	P	PTS
1	Team A	8	28
2	Team B	8	24
3	Team C	8	20
4	Team D	8	16
5	Team E	8	12
6	Team F	8	8

A rough preview with your colours. After saving, every graphic reloads with them.

FIG 6.7 Colours and font, with the preview.

6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 21U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.



COMPETITION LOGO

Shown beside the title on the logos page, above the title banner (with `?logo=show`) and in the corner of the ladder. Add a light mode version for a light picture behind it and a dark mode version for a dark one. If you only add one, it is used for both. A PNG with a transparent background looks best.

LIGHT MODE

Elite Netball League.png

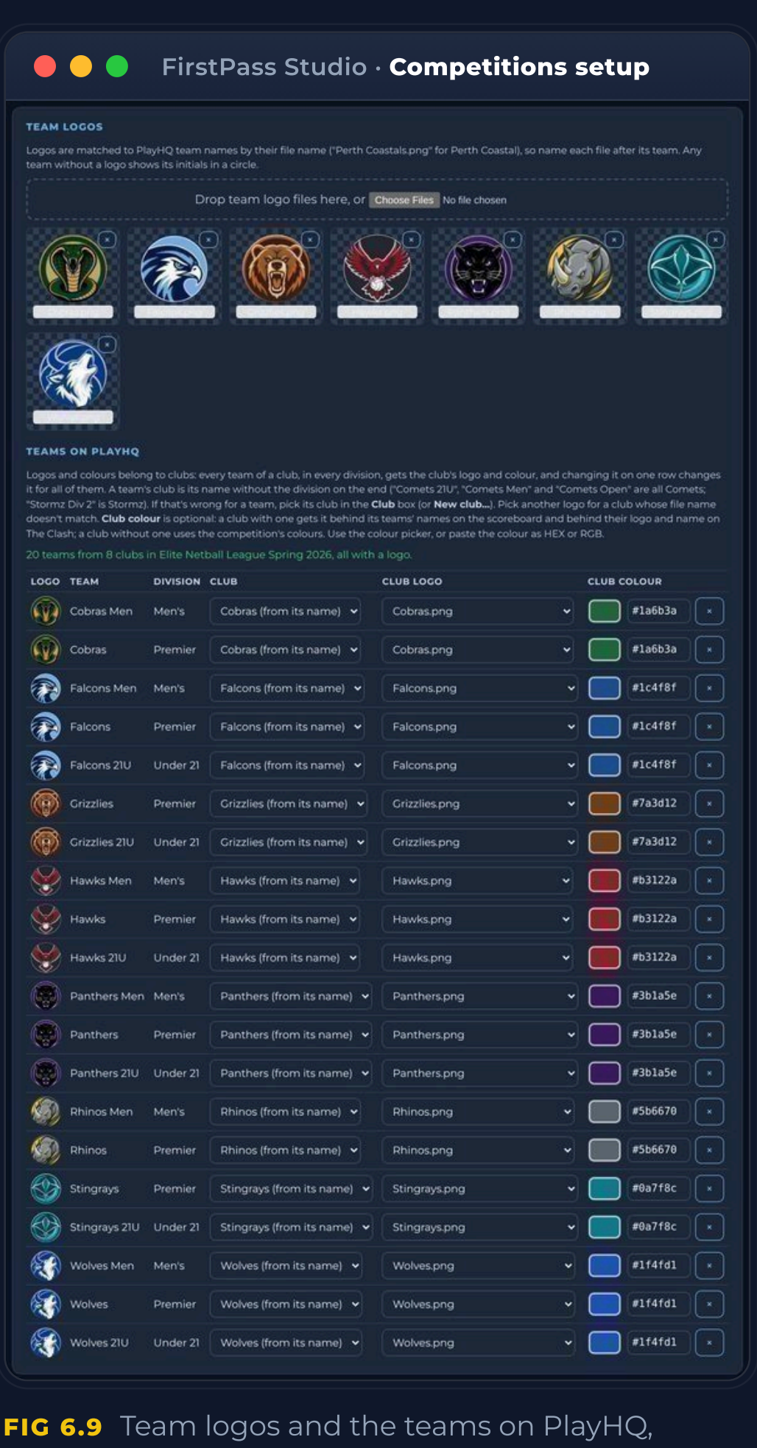
No file chosen

DARK MODE

No dark mode logo: the light mode one is used.

No file chosen

FIG 6.8 Competition logo, light and dark mode.



TEAM LOGOS

Logos are matched to PlayHQ team names by their file name ("Perth Coastals.png" for Perth Coastal), so name each file after its team. Any team without a logo shows its initials in a circle.

Drop team logo files here, or No file chosen

TEAMS ON PLAYHQ

Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, and changing it on one row changes it for all of them. A team's club is its name without the division on the end ("Comets 21U", "Comets Men" and "Comets Open" are all Comets; "Stormz Div 2" is Stormz). If that's wrong for a team, pick its club in the **Club** box (or **New club…**). Pick another logo for a club whose file name doesn't match. **Club colour** is optional: a club with one gets it behind its teams' names on the scoreboard and behind their logo and name on The Clash; a club without one uses the competition's colours. Use the colour picker, or paste the colour as HEX or RGB.

20 teams from 8 clubs in Elite Netball League Spring 2026, all with a logo.

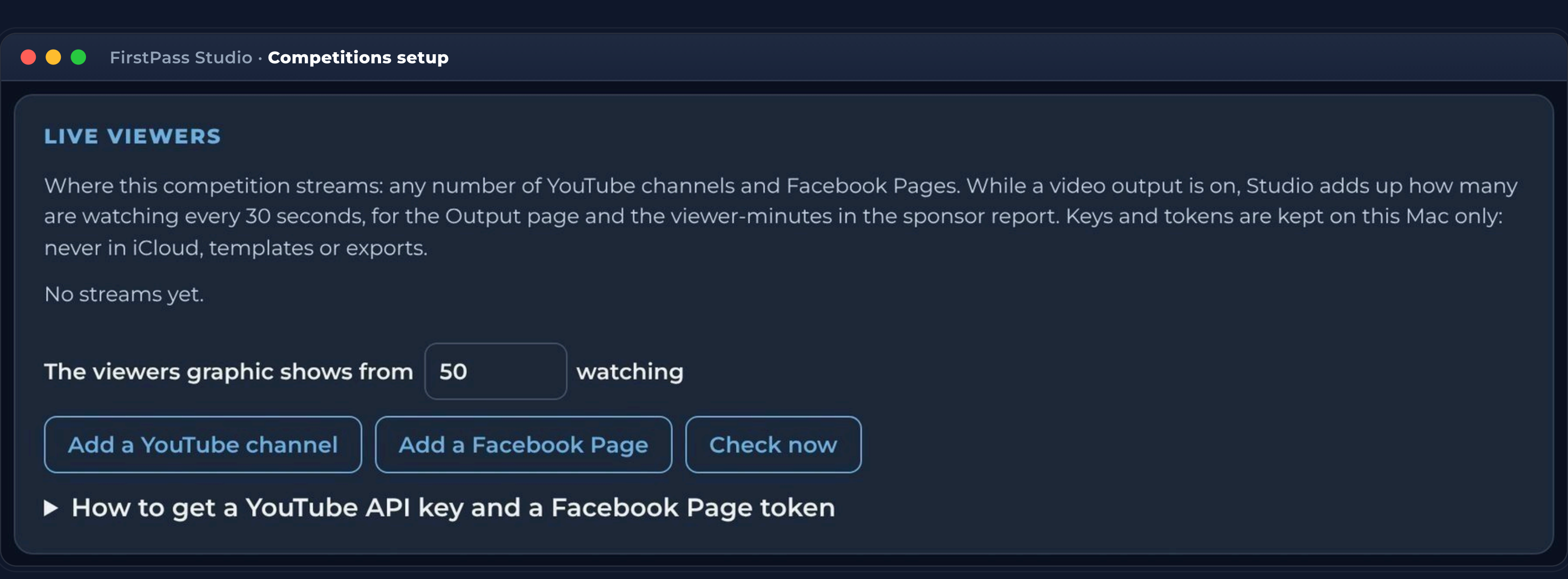
LOGO	TEAM	DIVISION	CLUB	CLUB LOGO	CLUB COLOUR
	Cobras Men	Men's	Cobras (from its name)	Cobras.png	#1a6b3a
	Cobras	Premier	Cobras (from its name)	Cobras.png	#1a6b3a
	Falcons Men	Men's	Falcons (from its name)	Falcons.png	#1c4f8f
	Falcons	Premier	Falcons (from its name)	Falcons.png	#1c4f8f
	Falcons 21U	Under 21	Falcons (from its name)	Falcons.png	#1c4f8f
	Grizzlies	Premier	Grizzlies (from its name)	Grizzlies.png	#7a3d12
	Grizzlies 21U	Under 21	Grizzlies (from its name)	Grizzlies.png	#7a3d12
	Hawks Men	Men's	Hawks (from its name)	Hawks.png	#b3122a
	Hawks	Premier	Hawks (from its name)	Hawks.png	#b3122a
	Hawks 21U	Under 21	Hawks (from its name)	Hawks.png	#b3122a
	Panthers Men	Men's	Panthers (from its name)	Panthers.png	#3b1a5e
	Panthers	Premier	Panthers (from its name)	Panthers.png	#3b1a5e
	Panthers 21U	Under 21	Panthers (from its name)	Panthers.png	#3b1a5e
	Rhinos Men	Men's	Rhinos (from its name)	Rhinos.png	#5b6670
	Rhinos	Premier	Rhinos (from its name)	Rhinos.png	#5b6670
	Stingrays	Premier	Stingrays (from its name)	Stingrays.png	#0a7f8c
	Stingrays 21U	Under 21	Stingrays (from its name)	Stingrays.png	#0a7f8c
	Wolves Men	Men's	Wolves (from its name)	Wolves.png	#1f4fd1
	Wolves	Premier	Wolves (from its name)	Wolves.png	#1f4fd1
	Wolves 21U	Under 21	Wolves (from its name)	Wolves.png	#1f4fd1

FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
>
> CHAPTER 6 · COMPETITIONS SETUP

6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.



LIVE VIEWERS

Where this competition streams: any number of YouTube channels and Facebook Pages. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page and the viewer-minutes in the sponsor report. Keys and tokens are kept on this Mac only: never in iCloud, templates or exports.

No streams yet.

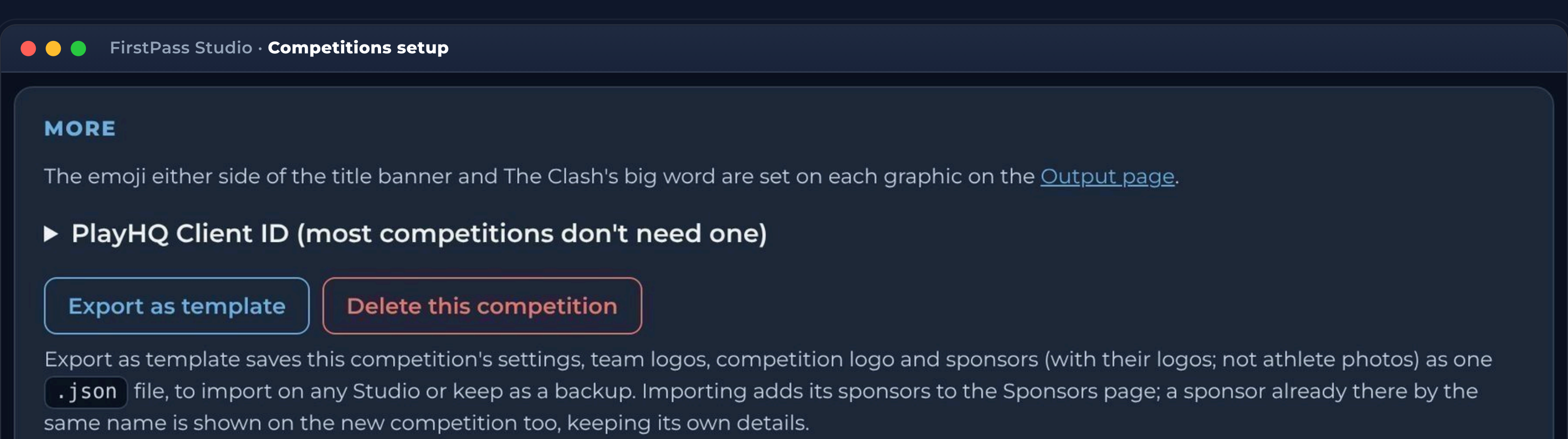
The viewers graphic shows from 50 watching

▸ How to get a YouTube API key and a Facebook Page token

FIG 6.10 Live viewers.

6.9 Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.



MORE

The emoji either side of the title banner and The Clash's big word are set on each graphic on the Output page.

▸ PlayHQ Client ID (most competitions don't need one)

Export as template saves this competition's settings, team logos, competition logo and sponsors (with their logos; not athlete photos) as one `.json` file, to import on any Studio or keep as a backup. Importing adds its sponsors to the Sponsors page; a sponsor already there by the same name is shown on the new competition too, keeping its own details.

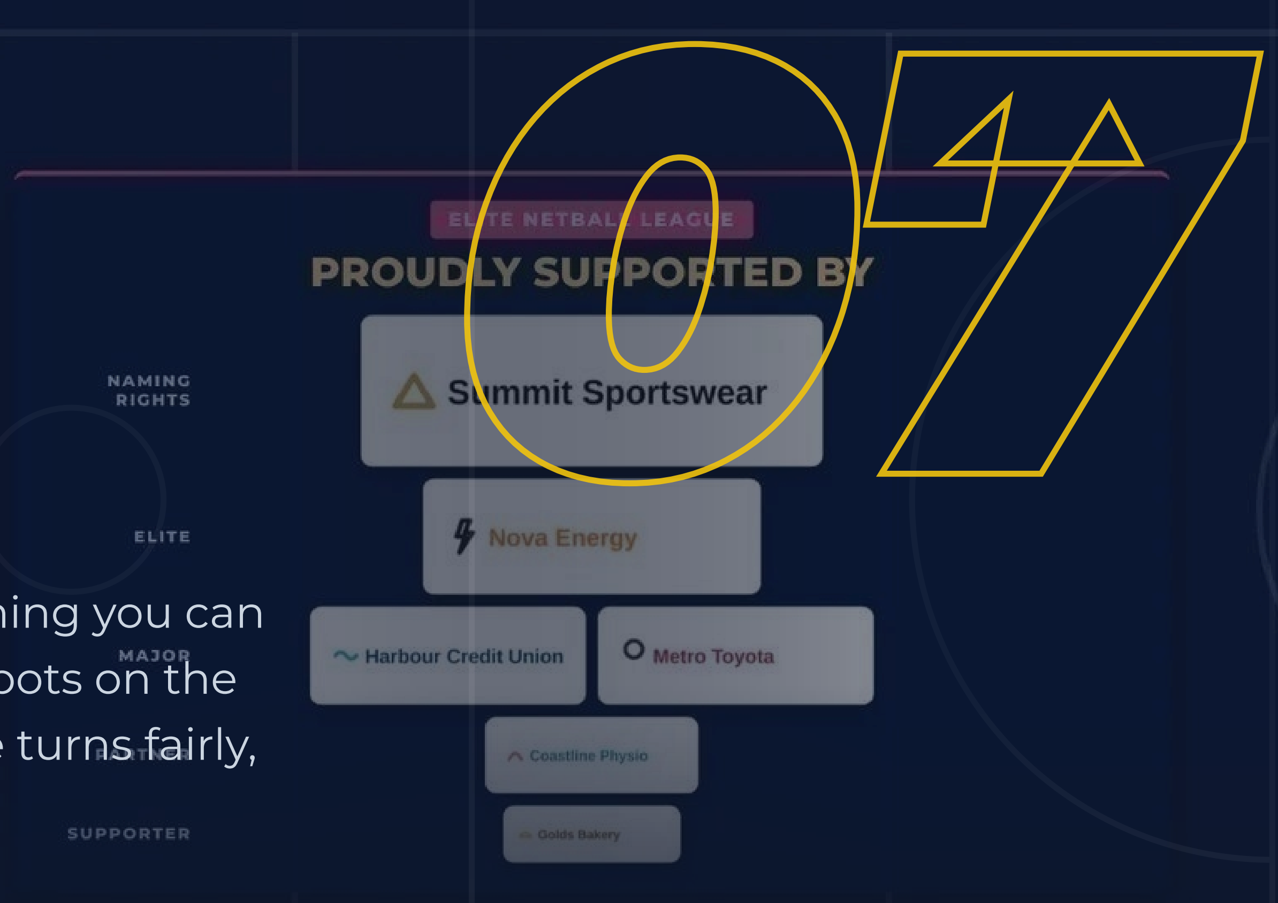
FIG 6.11 The More section: Client ID, export and delete.

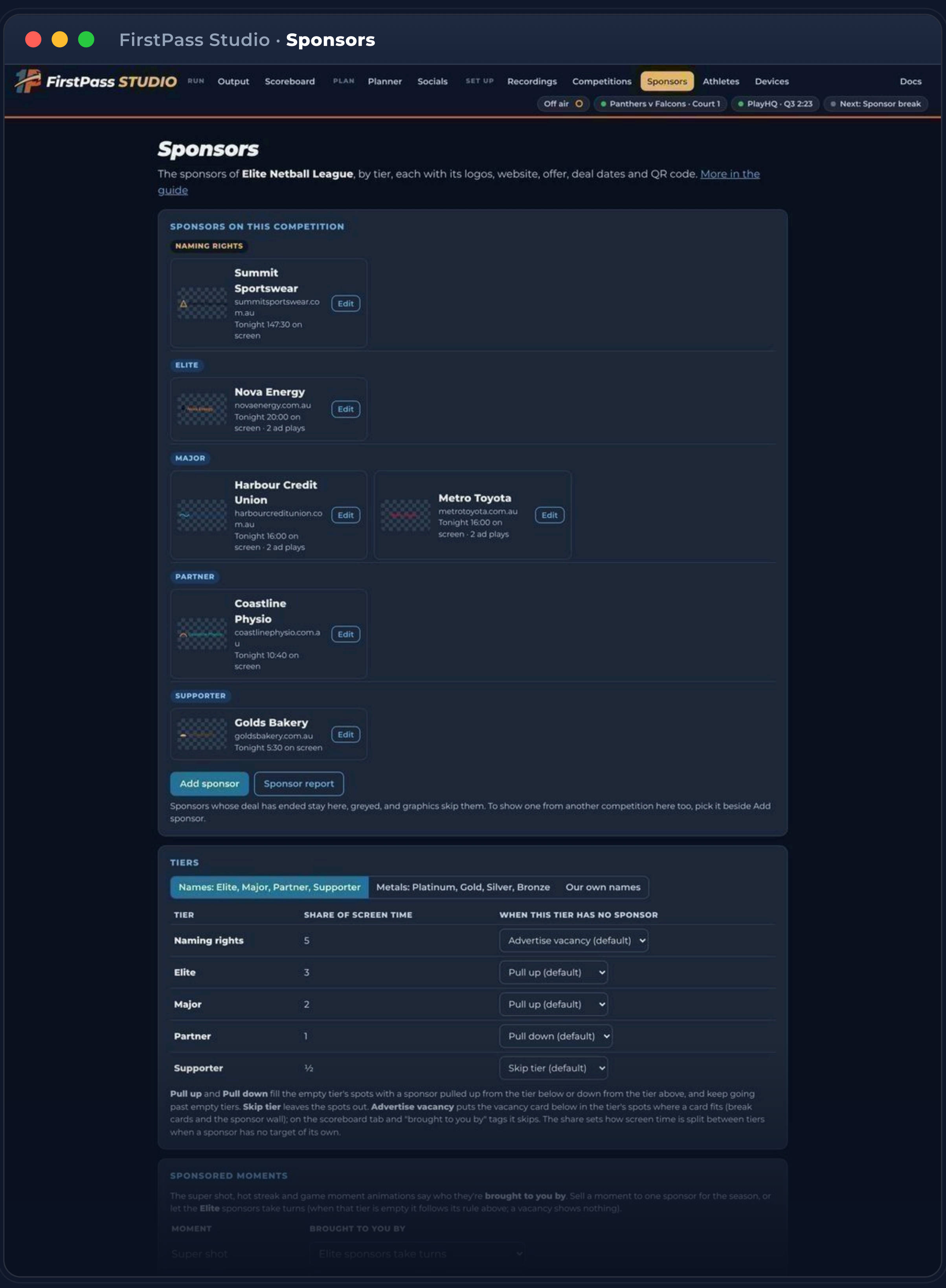
CHAPTER 7 OF 11

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1 Tiers · 2 A sponsor · 3 Spots and moments · 4 Proof of play





Sponsors

The sponsors of **Elite Netball League**, by tier, each with its logos, website, offer, deal dates and QR code. More in the guide

SPONSORS ON THIS COMPETITION

NAMING RIGHTS

Summit Sportswear summitsportswear.com.au Tonight 147:30 on screen · Edit

ELITE

Nova Energy novaenergy.com.au Tonight 20:00 on screen · 2 ad plays · Edit

MAJOR

Harbour Credit Union harbourcreditunion.com.au Tonight 16:00 on screen · 2 ad plays · Edit

Metro Toyota metrotoyota.com.au Tonight 16:00 on screen · 2 ad plays · Edit

PARTNER

Coastline Physio coastlinephysio.com.au Tonight 10:40 on screen · Edit

SUPPORTER

Golds Bakery goldsbakery.com.au Tonight 5:30 on screen · Edit

Add sponsor · Sponsor report

Sponsors whose deal has ended stay here, greyed, and graphics skip them. To show one from another competition here too, pick it beside Add sponsor.

TIERS

Names: Elite, Major, Partner, Supporter · Metals: Platinum, Gold, Silver, Bronze · Our own names

TIER	SHARE OF SCREEN TIME	WHEN THIS TIER HAS NO SPONSOR
Naming rights	5	Advertise vacancy (default)
Elite	3	Pull up (default)
Major	2	Pull up (default)
Partner	1	Pull down (default)
Supporter	½	Skip tier (default)

Pull up and **Pull down** fill the empty tier's spots with a sponsor pulled up from the tier below or down from the tier above, and keep going past empty tiers. **Skip tier** leaves the spots out. **Advertise vacancy** puts the vacancy card below in the tier's spots where a card fits (break cards and the sponsor wall); on the scoreboard tab and "brought to you by" tags it skips. The share sets how screen time is split between tiers when a sponsor has no target of its own.

SPONSORED MOMENTS

The super shot, hot streak and game moment animations say who they're **brought to you by**. Sell a moment to one sponsor for the season, or let the **Elite** sponsors take turns (when that tier is empty it follows its rule above; a vacancy shows nothing).

MOMENT	BROUGHT TO YOU BY
Super shot	Elite sponsors take turns

FIG 7.1 The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

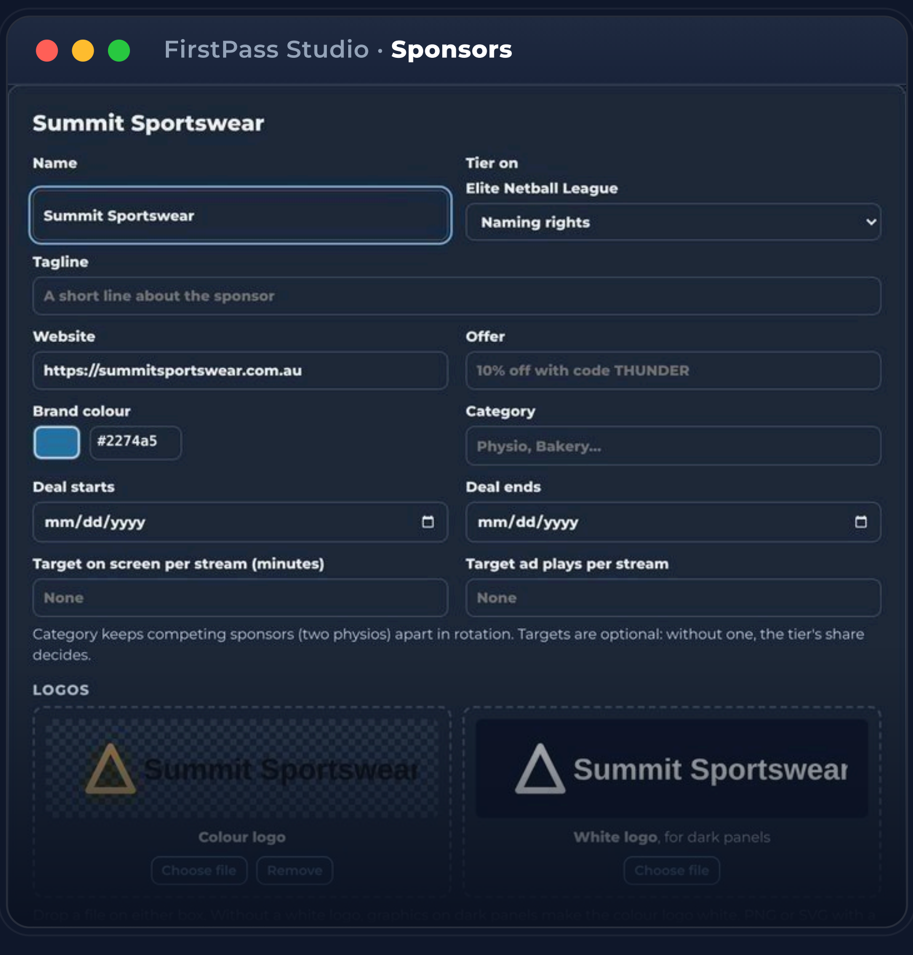
When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> **The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.**
>
> CHAPTER 7 · SPONSORS

7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.



Summit Sportswear

Name: Summit Sportswear

Tier on Elite Netball League: Naming rights

Tagline: A short line about the sponsor

Website: https://summitsportswear.com.au

Offer: 10% off with code THUNDER

Brand colour: #2274a5

Category: Physio, Bakery...

Deal starts: mm/dd/yyyy

Deal ends: mm/dd/yyyy

Target on screen per stream (minutes): None

Target ad plays per stream: None

Category keeps competing sponsors (two physios) apart in rotation. Targets are optional: without one, the tier's share decides.

LOGOS

Colour logo

White logo, for dark panels

FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

7.3 Spots and moments

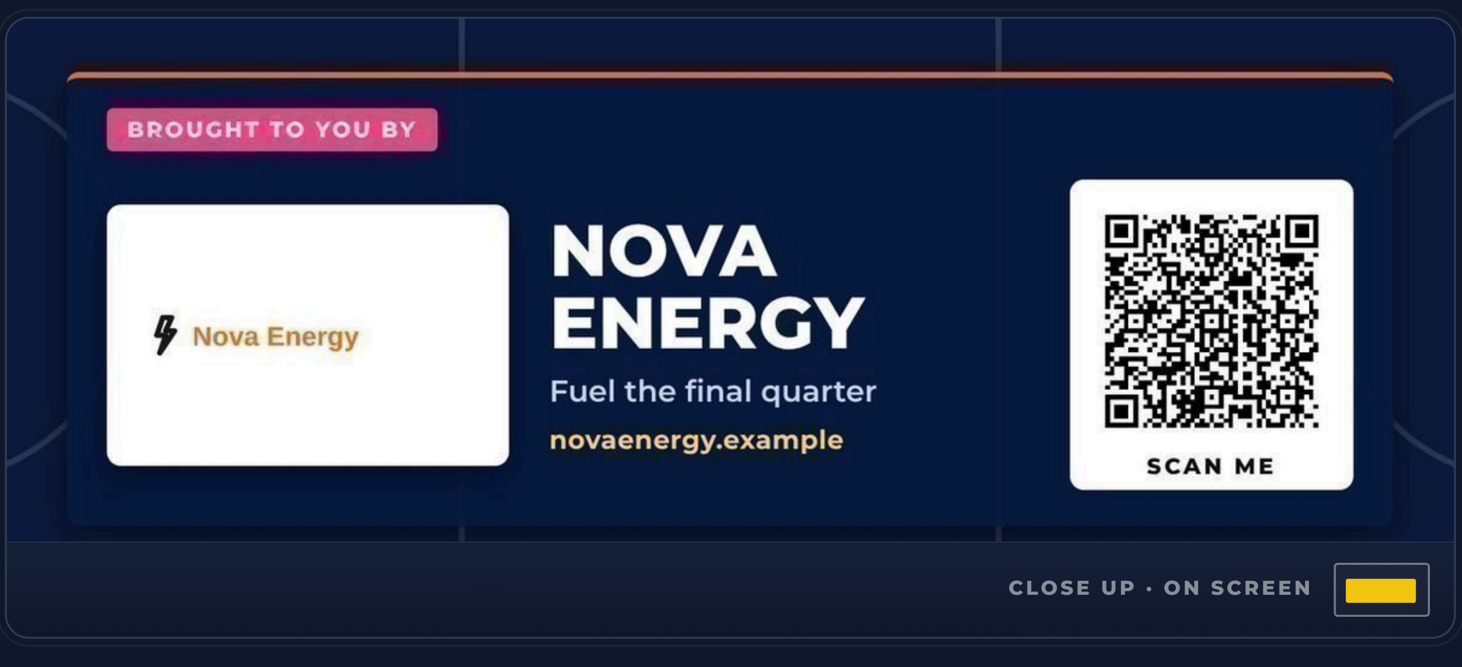


FIG 7.3 The sponsor card in a break, with its QR code.

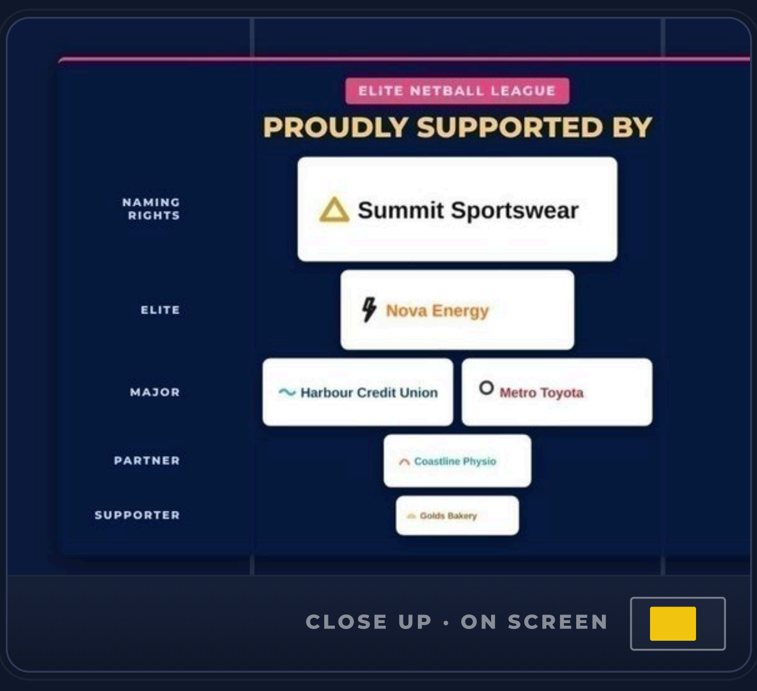
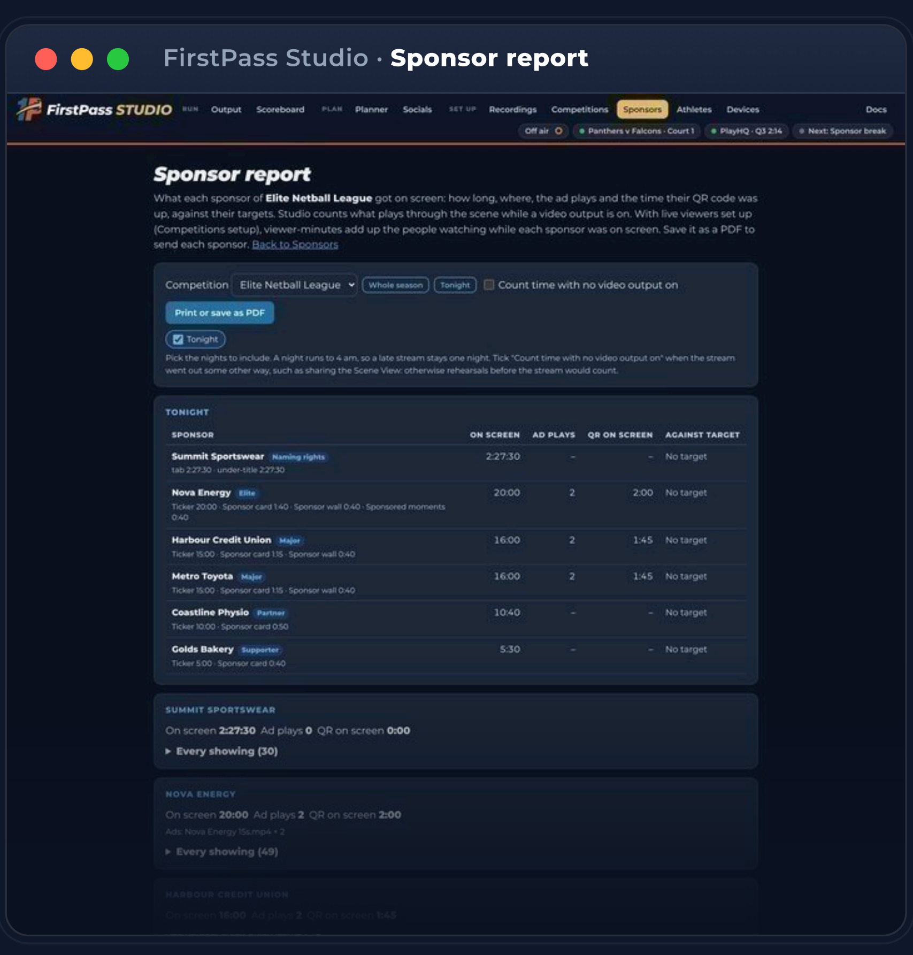


FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

7.4 Proof of play



Sponsor report

What each sponsor of **Elite Netball League** got on screen: how long, where, the ad plays and the time their QR code was up, against their targets. Studio counts what plays through the scene while a video output is on. With live viewers set up (Competitions setup), viewer-minutes add up the people watching while each sponsor was on screen. Save it as a PDF to send each sponsor. Back to Sponsors

Competition: Elite Netball League · ☐ Count time with no video output on

☑ Tonight

Pick the nights to include. A night runs to 4 am, so a late stream stays one night. Tick "Count time with no video output on" when the stream went out some other way, such as sharing the Scene View: otherwise rehearsals before the stream would count.

TONIGHT

SPONSOR	ON SCREEN	AD PLAYS	QR ON SCREEN	AGAINST TARGET
Summit Sportswear Naming rights tab 2:27:30 · under-title 2:27:30	2:27:30	–	–	No target
Nova Energy Elite Ticker 20:00 · Sponsor card 1:40 · Sponsor wall 0:40 · Sponsored moments 0:40	20:00	2	2:00	No target
Harbour Credit Union Major Ticker 15:00 · Sponsor card 1:15 · Sponsor wall 0:40	16:00	2	1:45	No target
Metro Toyota Major Ticker 15:00 · Sponsor card 1:15 · Sponsor wall 0:40	16:00	2	1:45	No target
Coastline Physio Partner Ticker 10:00 · Sponsor card 0:50	10:40	–	–	No target
Golds Bakery Supporter Ticker 5:00 · Sponsor card 0:40	5:30	–	–	No target

SUMMIT SPORTSWEAR

On screen **2:27:30** Ad plays **0** QR on screen **0:00**

▸ Every showing (30)

NOVA ENERGY

On screen **20:00** Ad plays **2** QR on screen **2:00**

Ads: Nova Energy 15s.mp4 × 2

▸ Every showing (49)

HARBOUR CREDIT UNION

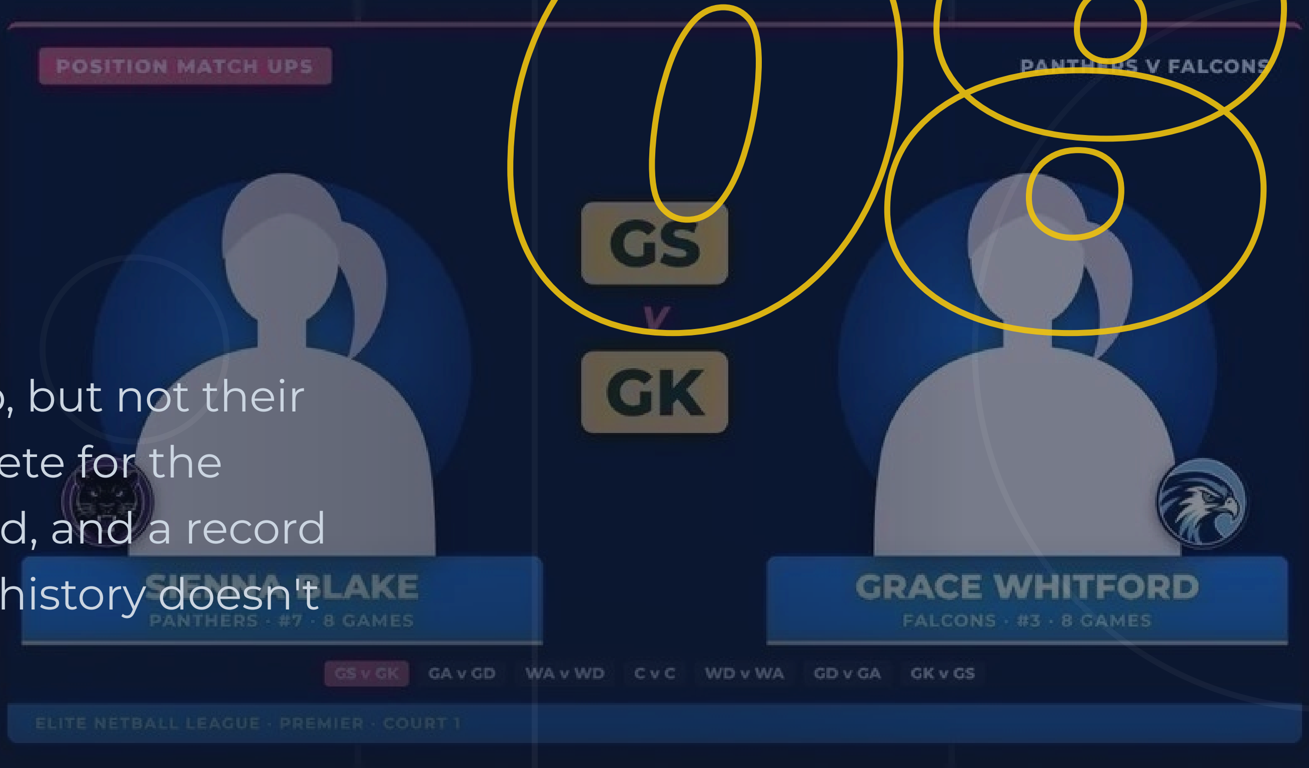
On screen **16:00** Ad plays **2** QR on screen **1:45**

FIG 7.5 The sponsor report for the sample night.

While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count if one is set up. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.

CHAPTER 8 OF 11

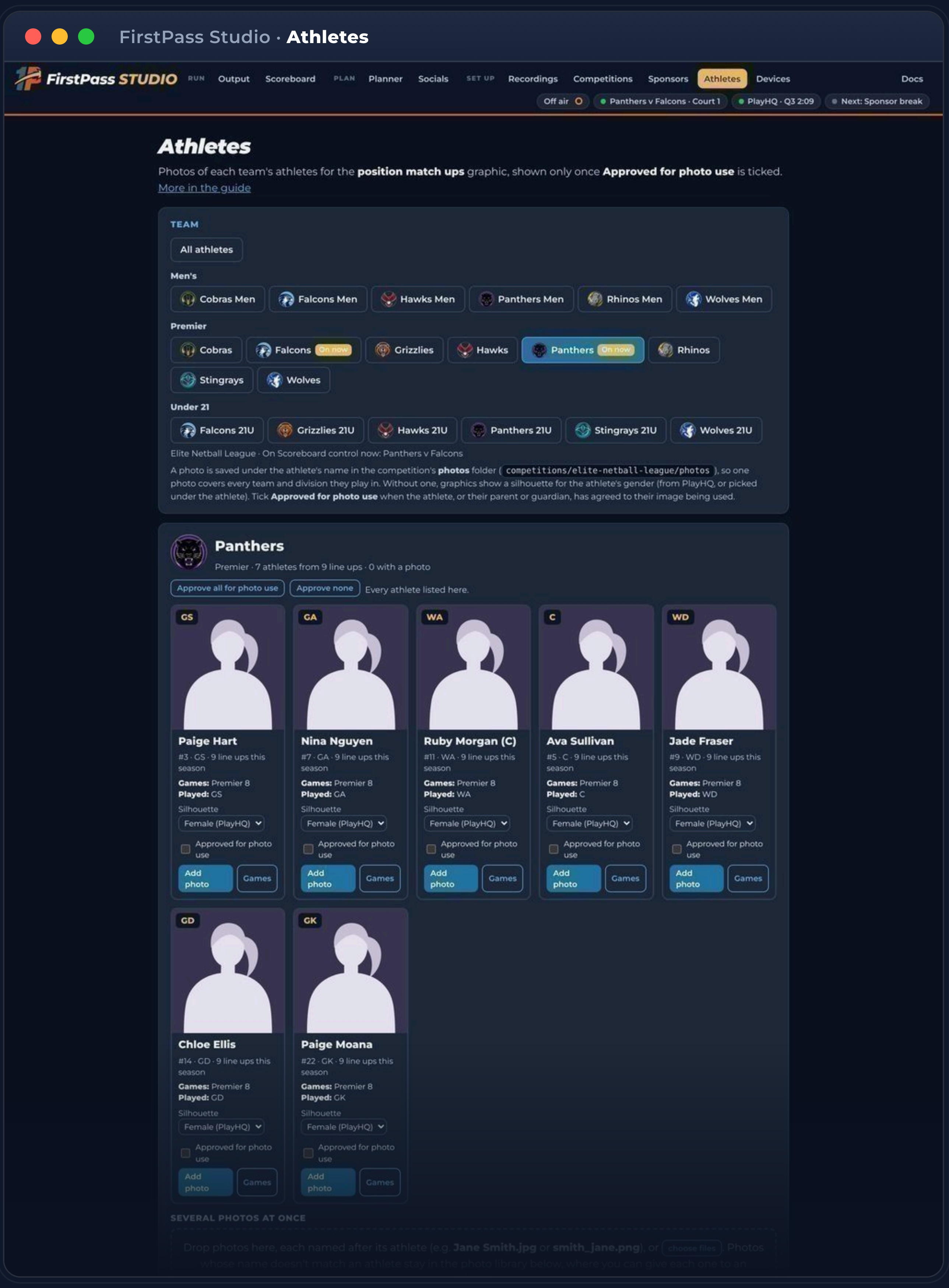
Athletes



PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

1 Photos

2 Positions and games played



Athletes

Photos of each team's athletes for the **position match ups** graphic, shown only once **Approved for photo use** is ticked. More in the guide

TEAM

Elite Netball League · On Scoreboard control now: Panthers v Falcons

A photo is saved under the athlete's name in the competition's **photos** folder (`competitions/elite-netball-league/photos`), so one photo covers every team and division they play in. Without one, graphics show a silhouette for the athlete's gender (from PlayHQ, or picked under the athlete). Tick **Approved for photo use** when the athlete, or their parent or guardian, has agreed to their image being used.

Panthers

Premier · 7 athletes from 9 line ups · 0 with a photo

Every athlete listed here.

GS **Paige Hart**
#3 · GS · 9 line ups this season
Games: Premier 8
Played: GS
Silhouette: Female (PlayHQ)
☐ Approved for photo use

GA **Nina Nguyen**
#7 · GA · 9 line ups this season
Games: Premier 8
Played: GA
Silhouette: Female (PlayHQ)
☐ Approved for photo use

WA **Ruby Morgan (C)**
#11 · WA · 9 line ups this season
Games: Premier 8
Played: WA
Silhouette: Female (PlayHQ)
☐ Approved for photo use

C **Ava Sullivan**
#5 · C · 9 line ups this season
Games: Premier 8
Played: C
Silhouette: Female (PlayHQ)
☐ Approved for photo use

WD **Jade Fraser**
#9 · WD · 9 line ups this season
Games: Premier 8
Played: WD
Silhouette: Female (PlayHQ)
☐ Approved for photo use

GD **Chloe Ellis**
#14 · GD · 9 line ups this season
Games: Premier 8
Played: GD
Silhouette: Female (PlayHQ)
☐ Approved for photo use

GK **Paige Moana**
#22 · GK · 9 line ups this season
Games: Premier 8
Played: GK
Silhouette: Female (PlayHQ)
☐ Approved for photo use

SEVERAL PHOTOS AT ONCE

Drop photos here, each named after its athlete (e.g. **Jane Smith.jpg** or **smith_jane.png**), or choose files Photos whose name doesn't match an athlete stay in the photo library below, where you can give each one to an

FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.

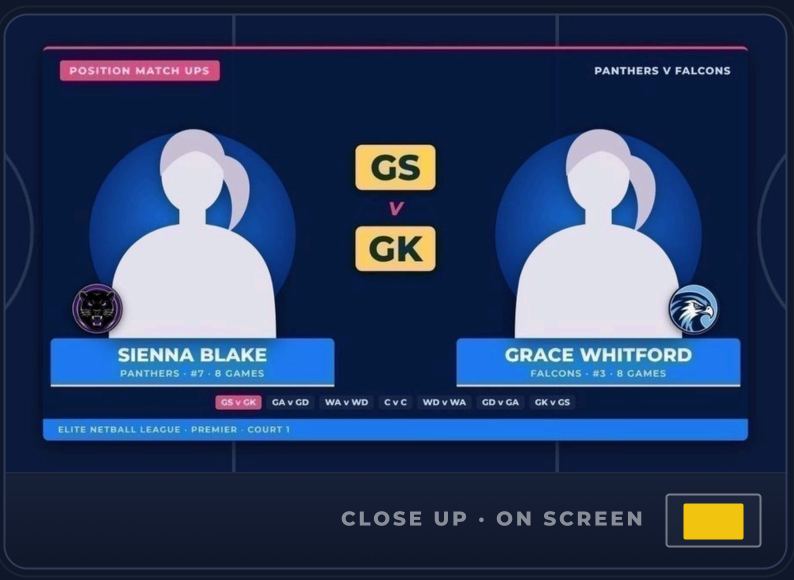


FIG 8.2 Position match ups use the photos, pairing by pairing.

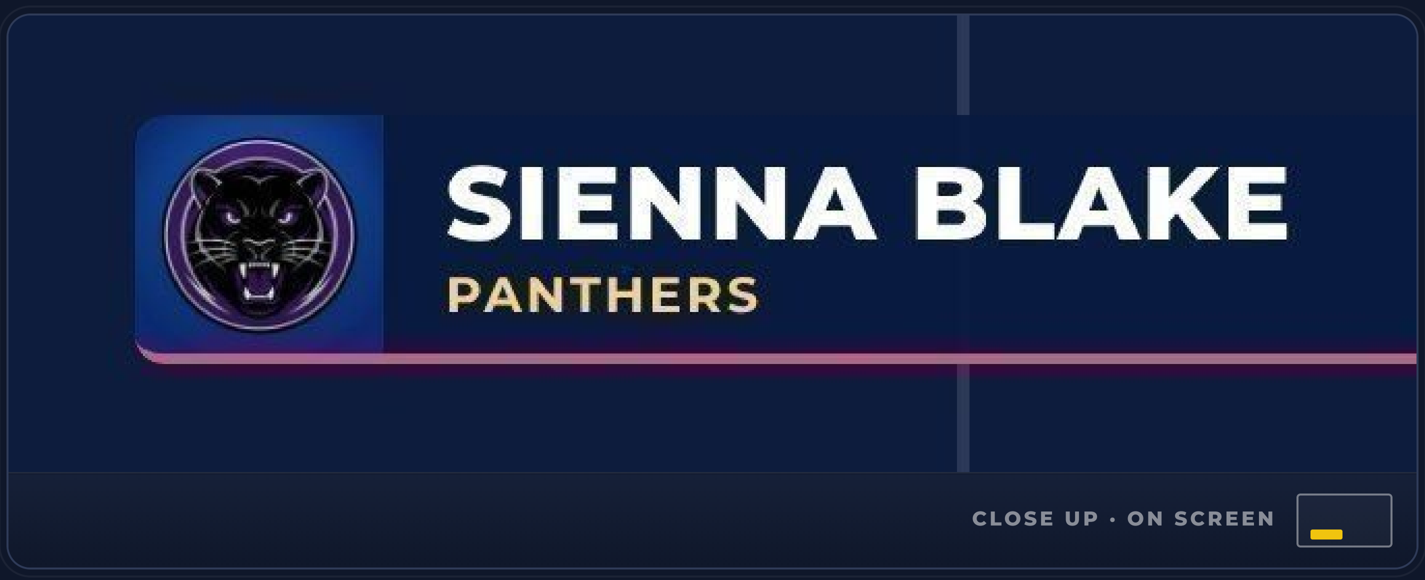
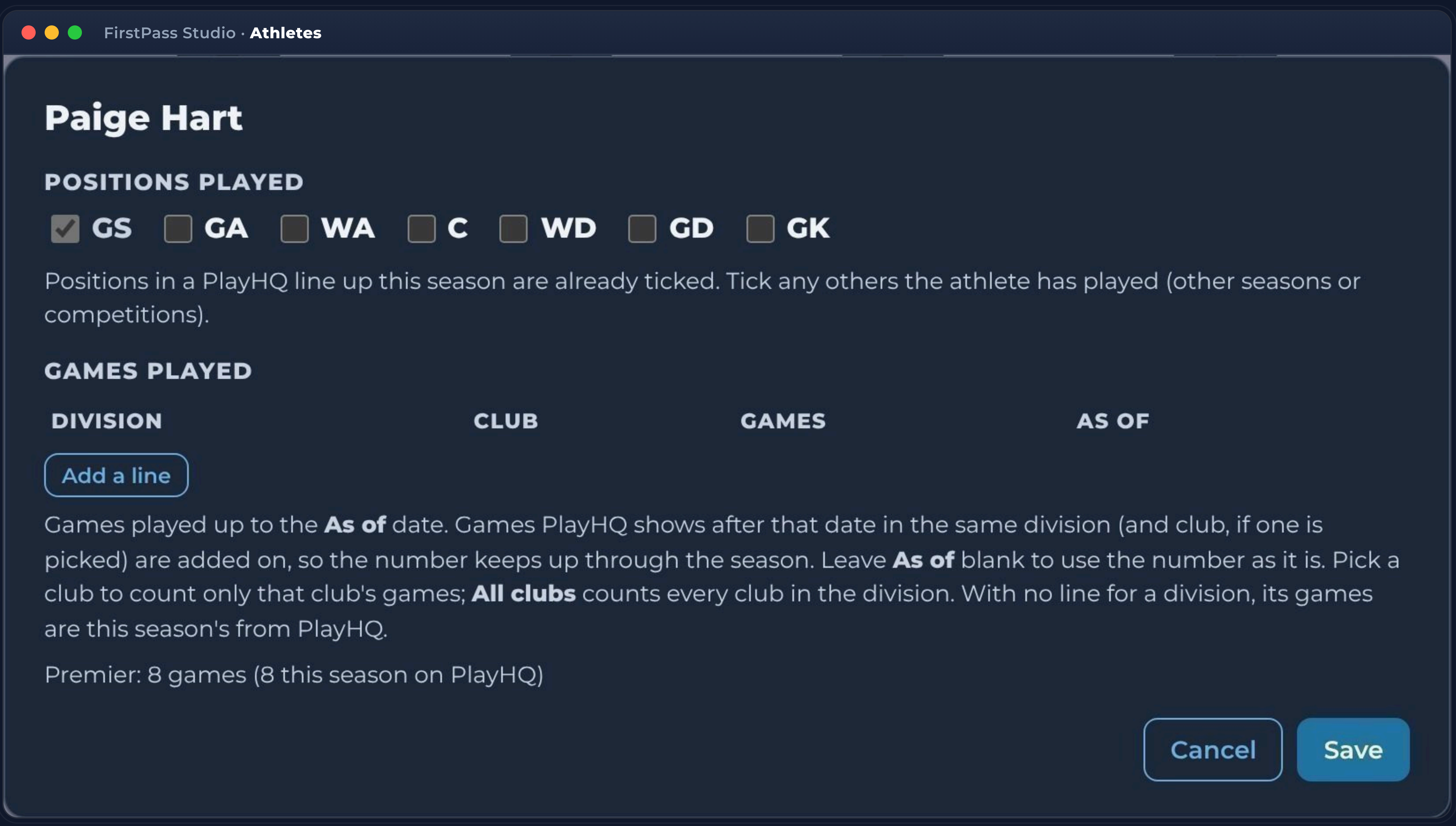


FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***
>
> CHAPTER 8 · ATHLETES

8.2 Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.



Paige Hart

POSITIONS PLAYED

☑ GS ☐ GA ☐ WA ☐ C ☐ WD ☐ GD ☐ GK

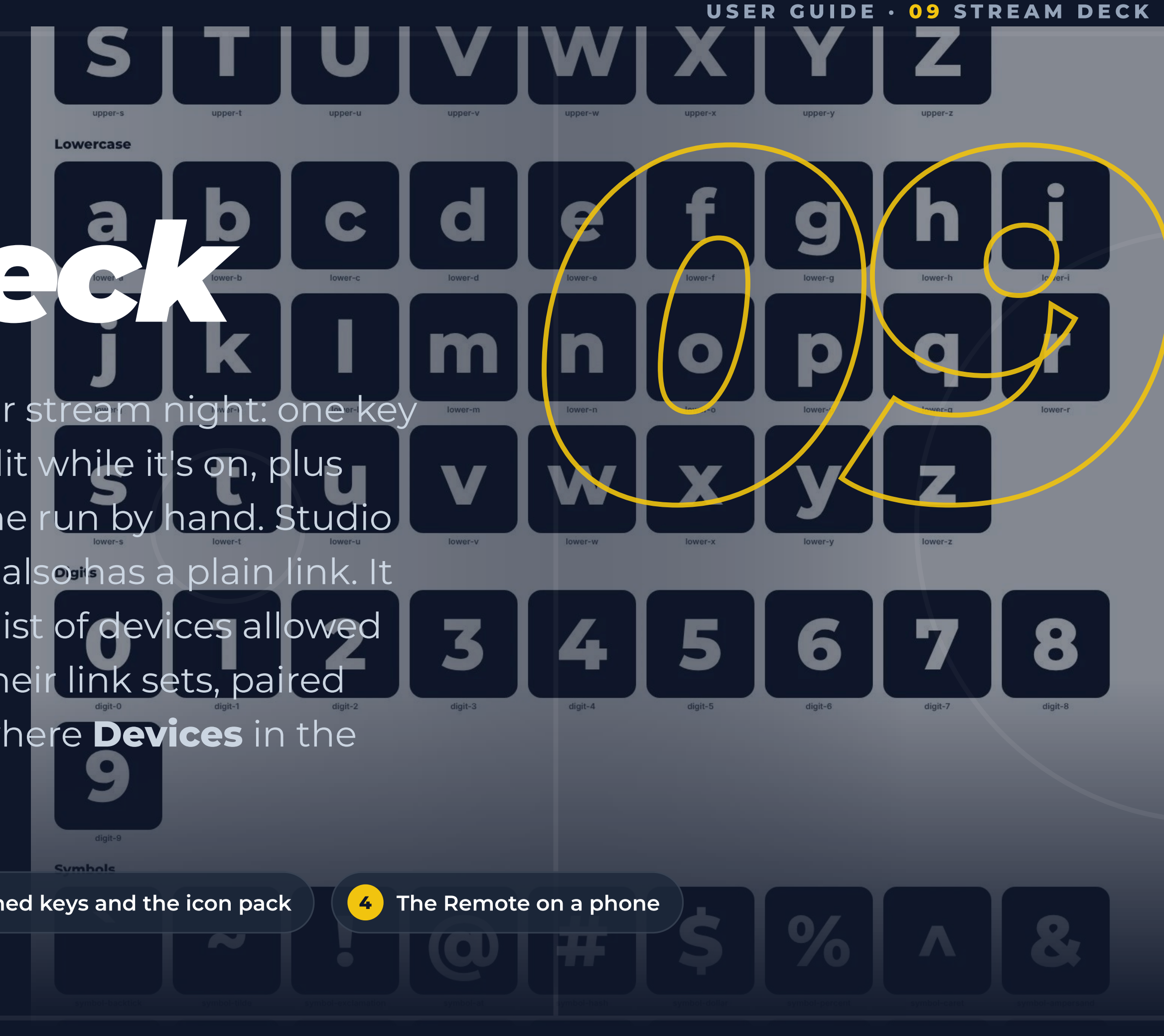
Positions in a PlayHQ line up this season are already ticked. Tick any others the athlete has played (other seasons or competitions).

GAMES PLAYED

DIVISION	CLUB	GAMES	AS OF

Games played up to the **As of** date. Games PlayHQ shows after that date in the same division (and club, if one is picked) are added on, so the number keeps up through the season. Leave **As of** blank to use the number as it is. Pick a club to count only that club's games; **All clubs** counts every club in the division. With no line for a division, its games are this season's from PlayHQ.

Premier: 8 games (8 this season on PlayHQ)

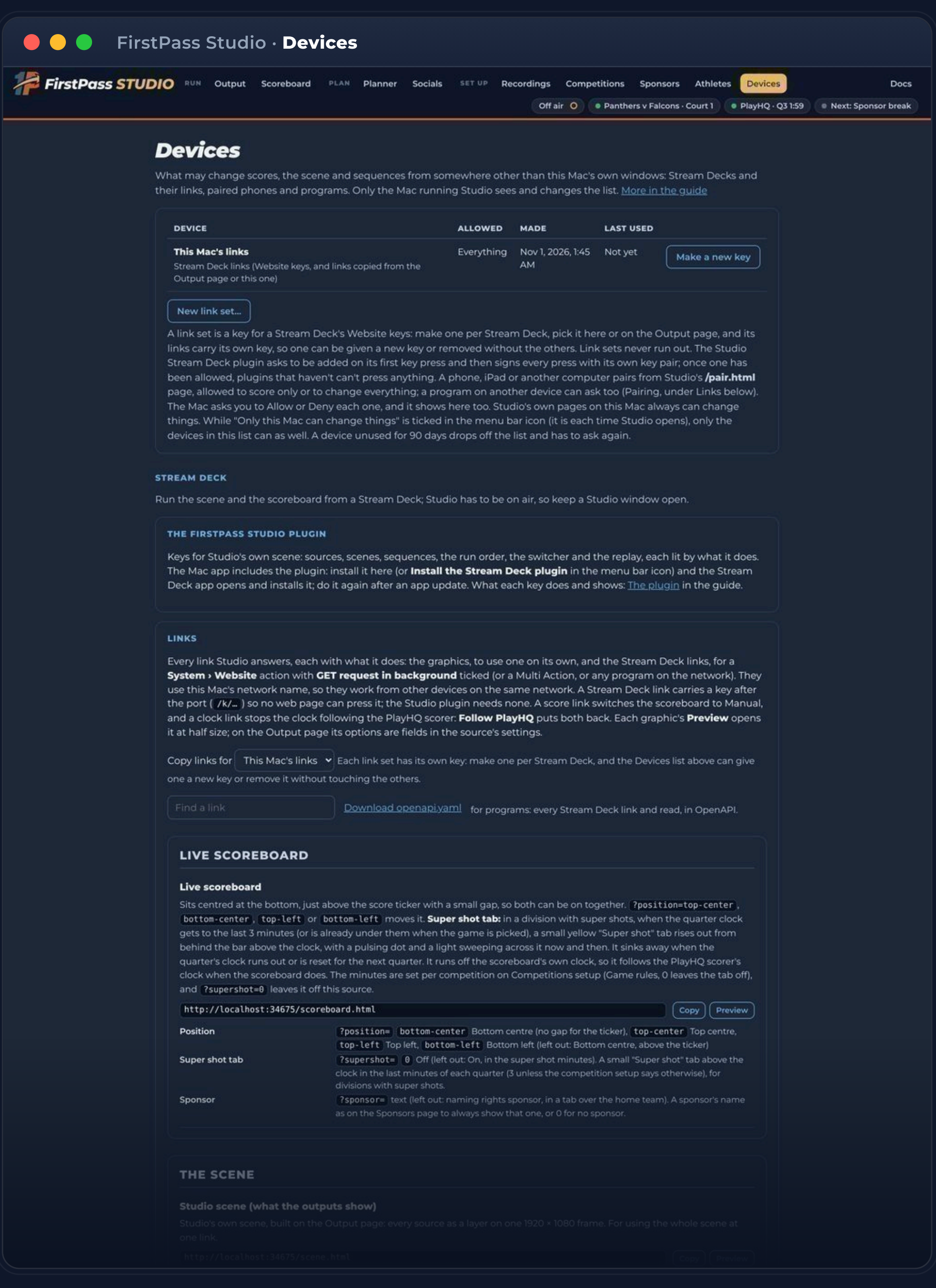


CHAPTER 9 OF 11

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link. It all lives on the **Devices** tab, under the list of devices allowed to change things (Stream Decks and their link sets, paired phones and programs), which is also where **Devices** in the menu bar icon goes.

1. The plugin
2. Links for everything else
3. Timed keys and the icon pack
4. The Remote on a phone



Devices

What may change scores, the scene and sequences from somewhere other than this Mac's own windows: Stream Decks and their links, paired phones and programs. Only the Mac running Studio sees and changes the list. More in the guide

DEVICE	ALLOWED	MADE	LAST USED	
This Mac's links Stream Deck links (Website keys, and links copied from the Output page or this one)	Everything	Nov 1, 2026, 1:45 AM	Not yet	Make a new key

New link set…

A link set is a key for a Stream Deck's Website keys: make one per Stream Deck, pick it here or on the Output page, and its links carry its own key, so one can be given a new key or removed without the others. Link sets never run out. The Studio Stream Deck plugin asks to be added on its first key press and then signs every press with its own key pair; once one has been allowed, plugins that haven't can't press anything. A phone, iPad or another computer pairs from Studio's **/pair.html** page, allowed to score only or to change everything; a program on another device can ask too (Pairing, under Links below). The Mac asks you to Allow or Deny each one, and it shows here too. Studio's own pages on this Mac always can change things. While "Only this Mac can change things" is ticked in the menu bar icon (it is each time Studio opens), only the devices in this list can as well. A device unused for 90 days drops off the list and has to ask again.

STREAM DECK

Run the scene and the scoreboard from a Stream Deck; Studio has to be on air, so keep a Studio window open.

THE FIRSTPASS STUDIO PLUGIN

Keys for Studio's own scene: sources, scenes, sequences, the run order, the switcher and the replay, each lit by what it does. The Mac app includes the plugin: install it here (or **Install the Stream Deck plugin** in the menu bar icon) and the Stream Deck app opens and installs it; do it again after an app update. What each key does and shows: The plugin in the guide.

LINKS

Every link Studio answers, each with what it does: the graphics, to use one on its own, and the Stream Deck links, for a **System › Website** action with **GET request in background** ticked (or a Multi Action, or any program on the network). They use this Mac's network name, so they work from other devices on the same network. A Stream Deck link carries a key after the port (`/k/…`) so no web page can press it; the Studio plugin needs none. A score link switches the scoreboard to Manual, and a clock link stops the clock following the PlayHQ scorer: **Follow PlayHQ** puts both back. Each graphic's **Preview** opens it at half size; on the Output page its options are fields in the source's settings.

Copy links for [This Mac's links] Each link set has its own key: make one per Stream Deck, and the Devices list above can give one a new key or remove it without touching the others.

Find a link Download openapi.yaml for programs: every Stream Deck link and read, in OpenAPI.

LIVE SCOREBOARD

Live scoreboard

Sits centred at the bottom, just above the score ticker with a small gap, so both can be on together. `?position=top-center`, `bottom-center`, `top-left` or `bottom-left` moves it. **Super shot tab:** in a division with super shots, when the quarter clock gets to the last 3 minutes (or is already under them when the game is picked), a small yellow "Super shot" tab rises out from behind the bar above the clock, with a pulsing dot and a light sweeping across it now and then. It sinks away when the quarter's clock runs out or is reset for the next quarter. It runs off the scoreboard's own clock, so it follows the PlayHQ scorer's clock when the scoreboard does. The minutes are set per competition on Competitions setup (Game rules, 0 leaves the tab off), and `?supershot=0` leaves it off this source.

`http://localhost:34675/scoreboard.html` Copy Preview

Position	`?position=` `bottom-center` Bottom centre (no gap for the ticker), `top-center` Top centre, `top-left` Top left, `bottom-left` Bottom left (left out: Bottom centre, above the ticker)
Super shot tab	`?supershot=` `0` Off (left out: On, in the super shot minutes). A small "Super shot" tab above the clock in the last minutes of each quarter (3 unless the competition setup says otherwise), for divisions with super shots.
Sponsor	`?sponsor=` text (left out: naming rights sponsor, in a tab over the home team). A sponsor's name as on the Sponsors page to always show that one, or 0 for no sponsor.

THE SCENE

Studio scene (what the outputs show)

Studio's own scene, built on the Output page: every source as a layer on one 1920 × 1080 frame. For using the whole scene at one link.

`http://localhost:34675/scene.html`

FIG 9.1

The Devices tab: the devices allowed to change things at the top, then the plugin, then Links: every link Studio answers, each with what it does, Copy and its settings.

9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Each kind of key has its own colour and picture, so you can see what a key does at a glance. The numbered disc in the top right corner shows which output it acts on, in that output's colour.

KEY	LOOKS	PRESS	SHOWS
Source	Teal, an eye (pink with a speaker for video sound)	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; lit while it's on, however it was turned on
Scene	Blue, stacked layers	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	Purple, a film strip	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Run order	Yellow, a numbered list	Play next (default), skip what's next, load a saved order, clear the list	What plays next, like **Play next ▶ / Quarter break / +2 more**; lit while something is queued, **empty** when nothing is; a Load key names its order and how many items it holds
Switcher	Red, a camera; no disc	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air; a yellow dot while the switcher isn't answering
Replay	Grey, a replay arrow; no disc	Play or pause, next or previous moment, nudge, stop the 4K replay or practice game running	What it acts on (**4K replay**, **Practice**, **Demo**); lit while one runs

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). Its disc shows the output's number in its colour (Output 1 orange, 2 pink, 3 blue, 4 green, as on the Output page), so two keys for the same source on different outputs can't be mixed up. A key for a removed output loses its disc and says **no output**.

> **“Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.**
>
> CHAPTER 9 · STREAM DECK

9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Devices tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Devices tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices can take that one away without touching your other keys. The Devices tab lists every one under **Links**, with what it does, the settings it takes and **Copy**; **Find** narrows the list, a graphic has **Preview** and a status read has **Try**. The useful ones:

GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Run order	Play what's next, skip it, clear the list, add a sequence or scene, load a saved order.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.

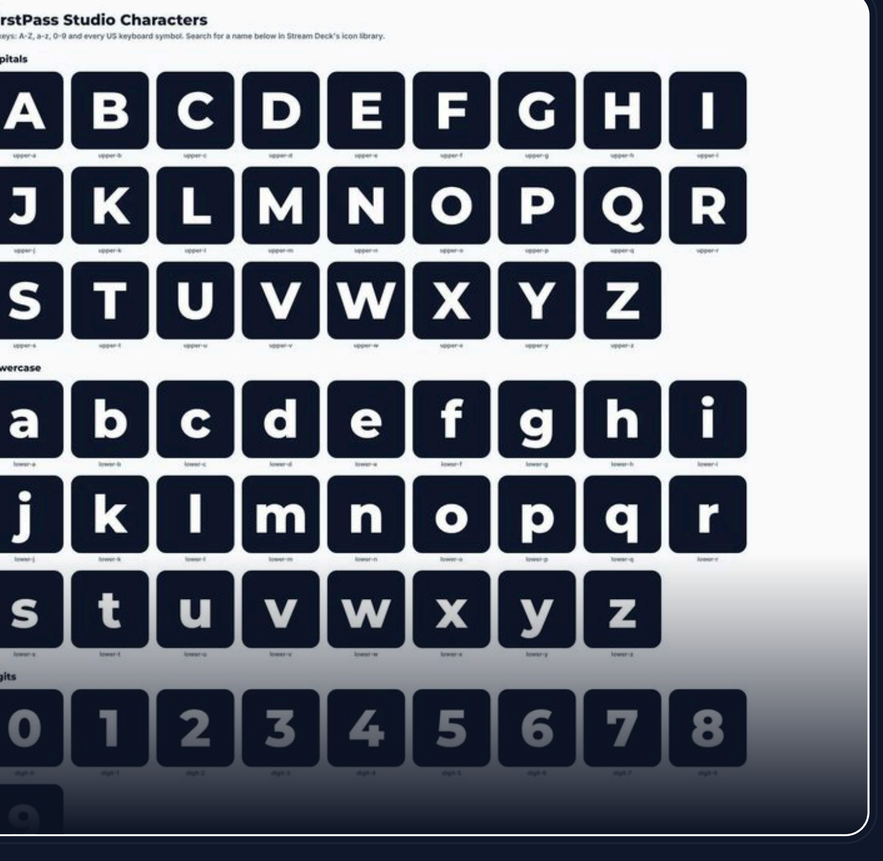
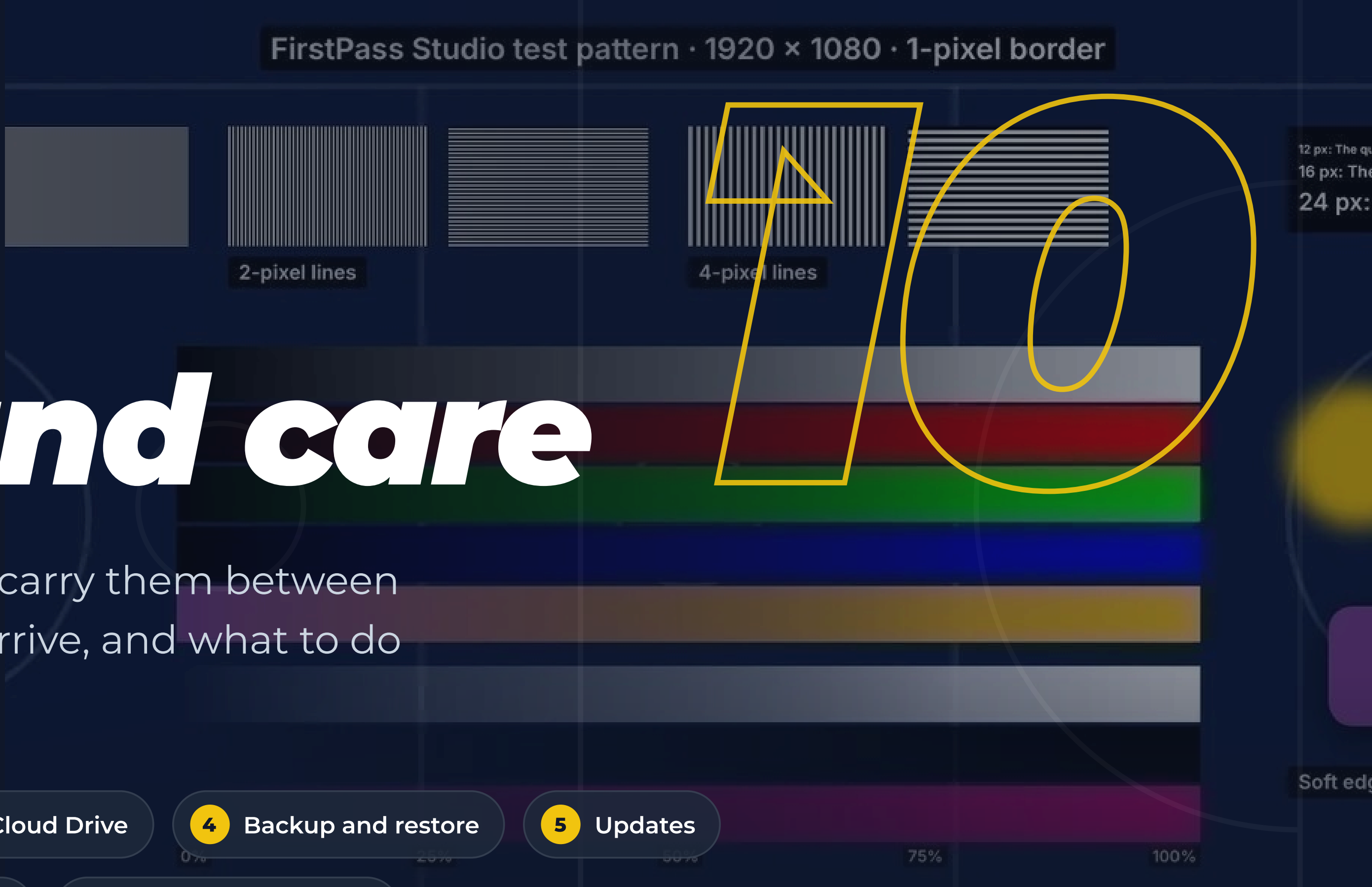

FirstPass Studio Characters
94 keys: A–Z, a–z, 0–9 and every US keyboard symbol. Search for a name below in Stream Deck's icon library.
Capitals
Lowercase
Digits

FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.

9.4 The Remote on a phone

No Stream Deck to hand, or away from the desk? The **Remote** puts the status board and a Stream Deck's sequence keys on a phone. On the Mac, press **Open on your phone** at the top of the Socials tab, pick **Remote** and scan the QR code with the phone's camera, on the same Wi-Fi; the first time, choose **Allow everything** on the Mac when it asks, then add the page to the Home Screen to open it like an app (it's the Studio link with `/remote.html` on the end). At the top it says what's on at a glance: on air or off air and which outputs, the game's score, clock and whether PlayHQ is scoring it, the sequence playing and any about to start. Under that, **Play next ▶** and **Skip** for the run order, **Next step**, **Pause** and **Stop** for the sequence playing, and a key for every sequence in its colour, lit while it plays: press it to play the sequence on Output 1, press it again to stop it (which does its Then, as on the Output page). While the scoreboard is on **Manual** (a score press on the Mac or a Stream Deck switches it), a **Scoring by hand** card comes up with each team's score and +1, +2 (where super shots are worth 2) and −1 keys, the clock's Start and Stop and Next quarter; it goes away once the scoreboard follows PlayHQ again. A phone paired to score only can use those keys too.


FirstPass Studio test pattern · 1920 × 1080 · 1-pixel border
2-pixel lines
4-pixel lines
10

CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

1 The 1P menu
2 The data folder
3 Settings in iCloud Drive
4 Backup and restore
5 Updates
6 Links and other devices
7 When something isn't right
8 Data and bandwidth

10.1 The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices**; **User guide**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up...** and **Restore...**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Appearance** (Follow the Mac, Light, Dark), **Only this Mac can change things**, **What's new...** and Quit.

This guide comes with Studio: **User guide** in the menu bar icon, **Help > FirstPass Studio User Guide** while a Studio window is in front open it in a window of its own, and it is the **Docs** tab (⌘0) in any Studio window, with no internet needed. Phones and other computers on the network can read it too, at the Studio link with `/guide/` on the end.

10.2 The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

10.3 Settings in iCloud Drive

Settings folder > Move to iCloud Drive... moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac...** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac...** undoes the move. Studio goes off air for a moment for each and comes straight back.

10.4 Backup and restore

Back up... saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore...** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

10.5 Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

> “ ***Updates download by themselves and install when the app quits, never mid-stream.***
>
> CHAPTER 10 · SETTINGS AND CARE

10.6 Links and other devices

Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

Secure phone link. A phone's browser lets a page use its microphone (a phone as a backup commentator headset) or its share sheet only over https, which the `.local` link isn't. Turned on in the **Secure phone link** card on the Devices tab, Studio also answers at a name of its own on the internet, such as `https://k3f9x2.studio.firstpass.pivt.au:8443`, with a free Let's Encrypt certificate it renews by itself, and every **Open on your phone** QR code uses it; a phone pairs once more on the new link. The graphics, the Stream Deck and the http links carry on the same. It needs the internet at the venue, and a few routers refuse a name on the internet that points at their own network: then the https link doesn't open there and the http one still does. It's set up once for PIVT, then each Studio Mac is given the same three things:

1. In AWS, **Route 53 › Hosted zones › Create hosted zone**: domain `studio.firstpass.pivt.au`, Public. Note its **Hosted zone ID** (starts with Z) and its 4 NS values.
2. In GoDaddy's DNS for **pivt.au**, add 4 records: type **NS**, name `studio.firstpass`, one NS value each. It can take up to an hour to work.
3. In **IAM**, a policy `FirstPassStudioDNS` allowing `route53:ChangeResourceRecordSets` and `route53:ListResourceRecordSets` on that hosted zone only, and `route53:GetChange`; a user `firstpass-studio-dns` with it attached and no console access; then **Security credentials › Create access key** (Application running outside AWS). Keep the secret somewhere safe: AWS shows it once.
4. On each Studio Mac, paste the hosted zone ID, the access key ID and the secret into the card and press **Turn on**. It takes about a minute the first time, and the card says each step and, if one fails, why and what to check. The secret goes in the Mac's Keychain.

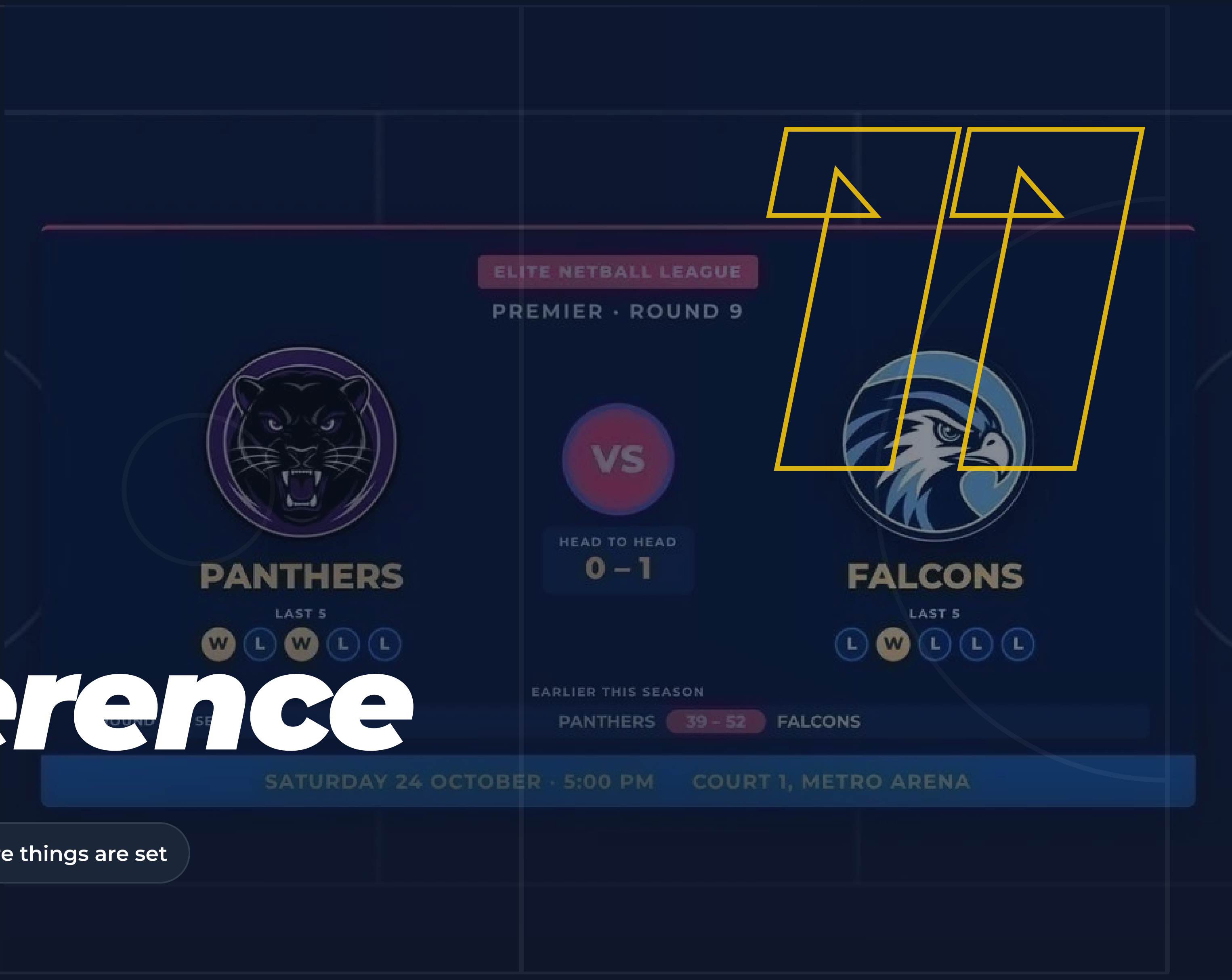
FirstPass HQ. A club that records FirstPass stats keeps them in its own FirstPass HQ, and Studio signs in to each club, once per Mac, in the **FirstPass HQ** card on the Devices tab. Press **Sign in** beside the club: its Google sign-in page opens in your browser, you sign in with your own account, and the browser says **Signed in**; nothing is typed into Studio. A club admin first gives your account the **Studio** role in HQ: without it Studio keeps nothing and the card says so. The sign-in stays in the Mac's Keychain, never in iCloud Drive or a backup, and is checked again every half hour. **Sign out**, or the club removing or pausing your role, deletes it and every FirstPass stat Studio kept for that club. FirstPass stats are about athletes, many under 18, so they stay private: only this Mac, and phones or browsers paired over the Secure phone link and opened on it, can see them (a device paired at the `.local` link pairs once more on the secure link), whatever **Only this Mac can change things** says. While a club is signed in, Studio reads the FirstPass stats of the game on the scoreboard as it is played, when that club recorded it in HQ, and keeps them up to date on its own. Studio deletes a game's stats 30 days after it, and never writes them to the debug log.

10.7 When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering, or (PlayHQ chip purple, **PlayHQ not live**) the scorebench is off line. Fix the scorebench, or switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

10.8 Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game while it's on (or as set, down to one a second; every 30 seconds just before it, every 5 minutes otherwise), the ticker's other courts every 30 seconds, the fixture and ladders about once a minute while a game is on and every 5 minutes when none is, and the rest once when a graphic loads. **Refresh list** on Scoreboard control reads the night's games from PlayHQ there and then. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth. The PlayHQ line above the score on the Output page's status board shows it live: the ping, the bandwidth over the last minute and the data downloaded since Studio went on air.



CHAPTER 11 OF 11

Quick reference

1. Keyboard
2. Stream night in nine lines
3. Where things are set

11.1 Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

11.2 Stream night in nine lines

01 Open Studio from the 1P menu and keep the Output page open all night.

02 Check the competition in use and the court on Competitions setup.

03 On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

04 Turn on the video output your switcher takes and show one graphic to see it land.

05 Play a few seconds of music and an animation with sound.

06 Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

07 Go to the Live game scene and let the sequences run the breaks.

08 Pick finals winners as they happen; turn on lower thirds when people appear.

09 Press Stream close, turn the output off, and send the sponsor report.

11.3 Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links, and the devices allowed to change things	Devices
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

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